



BASE24[®]

ANSI Host Interface Manual

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Preface

This manual describes the ANSI Host Interface process, which is used to establish a communications link between BASE24 and a host.

The ANSI Host Interface process uses an external message based on the American National Standards Institute (ANSI) X9.2–1980 standard. The ISO Host Interface process, which uses an external message based on the the International Organization for Standardization (ISO) 8583 standard, is documented in the ***BASE24 ISO Host Interface Manual***.

Audience

This manual is intended as a reference source for those persons responsible for the interface between BASE24 and a host. It provides a comprehensive description of all the functions performed by the ANSI Host Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 ISO Host Interface Manual*** discusses the ISO Host Interface. The ISO Host Interface allows communications with a host using the International Organization for Standardization (ISO) message format.

- The ***BASE24 Base Files Maintenance Manual*** documents BASE24 file maintenance screen fields, including those for the Host Configuration File (HCF) and Key File (KEYF). These files are used in configuring the ANSI Host Interface.
- The ***BASE24 Transaction Security Manual*** describes in detail how to configure key management and message authentication.
- The ***BASE24 Network Control Operators Manual*** documents the BASE24 file maintenance screen fields for the Log Message Configuration File (LMCF). The LMCF can be used to modify the log messages generated by the ANSI Host Interface process.
- The ***BASE24 Logical Network Configuration File Manual*** documents the Logical Network Configuration File (LCONF), the entry program used to access the LCONF, and the assigns and params that are stored in the LCONF. The LCONF is used in configuring the ANSI Host Interface.
- The ***BASE24 Text Command Reference Manual*** documents in detail the text commands that can be sent to the ANSI Host Interface process.
- The ***BASE24 Log Message Reference Manual*** describes how to access log message documentation. The documentation includes causes and remedies for ANSI Host Interface log messages.

Software

This manual documents standard processing for the following BASE24 products:

- BASE24-atm Release 5.0
- BASE24-pos Release 5.0

Customer-specific code modifications (i.e., RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24 ANSI Host Interface Manual*** is divided into the following sections and appendixes:

Section 1, “Introduction,” provides an introduction to the BASE24 ANSI Host Interface process, identifies the files used by the ANSI Host Interface, and identifies the internal and external message types supported by the ANSI Host Interface.

Section 2, “Configuration and Control,” provides information about the configuration and control of the ANSI Host Interface.

Section 3, “Network Management and Operations,” provides an overview of the network management and operations for the ANSI Host Interface. It discusses monitoring ANSI Host Interface operation and the network management messages that are used in communicating with the host.

Section 4, “Store-and-Forward Processing,” provides an overview of store-and-forward processing and store-and-forward message flows.

Section 5, “BASE24-atm Financial Message Handling,” provides an overview of BASE24-atm financial transaction processing and message flows.

Section 6, “BASE24-pos Financial Message Handling,” provides an overview of BASE24-pos financial transaction processing and message flows.

Appendix A, “Processing Information,” contains ANSI Host Interface processing information. It describes the steps that the ANSI Host Interface performs at startup and as a result of a warmboot request, and identifies the text commands that the ANSI Host Interface can process.

Appendix B, “Network Management Messages,” describes the Network Management Message (NMM) used by the ANSI Host Interface when sending network management messages to a host.

Appendix C, “BASE24-atm Standard External Messages,” describes the BASE24-atm Standard External Message (SEM) used by the ANSI Host Interface when sending BASE24-atm messages to a host.

Appendix D, “BASE24-pos Standard External Messages,” describes the BASE24-pos Standard External Message (PSEM) used by the ANSI Host Interface when sending BASE24-pos messages to a host.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-DH010-01 for the ***BASE24 ANSI Host Interface Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 12/93 for December, 1993) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Section 1

Introduction

This section introduces the ANSI Host Interface and provides an overview of ANSI Host Interface functionality. It also identifies the BASE24 files used by the ANSI Host Interface, and the message types supported.

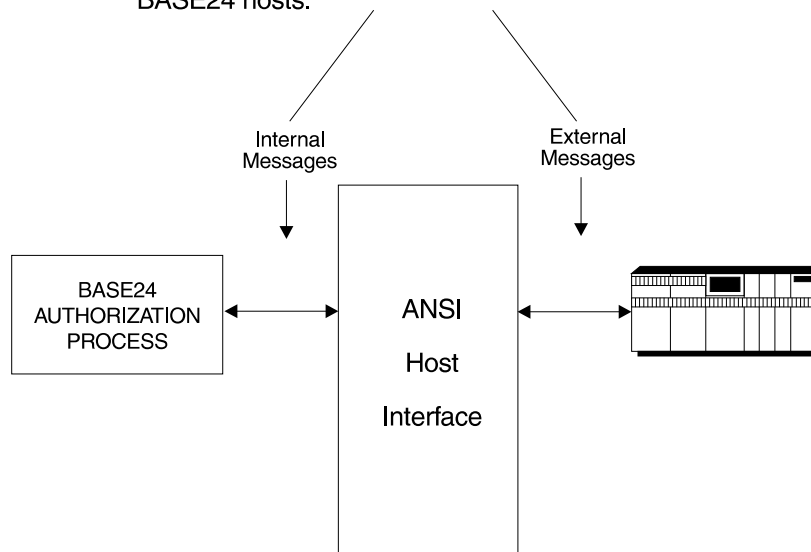
ANSI Host Interface Overview

The ANSI Host Interface process, shown in the illustration below, is responsible for the online message transfer between BASE24 and its hosts. This process performs the following functions:

- Controls message traffic to and from BASE24 hosts, enforcing an orderly message exchange.
- Translates messages from the internal message format to an external message format recognizable to the host.
- During periods when the host is unavailable, stores certain types of messages bound for the host for later transmittal.

ANSI Host Interface Process

The BASE24 ANSI Host Interface process translates messages between BASE24's internal message format and an external format recognizable to BASE24 hosts.



BASE24 supports communication with hosts using two external message formats: one based on the American National Standards Institute (ANSI) X9.2–1980 standard and one based on the International Organization for Standardization (ISO) 8583 standard. The ISO-based format is newer and more flexible than the ANSI-based format.

Because of the substantial differences in the format and processing of the BASE24 ANSI and ISO external messages, two separate Host Interface processes are required. The Host Interface processing described in this manual applies only to hosts using the ANSI-based external message. For a description of the ISO Host Interface process used to support the ISO-based external message, refer to the ***BASE24 ISO Host Interface Manual***.

Hosts

A host is an external mainframe computer system with which BASE24 interacts—either online or by batch tape—to authorize transactions and/or update cardholder records.

The ANSI Host Interface process is responsible for the online interaction between BASE24 and its hosts. The BASE24 Refresh and Super Extract processes are responsible for the batch tape interaction between BASE24 and its hosts.

Although the term *host* generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing.

Data Processing Centers

BASE24 refers to hosts as *data processing centers* (DPCs). Generally, the terms *DPC* and *host* are synonymous. However, the term *host* implies either a host computer or the organization responsible for a host computer, and the term *DPC* always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with BASE24, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration, a DPC represents a single logical host. A host site can establish multiple DPCs; however, BASE24 considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and defined to BASE24 via the Host Configuration File (HCF). Each DPC must have a record in the HCF in order to be recognized by BASE24. The HCF contains options at the DPC level that allow hosts to specify individual processing for each DPC. The HCF also defines the stations used for communication between BASE24 and the DPC.

Stations

Stations provide the means by which communication takes place between BASE24 and a DPC. All messages sent to a DPC are actually sent to the stations defined for the DPC. Likewise, all messages received from a DPC are actually received from stations.

Stations and their lines must be defined to BASE24 via the BASE24 Network Environment Source File (NEFS). Stations and lines are tied to their respective DPCs through the Host Configuration File (HCF). At least one station is required for each DPC, and a maximum of 32 stations are allowed per DPC. For information on defining stations and lines to BASE24, refer to the ***BASE24 Network Environment Source File Manual***.

Note: All protocol-level functions between BASE24 and the stations are actually handled by the Nucleus using Tandem's communications control software. The ANSI Host Interface process sends and receives messages via the Nucleus. It is the Nucleus' responsibility to maintain the communications links with the stations according to the protocols under which the stations are defined.

Issuer and Acquirer Hosts

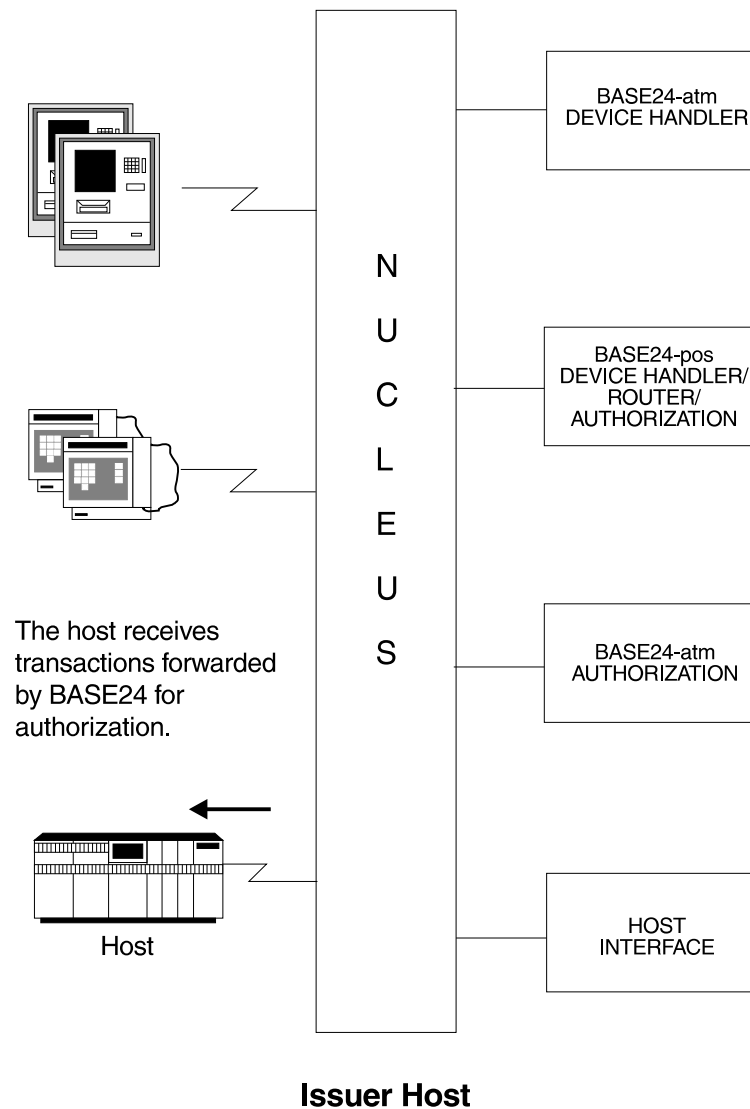
In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorize and process transactions.

Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorizing cardholder transactions and keeping track of cardholder accounts. *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

Issuer Hosts

Hosts acting on behalf of an issuer in a BASE24 message exchange are referred to as *issuer hosts*. Issuer hosts authorize and/or post transactions sent to them by BASE24, as shown below.

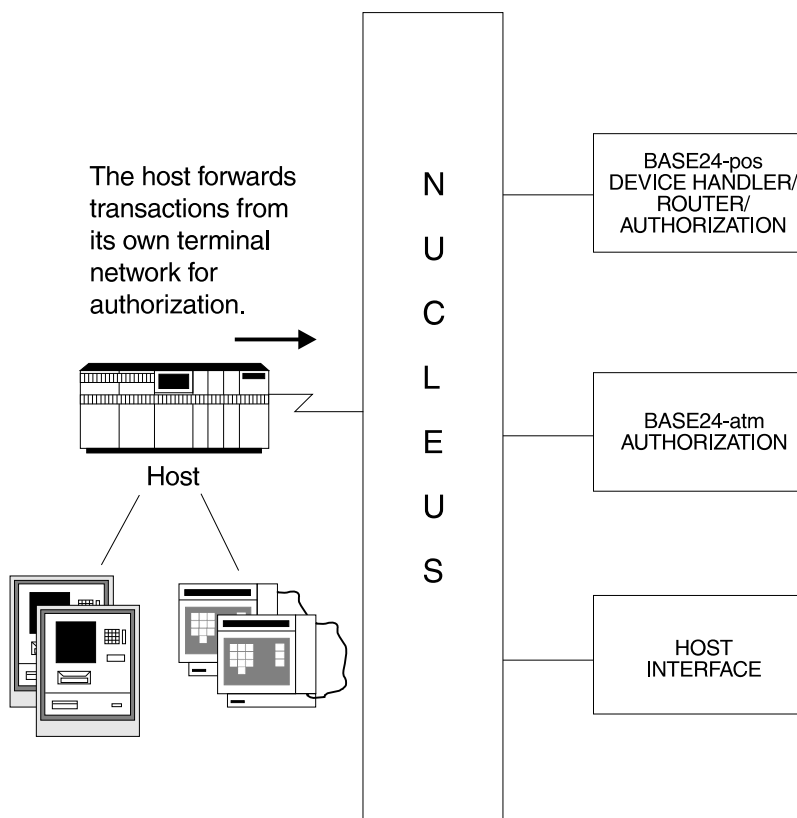
Note: Because of their general position in the transaction path, issuer hosts are also referred to as *back-end hosts*.



Acquirer Host

Hosts acting on behalf of an acquirer in a BASE24 message exchange are referred to as *acquirer hosts*. Acquirer hosts support ATM or POS networks external to BASE24 and forward transactions to BASE24 for authorization, as shown below.

Note: Because of their general position in the transaction path, acquirer hosts are also referred to as *front-end hosts*.



Acquirer Host

Issuer and Acquirer Host Support

BASE24 can function in the role of an issuer or an acquirer. When BASE24 sends financial transactions to an issuer host for authorization, BASE24 represents the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to BASE24 for authorization from an acquirer host, BASE24 represents the card issuer, even though BASE24 may only forward the transaction to an issuer host for authorization.

Hosts can also function on behalf of both card issuers (or account owners) and transaction acquirers, as would be the case when an institution issues cards and supports its own network of ATM or POS devices. In order to allow for the different processing requirements, ACI provides two ANSI Host Interface modules: the Standard Issuer Host Interface module and the Acquirer Host Interface module.

The Standard Issuer Host Interface module supports communications with issuer hosts only. It can send transaction requests to a host and accept transaction responses from a host, but it cannot accept transaction requests from a host.

The Acquirer Host Interface module supports communications with both issuer and acquirer hosts. It can send transaction requests and responses to a host and accept transaction requests and responses from a host.

The Standard External Message

The ANSI Host Interface uses a standard external message to communicate with its hosts. This message is based on the American National Standards Institute (ANSI) X9.2–1980 standard. Each product has its own message format or formats. Specifications for this message can be found in the following appendixes:

- Appendix B describes the Network Management Message (NMM) and the Network Management Message for Dynamic Key Management (NMMKEY) used by the ANSI Host Interface when sending network management messages to a host.
- Appendix C describes the BASE24-atm Standard External Message (SEM) used by the ANSI Host Interface when sending BASE24-atm messages to a host.
- Appendix D describes the BASE24-pos Standard External Message (PSEM) used by the ANSI Host Interface when sending BASE24-pos messages to a host.

Store-and-Forward Function

A store-and-forward function is incorporated into the ANSI Host Interface process to collect transaction messages during periods when a host is unavailable.

When a host is unavailable, messages required as notification to the host are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored. For more information on store-and-forward processing, refer to section 4.

Multiple Product Support

The same ANSI Host Interface process can be set up to process host messages for multiple BASE24 products. If an institution purchases multiple BASE24 products, the appropriate product code is bound into the ANSI Host Interface to allow it to recognize and process transaction messages for each product.

Message Processing

Due to the variances in processing that must occur in handling messages for different products, the ANSI Host Interface processes messages differently based on the message type and product ID for the message. For more information on how messages flow back and forth between the host and BASE24, refer to the following sections, based on the type of message:

- Network management message flows are documented in section 3.
- Store-and-forward message flows are documented in section 4.
- BASE24-atm message flows are documented in section 5.
- BASE24-pos message flows are documented in section 6.

File Usage

The ANSI Host Interface process uses a number of files in processing. The majority of the files used by the ANSI Host Interface are control files. Control files contain configuration information and parameters that determine how the ANSI Host Interface process operates. The files that contain configuration information for the ANSI Host Interface include:

- Host Configuration File (HCF)
- Key File (KEYF)
- Log Message Configuration File (LMCF)
- Logical Network Configuration File (LCONF)

For more information on how the LCONF and the HCF are used to configure the ANSI Host Interface process, refer to section 2. For more information on how the KEYF is used to configure the ANSI Host Interface process, refer to the ***BASE24 Transaction Security Manual***. For more information on how the LMCF is used to configure the ANSI Host Interface process, refer to appendix A.

In addition to the control files above, the ANSI Host Interface process uses the Store-and-Forward File (SAF) to store information about individual transactions processed by the ANSI Host Interface. For more information on how the SAF is used by the ANSI Host Interface, refer to section 4.

Supported Message Types

The ANSI Host Interface process supports the external message types shown below for transactions sent between BASE24 and a host.

Type	Description	BASE24 Product	
		ATM	POS
0200	Financial Transaction Request	✓	✓
0205	Statement Print Request *	✓	
0210	Financial Transaction Request Response	✓	✓
0215	Statement Print Request Response *	✓	
0220	Financial Transaction Advice	✓	✓
0221	Financial Transaction Advice Repeat	✓	✓
0230	Financial Transaction Advice Response	✓	✓
0231	Financial Transaction Advice Response Repeat	✓	✓
0400	Reversal	✓	✓
0401	Reversal Repeat	✓	✓
0410	Reversal Response	✓	✓
0411	Reversal Response Repeat	✓	✓
0500	Settlement Record		✓
0700	Balance Inquiry Request	✓	
0710	Balance Inquiry Request Response	✓	
0800	Network Management Message	✓	✓
5400	Interchange Adjustment	✓	✓
5401	Interchange Adjustment Repeat		✓
5410	Interchange Adjustment Response	✓	✓

Type	Description	BASE24 Product	
		ATM	POS
5411	Interchange Adjustment Response Repeat	✓	✓
9220	Store-and-Forward Message	✓	✓
9221	Store-and-Forward Message Repeat	✓	✓
9230	Store-and-Forward Message Response	✓	✓
9231	Store-and-Forward Message Response	✓	✓
9510	Store-and-Forward Settlement Record		✓
9511	Store-and-Forward Settlement Record Response		✓

* 0205 and 0215 messages are for specialized support of statement print functions. The ANSI Host Interface process does not support statement print transactions generated by the BASE24-atm Self-Service Banking (SSB) product; however, it can process statement print transactions generated by Device Handlers that are not using the SSB product.

Amount Size Restrictions

The largest amounts allowed in the ANSI external message are 8 positions for BASE24-atm and 9 positions for BASE24-pos. Internally, however, BASE24 supports larger amounts.

In the case of BASE24-atm transactions being sent to a host, if amount 1 (AMT-1) in the Standard Internal Message (STM) is larger than 8 positions, the ANSI Host Interface process will fail the transaction back to its originator. Conversely, if amounts 2 and 3 (AMT-2 and AMT-3) in the STM are larger than 8 positions, the ANSI Host Interface will simply truncate the left-most (high-order) digits when moving the amounts from the internal message to the external message.

In the case of BASE24-pos transactions being sent to a host, if any amount in the POS Standard Internal Message (PSTM) is larger than 9 positions, the ANSI Host Interface process will simply truncate the left-most (high-order) digits when moving the message from the internal message to the external message.

Section 2

Configuration and Control

This section describes the configuration and control of the ANSI Host Interface process and includes the following:

- Process configuration information
- Descriptions of institution- and DPC-level controls
- A description of how stations are defined and used by the ANSI Host Interface
- Descriptions of the timers used by the ANSI Host Interface and the processing performed when each timer expires
- An overview of configuring transaction security

LCONF Assigns and Params

The ANSI Host Interface uses a number of Logical Network Configuration File (LCONF) assigns and params in its processing. These assigns and params are identified below. For more information on configuring the ANSI Host Interface via these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

Assigns

The following assigns identify the names of files that contain ANSI Host Interface configuration and control information, or that the ANSI Host Interface uses during processing. The ANSI Host Interface reads these assigns regardless of the products supported.

HCF — Identifies the name of the Host Configuration File (HCF).

KEYF — Identifies the name of the Key File (KEYF).

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

PLF — Identifies the name of the PCS Log File (PLF) and indicates BASE24-pcs is being used to monitor the network. If Logger is being used as the network logging facility and if the ANSI Host Interface finds this assign in the LCONF, the ANSI Host Interface sends any log messages to the BASE24-pcs system, in addition to sending the messages to Logger. If Tandem's Event Management Service (EMS) is being used as the network logging facility, ANSI Host Interface messages are automatically filtered to the BASE24-pcs system.

SAF — Identifies the name of the Store-and-Forward File (SAF).

SECURE-DEV1

SECURE-DEV2 — Identify the names and device subtypes of the security modules the ANSI Host Interface must open if BASE24 PIN management is performed via a security module. If BASE24 PIN management is performed via a security module, at least one of these assigns is required.

Params

The following params contain configuration information for the ANSI Host Interface. Unless otherwise noted, the ANSI Host Interface reads these params regardless of the products supported.

ATM-LN-CUTOVER — Specifies logical network cutover time for BASE24-atm. This param is only read if BASE24-atm support is bound in to the ANSI Host Interface.

LOGICAL-NET — Specifies the name of the logical network of which the ANSI Host Interface is a part.

POS-LN-CUTOVER — Specifies logical network cutover time for BASE24-pos. This param is only read if BASE24-pos support is bound in to the ANSI Host Interface.

REVERSE-BAL-INQ — Specifies whether balance inquiry transactions can be reversed. This param is only read if BASE24-atm support is bound in to the ANSI Host Interface.

SECURE-DEV-CONF-CMD100 — Specifies the command 100 text sent to Atalla Network Security Processors (NSPs) when the ANSI Host Interface is initialized or warmbooted.

SECURE-DEV-CONF-TXT — Specifies the configuration text sent to Atalla NSPs when the ANSI Host Interface is initialized or warmbooted.

SECURE-DEV-FAIL-NOTIFY — Specifies the interval, in number of transactions, at which the ANSI Host Interface logs messages once a security module is determined to be inoperative.

SECURE-DEV-MONITOR-THRESHOLD — Specifies the interval, in number of transactions processed by the security module defined in the SECURE-DEV1 assign, at which the security module defined in the SECURE-DEV2 assign is checked.

SECURE-DEV-TIM-LMT — Specifies the number of seconds the ANSI Host Interface waits for a response from the security module.

SECURE-DEV-TYP — Specifies the type of security modules named in the SECURE-DEV1 and SECURE-DEV2 assigns.

Configuring a DPC

Each DPC with which BASE24 communicates requires a record in the Host Configuration File (HCF). DPCs are assigned their DPC numbers and are tied to the ANSI Host Interface processes with which they communicate via the HCF.

A number of processing options are controlled at the DPC level, which enable institutions to set up DPCs with individual processing requirements. The processing options available at the DPC level are stored in the HCF, and are described below.

For more information on fields in the HCF, refer to the ***BASE24 Base Files Maintenance Manual***.

Specifying Timer Lengths

The ANSI Host Interface process uses internal timers to control host message traffic. The length of each of these timers is specified in the HCF or KEYF. Some timer intervals are specified once at the DPC level, while others are specified at the DPC and product level (e.g., you can specify one time limit for inbound BASE24-atm messages and a different time limit for inbound BASE24-pos messages).

The table below identifies each timer used by the ANSI Host Interface, along with the level at which it is defined and the file in which it is defined. Each of these timers is discussed later in this section.

Timer	DPC Level	DPC/Product Level
Network management message timer	HCF	
Extended network management message timer	HCF	
Store-and-forward message timer	HCF	
Wait-for-traffic timer	HCF	
Performance timer	HCF	
Outbound request timer		HCF
Inbound request timer		HCF

Timer	DPC Level	DPC/Product Level
Completion acknowledgment timer		HCF
PIN key exchange timer	KEYF	
Clear old key timer	KEYF	

Interspersing Store-and-Forward Messages

Institutions can specify how the host receives store-and-forward messages. Store-and-forward messages can be interspersed with request messages, or all store-and-forward messages can be sent to the host prior to any further requests.

Interspersing store-and-forward messages allows other transactions to be processed during store-and-forward activity, while receiving all store-and-forward messages prior to further requests may cause requests to be processed offline or be declined because of an inability to send them to the host.

Institutions specify the method of store-and-forward processing via the STORE AND FORWARD field on HCF screen 1.

Specifying Single-Threaded and Multithreaded Processing

DPC stations can be *single-threaded* or *multithreaded*. Single-threaded means that only one message can be outstanding to a station at a time. In this case, BASE24 must receive a response or the message must time out before another message can be sent to the station.

Multithreaded means that one or more messages can be outstanding to a single station at one time. In this case, a response to one message is not required before another message can be sent to the station.

Single-threading and multithreading is allowed for both issuer and acquirer hosts, although BASE24 only controls the threading for issuer hosts. Acquirer hosts can use single-threading or multithreading without regard for the DPC TYPE field setting.

Institutions specify the type of threading BASE24 uses for online messages via the DPC TYPE field on HCF screen 1. Because this setting is at the DPC level, all stations associated with that DPC are either single-threaded or multithreaded.

Note: The setting in the DPC TYPE field only affects the number of online messages outstanding to a station at one time. Store-and-forward messages are always sent single-threaded. That is, only one store-and-forward message can be outstanding to the host at a time. For more information on store-and-forward cycles, refer to section 4.

Sending Text-Level Acknowledgments to a DPC

Acquirer institutions must specify whether the host expects text-level acknowledgments from BASE24. If text-level acknowledgments are required, BASE24 returns a 0230 message for each 0220 or 0221 message; a 0410 message for each 0400 or 0401 message; and a 9230 message for each 9220 or 9221 message.

Institutions indicate whether text-level acknowledgments are required via the ACK TO DPC field on HCF screen 1.

Expecting Text-Level Acknowledgments from a DPC

Issuer institutions must indicate whether BASE24 should expect text-level acknowledgments. If text-level acknowledgments are to be sent, BASE24 expects a 0230 message for each 0220 message; a 0410 message for each 0400 message; a 5410 message for each 5400 message; and a 9230 message for each 9220 or 9221 message.

Institutions indicate whether they send text-level acknowledgments via the ACK FROM DPC field on HCF screen 1.

Note: The ANSI Host Interface always requires acknowledgments for its 9220 or 9221 store-and-forward messages. If text-level acknowledgments are not to be sent, the ANSI Host Interface requests a *logical acknowledgment* from the BASE24 Nucleus. A logical acknowledgment notifies the ANSI Host Interface that the message has been sent successfully to the host.

Specifying Maximum Timeouts

Institutions must indicate the number of consecutive timeouts of non-network management and non-store-and-forward messages that can occur at a station before the station is marked down. For example, if 3 is specified, the ANSI Host Interface marks the station down upon the third consecutive timeout. However, if only two consecutive timeouts have occurred, the station is not marked down.

Institutions control when a station is marked down because of timeouts via the MAX TIMEOUTS field on HCF screen 1.

Enabling Network Management

Institutions specify whether BASE24 is allowed to initiate network management messages (that is, logon, logoff, echo test). This option can limit BASE24's ability to log on and solicit communications with a host.

Institutions indicate whether BASE24 can initiate network management messages via the NMM ENABLED field on HCF screen 1.

Note: This option only controls whether or not BASE24 can initiate the network management actions; BASE24 always responds to the network management messages it receives.

Also, although key management messages are sent as 0800 messages, they are not controlled by this option. BASE24 initiates key management messages based on settings in the Key File (KEYF). For more information on configuring the KEYF, refer to the ***BASE24 Transaction Security Manual***.

Specifying Maximum Store-and-Forward Retries

Institutions set the maximum number of times the ANSI Host Interface attempts to send a store-and-forward message to a host before sending that message to the log and deleting it from the SAF.

This option can be used to stop processing a message when, due to the number of timeouts on a store-and-forward message, it becomes apparent that the host cannot process the message. Institutions control this option via the MAX SAF RETRY field on HCF screen 1.

Specifying Maximum Outbound Messages

BASE24 allows institutions to specify the maximum number of 0200, 0220, 0400, and 5400 messages that can be outstanding to an issuer host before BASE24 stops sending additional messages. Outstanding messages are those messages that have not timed out, but for which the required responses have not been received (for example, a 0210 for a 0200 or a 0230 for a 0220).

This option allows the ANSI Host Interface to prevent processing when it becomes apparent that the host cannot process a message within the time limits required because of the number of messages already outstanding to the host.

Once the number of outstanding messages reaches this limit, the ANSI Host Interface marks any new 0200 messages destined for the host as failed and returns them to their originating processes. The Host Interface process places any new 0220, 0400, and 5400 messages destined for the host in the SAF. At this point, the host does not receive additional messages until a response is received or a message times out.

Institutions control the number of outstanding request and advice messages to a host via the OUTBOUND field on HCF screen 1.

Specifying Maximum Inbound Requests

BASE24 also provides a parameter for limiting the number of outstanding 0200 messages to BASE24-atm and BASE24-pos Authorization processes. This parameter controls requests being sent from acquirer hosts. Once the number of outstanding 0200 messages reaches this limit, the ANSI Host Interface declines any new 0200 messages coming from the host and returns them as 0210 messages.

The number of outstanding requests to an Authorization process is controlled via the INBOUND field on HCF screen 1.

Note: This feature only affects 0200 messages. Other message types are not included in the count of outstanding messages, nor is their processing affected in any way.

Identifying Data Prefix Characters

BASE24 allows institutions to define certain characters they want included in front of the messages they receive from BASE24. An institution can specify up to nine characters to precede messages to their host system. Each time the ANSI Host Interface creates an external message to a DPC, it checks the HCF for data prefix characters. If the HCF contains data prefix characters for the DPC, the ANSI Host Interface places these characters at the front of the message.

Data prefix characters are placed in the HCF using the hexadecimal equivalents for the ASCII versions of the characters to be sent. For example, suppose a host required the following three characters in front of the message:

ESC 1 B
(escape)

To place these three values in an HCF record, the operator would have to enter hexadecimal equivalents (six characters) for the ASCII representations of these values, which would be:

1B 31 42

These values would be stored in the HCF as six characters (1B3142) and would be converted back to the three ASCII characters at the time the message is created.

Because BASE24 does not support transparent data communications, protocol characters must not be used as data prefix characters. Some characters to avoid are:

HEX	ASCII DISPLAY	HEX	ASCII DISPLAY
01	SOH	02	STX
03	ETX	04	EOT
05	ENQ	06	ACK
10	DLE	15	NAK
16	SYN	17	ETB
1F	ITB		

Identifying Authorization Processes

ANSI Host Interface processes handling requests from acquirer networks determine the Authorization processes to which to route inbound transaction requests using the AUTH PROCESS field on HCF screen 5 for BASE24-atm or HCF screen 7 for BASE24-pos.

Setting Up DPC Stations

Stations serve as the links between BASE24 and a host. All messages sent to a DPC are actually sent to the stations defined for that DPC. Likewise, all messages received from a DPC are actually received from the stations defined to that DPC.

Up to 32 stations can be defined for each DPC, although a recommended maximum is 10 to 12. A station must first be defined as a host station in the Network Environment File (NEF). The station's symbolic name can then be placed in the HCF to fully define the station.

The ANSI Host Interface reads the HCF at initialization and builds a table of the stations defined for each DPC.

Station Types

Hosts can define two types of stations for use by the ANSI Host Interface process: send/receive stations and receive-only/no output stations. These station types are described below. Station types are defined via the TYPE fields on HCF screen 2.

Note: The ANSI Host Interface does not support the use of statement print send/receive stations. Statement print send/receive stations can only be used by the ISO Host Interface.

Send/Receive Stations

Send/receive stations allow for both incoming and outgoing messages. The ANSI Host Interface process can send messages to and receive messages from the host using this type of station.

Receive-Only/No Output Stations

Receive-only/no output stations only allow for messages coming in to BASE24. The ANSI Host Interface process never uses receive-only/no output stations for sending messages to the host (this includes network management messages). Receive-only/no output stations are used to avoid line contention when using a point-to-point or SNA protocol. In this case, communications are received on one station and sent from a different station.

Note: For single-threading, messages and their responses must be sent and received via the same stations. For example, if a request is sent to station A, the response to that request must also be returned via station A. The response cannot be returned via a different station. Therefore, receive-only/no output stations should not be defined for single-threaded DPCs.

Setting Up the HCF Station Table

Stations are numbered from left to right, top to bottom on the HCF file maintenance screens. This numbering scheme is of particular importance to hosts with multiple stations on their lines.

When selecting a station to which to send a message, the ANSI Host Interface goes through its station table from start to finish until it finds a station from which the message can be sent.

If lines have multiple stations on them, the stations from each line should be alternated when setting up the HCF station table, and not numbered sequentially. Alternating the stations balances the transaction load so that if one station is chosen from one line, the next station chosen is on a different line.

Selecting Stations for Sending Messages

To select the station to which to send a message, the ANSI Host Interface begins with the first station in the HCF station table for that DPC and evaluates each station based on the following criteria:

1. It must be a send/receive station and the station must be marked up. The ANSI Host Interface never sends messages to receive-only/no output stations.
2. If the DPC is single-threaded, the station cannot be busy; that is, it must not have a message pending.
3. If the DPC is multi-threaded, the station must have the least number of messages pending in relation to other stations.

The ANSI Host Interface sends the message to the first station that meets the above criteria. To send the next message, the ANSI Host Interface performs its station selection in the same manner, starting with the next station in the table. For example, if the last message was sent to the second station in the table, the ANSI Host Interface would begin its station selection with the third station.

Configuring Timers

The ANSI Host Interface uses internal timers to control host message traffic. The length of each of these timers is specified in the HCF.

Network Management Message Timer

A network management message timer is set each time an 0800 message is sent to a DPC station. The network management message timer limits the amount of time that BASE24 waits for a response after transmitting an 0800 message. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on HCF screen 1.

The actions taken when a network management message timer expires depend on the type of message for which the timer was set. If the 0800 message was a logon, logoff, or echo-test message, the ANSI Host Interface performs as follows:

1. Marks the station down.
2. Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ANSI Host Interface process marks the DPC down.
3. Sets an extended network management message timer for the station.

If the 0800 message was a dynamic key management (DKM) message, the ANSI Host Interface performs as follows:

1. Adds one to the count of consecutive timeouts accumulated for the station.
2. Checks the number of consecutive timeouts accumulated for the station against the maximum allowed. The maximum allowed is specified in the MAX TIMEOUTS field on HCF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, processing continues with the next step (step 3).
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ANSI Host Interface process:
 - 1) Marks the station down.
 - 2) Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ANSI Host Interface process marks the DPC down.

3. If the message that timed out was a repeat key or verify key message, the ANSI Host Interface logs a message indicating that the message timed out.
4. If the message that timed out was a new key or a change key message, the ANSI Host Interface performs as follows:
 - a. Logs a message indicating that the message expired and that the message will be automatically retried.
 - b. Sets an extended network management message timer for the station. (The new key or change key message will be retried when this timer expires.)

Extended Network Management Message Timer

An extended network management message timer is set each time a DPC station is marked down and each time a new key or change key message times out. The extended network management message timer specifies the following intervals:

- The amount of time that BASE24 waits after a station is marked down before attempting to reestablish communications with the station.
- The amount of time that BASE24 waits after a new key or change key message times out before attempting to resend the message.

The number of seconds to use for this timer is taken from the EXTENDED NETWORK field on HCF screen 1.

If an extended network management message timer that was set when a station was marked down expires before any messages are received from the station, the ANSI Host Interface process:

1. Sends an echo test message to the station.
2. Sets a network management message timer for the echo test message.

If an extended network management message timer that was set after a new key or change key message timed out expires, the ANSI Host Interface determines whether there is a station available to which to send the message.

If a station is available, the ANSI Host Interface sends the new key message or change key message to that station and sets a network management message timer to wait for the response to the new key or change key message.

If no stations are available, the ANSI Host Interface sets another extended network management message timer. When this timer expires, the ANSI Host Interface attempts to send the message again.

Store-and-Forward Message Timer

A store-and-forward message timer is set each time a store-and-forward message (9220 or 9221) is sent to a DPC. The store-and-forward message timer limits the amount of time that BASE24 waits for an acknowledgment after transmitting a store-and-forward message. The number of seconds to use for this timer is taken from the STORE AND FORWARD field on HCF screen 1.

The ANSI Host Interface only expects text-level acknowledgments to store-and-forward messages if the ACK FROM DPC field on HCF screen 1 contains the value Y. If text-level acknowledgments are not required from the DPC, the ANSI Host Interface expects a logical acknowledgment from the BASE24 Nucleus instead. In this case, the Nucleus returns a logical acknowledgment once the message has been sent successfully to the DPC.

If this timer expires before an appropriate acknowledgment is received in response, the ANSI Host Interface:

1. Adds one to the SAF retries count (the number of times the SAF record has been sent to the host).
2. Reads the first record for this DPC from the SAF.
3. Checks the SAF retries count against the maximum allowed by the MAX SAF RETRIES field on HCF screen 1.
 - a. If the SAF retries count is equal to the maximum allowed, the ANSI Host Interface:
 - 1) Writes the SAF record to its log.
 - 2) Clears the SAF retries count.
 - 3) Deletes the record from the SAF.
 - b. If the SAF retries count is less than the maximum allowed, the ANSI Host Interface continues with the next step (step 4).

4. Updates the message type, if necessary.
 - a. If the TYP field in the external message is 0220, 0400, or 5400, the Host Interface process changes it to 0221, 0401, or 5401, respectively.
 - b. If the TYP field in the external message is 0221, 0401, or 5401, the Host Interface process leaves it as is.
5. Adds one to the number consecutive timeouts for this station.
6. If the number of consecutive timeouts for this station exceeds the number specified in the MAX TIMEOUTS field on HCF screen 1, the ANSI Host Interface marks the station down and starts an extended network management message timer for the station. When this timer expires, the ANSI Host Interface will attempt to reestablish communications with the station.
7. Locates an available send/receive station.
 - a. If there are no send/receive stations available, the Host Interface process:
 - 1) Drops the 9220/9221 message (the SAF record is still on file).
 - 2) Marks the DPC down.
 - b. If there is a send/receive station available, the Host Interface:
 - 1) Sets the TYP field to 9221.
 - 2) Sends the message to the station.
 - 3) Sets a store-and-forward message timer.

Wait-for-Traffic Timer

A wait-for-traffic timer is set each time a station is marked up and each time a previously set wait-for-traffic timer expires. The wait-for-traffic timer specifies the amount of time that BASE24 waits for traffic on the line before sending an echo test message. The number of seconds to use for this timer is taken from the WAIT FOR TRAFFIC field on HCF screen 1.

The ANSI Host Interface sets wait-for-traffic timers for its send/receive stations only. wait-for-traffic timers are not set for receive-only/no output stations.

When this timer expires, the ANSI Host Interface:

1. Determines whether the DPC is logged off.
 - a. If the DPC is logged off, no further action is taken.
 - b. If the DPC is not logged off, the ANSI Host Interface proceeds to step 2.

2. Determines whether there has been any message traffic to or from the station since the timer was set.
 - a. If there has been message traffic, the ANSI Host Interface simply sets another wait-for-traffic timer.
 - b. If there has not been message traffic, the ANSI Host Interface:
 - 1) Sends an echo test message to the station.
 - 2) Sets a network management message timer for the echo test message.

Performance Timer

A performance timer is set for each DPC at the start of that DPC's new performance statistics accumulation period. The performance timer specifies the interval the ANSI Host Interface uses for gathering performance statistics. The number of minutes to use for this timer is taken from the PERFORMANCE field on HCF screen 1.

For each DPC, the ANSI Host Interface tracks the number of requests sent to the DPC, the number and percentage of requests that time out, and the average response time. These statistics are tracked on an interval basis and are available for the most current four intervals (interval zero through interval three, with interval zero being the most current). The performance timer triggers the end of one interval and the beginning of the next.

When a performance timer expires, the ANSI Host Interface:

1. Moves the performance statistics from interval two to interval three.
2. Moves the performance statistics from interval one to interval two.
3. Moves the performance statistics from interval zero to interval one.
4. Zeros out the performance statistics in interval zero to begin accumulating statistics again.
5. Sets a performance timer for the DPC.

Note: These statistics are accessed via a Performance command entered from the Network Control Supervisor (NCS) screen. For information concerning the entry of the Performance command and the display of the statistics, refer to the description of the Performance command in the ***BASE24 Text Command Reference Manual***.

Outbound Request Timer

An outbound request timer is set each time a 0200 message is sent to a DPC. The outbound request timer limits the amount of time that BASE24 waits for a response from the host after transmitting a request message. The number of seconds to use for this timer is taken from the OUTBOUND field on the product-specific HCF screen.

If this timer expires before a 0210 response is received to the message, the ANSI Host Interface performs as follows.

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the ANSI Host Interface performs as follows.

Note: The outbound request timer is also used for 0205 messages.

 - 1) Retrieves the original message from memory.

The ANSI Host Interface retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - 2) Sets the RTE.STAT field in the STM to 11, to indicate the destination was not available.

This identifies the 0200 or 0205 message as a failure.
 - 3) If the message is a 0200 request, sets the RQST.COMPL-REQ field in the STM to 2, to cause the Authorization process to generate a 0220 message if it subsequently stands in and authorizes the transaction.
 - 4) Sends the STM to the process identified in the ORIG-PRO-NAME field of the STM.
 - b. If the message is a BASE24-pos message, the ANSI Host Interface performs as follows:
 - 1) Retrieves the 0200 message from memory.

The ANSI Host Interface retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - 2) Sets the RTE.STAT field in the PSTM to 11, indicating the destination was not available.

This identifies the 0200 message as failed.

- 3) Sets the COMPL-REQ field in the PSTM to 2, causing Authorization to generate a 0220 message if it subsequently stands in and authorizes the transaction.
 - 4) Sends the PSTM to the process identified in the ORIG-PRO-NAME field of the PSTM.
2. Adds one to the count of consecutive timeouts accumulated for the station.
3. Adds one to the number of request timeouts for the DPC.

The ANSI Host Interface tracks this information as part of its DPC performance statistics.

4. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAX TIMEOUTS field on HCF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ANSI Host Interface:
 - 1) Marks the station down.
 - 2) Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ANSI Host Interface marks the DPC down.
 - 3) Sets an extended network management message timer for the station.

Inbound Request Timer

An inbound request timer is set each time a 0100 or 0200 message is sent to a BASE24-atm or BASE24-pos Authorization process. The inbound request timer limits the amount of time that the ANSI Host Interface waits for a response from a BASE24 Authorization process after transmitting a request message. The number of seconds to use for this timer is taken from the INBOUND field on the product-specific HCF screen.

If an inbound request timer expires before a response message is received, the ANSI Host Interface:

1. Performs product-specific processing.
 - a. If the message is a BASE24-atm message, the ANSI Host Interface performs as follows:
 - 1) Retrieves the original message from memory.

The ANSI Host Interface retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
 - 2) Updates the TYP field in the SEM.
 - a) If the TYP is equal to 0700, the Host Interface process changes it to 0710.
 - b) If the TYP is equal to 0200, the Host Interface process changes it to 0210.
 - 3) Sets the RESP-CDE.RESP field in the SEM to 54 to indicate that the transaction was denied because of an inability to process it.
 - 4) Sets the ROUTE.STAT field in the SEM to 11, to indicate the destination was not available.
 - 5) Sends the SEM to the host.
 - b. If the message is a BASE24-pos message, the ANSI Host Interface performs as follows:
 - 1) Sets the RESP-CDE.RESP field in the PSEM to 100, indicating that the transaction is declined because of an inability to process it.
 - 2) Sets the ROUTE.STAT field in the PSEM to 11, indicating the destination was not available.
 - 3) Sends the PSEM to the host.
2. Adds one to the number of request timeouts for BASE24.

Completion Acknowledgment Timer

A completion acknowledgment timer is set each time BASE24 sends a 0220, 0400, or 5400 message to a DPC. The completion acknowledgment timer limits the amount of time that BASE24 waits for an acknowledgment from the host after transmitting an advice message. The number of seconds to use for this timer is taken from the COMPLETION ACK field on the product-specific HCF screen.

The ANSI Host Interface only expects text-level acknowledgments to completion messages if the ACK FROM DPC field on HCF screen 1 contains the value Y. If text-level acknowledgments are not required from the DPC, the ANSI Host Interface expects an acknowledgment from the BASE24 Nucleus instead. In this case, the Nucleus returns an acknowledgment once the message has been sent successfully to the DPC.

If a completion acknowledgment timer expires before an appropriate acknowledgment is received, the ANSI Host Interface performs as follows.

1. Retrieves the original message from memory.
The ANSI Host Interface retains a copy of the internal message for each message it sends to the host until it is notified that the message was received.
2. Changes the TYP field from 0220, 0400, or 5400 to 0221, 0401, or 5401, respectively.
3. Places the SEM in the SAF.
4. Adds one to the count of consecutive timeouts accumulated for the station.
5. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAX TIMEOUTS field on HCF screen 1.
 - a. If the current number of consecutive timeouts is less than the maximum allowed, no further action is taken.
 - b. If the current number of consecutive timeouts is equal to the maximum allowed, the ANSI Host Interface:
 - 1) Marks the station down.
 - 2) Sets an extended network management message timer for the station.
6. Verifies that the DPC still has a station available for sending messages. If no send/receive stations are left up for the DPC, the ANSI Host Interface marks the DPC down.

7. Adds one to the number of request timeouts for the DPC.

The ANSI Host Interface tracks this information as part of its DPC performance statistics.

Key Exchange Timer

Key exchange timers are used to automatically initiate PIN key exchanges. If BASE24 is configured for dynamic key management, one key exchange timer is set for each unique PIN key being used by the ANSI Host Interface process. Key exchange timers are set or reset at the following times:

- When the ANSI Host Interface is initialized or warmbooted
- When the key for which the timer has been set is exchanged for any reason

The ANSI Host Interface sets and uses key exchange timers for PIN keys when the PIN KEY TIMER VALUE field on KEYF screen 3 is set to a nonzero value. PIN key exchange timers limit the amount of time that BASE24 uses a particular PIN key to encrypt and decrypt PINs sent between BASE24 and the host. The amount of time to use for these timers is taken from the PIN KEY TIMER INTERVAL and PIN KEY TIMER VALUE fields on KEYF screen 3.

If a key exchange timer expires, the ANSI Host Interface performs as follows:

1. Determines the key that needs to be exchanged.
2. Changes the status of the DPC to indicate that a key exchange is pending.
3. Determines the type of message to send based on the value in the KEY PROCESSING TYPE field on KEYF screen 3 and the key direction, as described below:
 - a. If the KEY PROCESSING TYPE field contains the value N (no key processing), then no further steps are taken.
 - b. If the KEY PROCESSING TYPE field contains the value M (master processor), the message type is a new key message.
 - c. If the KEY PROCESSING TYPE field contains the value S (slave processor), the message type is a change key message.
 - d. If the KEY PROCESSING TYPE field contains the value C (co-network processor) and the key direction is outbound, the message type is a new key message.

- e. If the KEY PROCESSING TYPE field contains the value C (co-network processor) and the key direction is inbound, the message type is a change key message.
4. Determines whether a key exchange is already in progress for the specific key. If a key exchange is already in progress for the specific key, no further steps are taken. If a key exchange is not in progress for the specific key, processing continues with the next step (step 5).
5. Determines whether there is a station available to which to send the message. Depending on whether a station is available, the ANSI Host Interface performs as follows:
 - a. If no stations are available, the ANSI Host Interface sets an extended network management message timer. When this timer expires, the ANSI Host Interface will attempt to send the message again. No further processing is performed at this time.
 - b. If a station is available, the ANSI Host Interface performs according to the type of message, as described below:
 - 1) If the message is a new key message, the ANSI Host Interface sends the message to the selected station. The ANSI Host Interface sets a network management message timer to wait for the response. The ANSI Host Interface also sets a flag to indicate that a key exchange is in progress and deletes the clear old key timer for the key being exchanged (if one is still set).
 - 2) If the message is a change key message, the ANSI Host Interface sends the message to the selected station. The ANSI Host Interface sets a network management message timer to wait for the response.
6. Adds one to the count of messages outstanding to the station.

Note: Upon receiving a response from the DPC, the ANSI Host Interface clears the flag indicating a key exchange is pending and sets a new key exchange timer.

Clear Old Key Timer

The clear old key timer limits the amount of time that BASE24 attempts to use the old key to verify a PIN after a key exchange has taken place.

The INBOUND PIN KEY and OUTBOUND PIN KEY fields on KEYF screen 2 can each contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key is currently in use. At the time of a successful key exchange, the CURRENT INDEX is updated to point to the new key and the key that was in use becomes the old key.

When processing a transaction, BASE24 uses the new key for PIN processing. If the security device cannot translate or verify the PIN because of a key error, the transaction is tried again using the old key, if one exists. This timer controls how long the old key exists. When the timer expires, the old key is cleared from the KEYF.

A clear old key timer is set under the following circumstances:

- When BASE24 receives a duplicate new key message from the host (carrying a key that was already received in a previous new key message and stored in the KEYF.)
- When BASE24 approves a new key message from the host.
- When BASE24 receives an approved new key response message from the host (i.e., BASE24 sent the host a new key, and the host has responded that the new key has been implemented successfully).

When a clear old key timer expires, the ANSI Host Interface performs as follows.

Note: KEYF information is read into memory by the ANSI Host Interface at initialization, and the ANSI Host Interface performs the following steps on the KEYF information in memory and in the KEYF file.

1. Determines the type of key for which the timer was set.
2. Determines the key to clear using the value in the CURRENT INDEX field.
Since the value in the CURRENT INDEX field points to the newest key, the ANSI Host Interface selects the key not being pointed to by the CURRENT INDEX field as the one to clear.
3. Sets the selected key value and its check digits in memory to ASCII zeros.
4. Sets the selected key value and its check digits in the KEYF to binary zeros.

Stale Transaction Timer

Issuer hosts can specify the maximum length of time that the BASE24 Nucleus can take to send a message before returning the message to the ANSI Host Interface as *stale*.

This option is particularly useful for 0200 messages because it allows the ANSI Host Interface to prevent processing when it becomes apparent that the host will not have time to respond to a message before the request times out.

If a message stays queued by the Nucleus for this length of time, the Nucleus returns the message to the ANSI Host Interface as a failed message rather than send it to the host. The ANSI Host Interface then returns the message to its originating process as a failed message.

This length of time is derived using the QUEUE SUBTRACT field on the product-specific HCF screens. There is one QUEUE SUBTRACT field each for BASE24-atm and BASE24-pos. The QUEUE SUBTRACT value is subtracted from each message's corresponding timer value to determine the maximum length of time to allow it to be queued.

For 0200 messages, the QUEUE SUBTRACT value is subtracted from the OUTBOUND value on the product-specific HCF screen to arrive at the maximum queuing time. For 0220, 0400, and 5400 messages, the QUEUE SUBTRACT value is subtracted from the COMPLETION ACK value on the product-specific HCF screen to arrive at the maximum queuing time.

Configuring Transaction Security

The transaction security requirements of any organization can vary greatly according to the environment in which the organization is operating. The ANSI Host Interface process supports transaction security between BASE24 and the host in the following areas:

- PIN encryption ensures that PIN information is maintained in a confidential manner.
- PIN verification ensures that the rightful cardholder is performing the transaction.

These transaction security measures each require that the ANSI Host Interface process and the host use keys to encrypt, decrypt, and verify information in the message. These keys should be changed regularly to maintain proper security. The monitoring and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management (DKM). The ANSI Host Interface process supports DKM between the host and BASE24.

There are many issues involved in configuring transaction security and DKM. Because of the complexities of establishing security, these issues are discussed in the ***BASE24 Transaction Security Manual***. For more information on establishing transaction security and DKM, refer to that manual.

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Section 3

Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications lines between BASE24 and the host, and those used to manage PIN keys between BASE24 and the host.

The ANSI Host Interface supports four types of network management messages:

- Logon messages
- Echo test messages
- Logoff messages
- Dynamic key management (DKM) messages

Network management messages are sent as 0800 messages and require 0800 messages in response.

This section describes the message flows that occur when network management messages are sent.

Monitoring DPCs and Stations

The ANSI Host Interface process keeps track of the status of all stations and DPCs with which it communicates via its DPC table. This DPC table is built at initialization and is updated continuously during processing.

Stations are marked *up* and *down* in the table depending on whether the ANSI Host Interface can communicate with them. For instance, if a station fails to respond appropriately to a message, the ANSI Host Interface checks the maximum timeouts allowed for the DPC's stations. If the maximum timeouts allowed is reached, the ANSI Host Interface marks the station down and takes the necessary network management actions to reestablish the link with the station. Once the link is reestablished, the ANSI Host Interface marks the station up.

The ANSI Host Interface also tracks the status of DPCs. A DPC is marked as *up* if the ANSI Host Interface can send messages to and receive messages from at least one of the DPC's stations. If all of the stations to a DPC are marked down, the DPC itself is marked down. Additionally, the ANSI Host Interface tracks such things as whether the DPC is logged off, whether store-and-forward processing is in progress for the DPC, and whether a SAF extract is in progress for the DPC.

The ANSI Host Interface keeps track of station and DPC information at all times and logs a message when the status of the DPC changes (for example, from up to down or at the start of store-and-forward processing).

Note: The actual communications link to each DPC station is maintained by the Nucleus. The ANSI Host Interface only tracks the logical ability of BASE24 to communicate with each station. If a station responds to a message, the ANSI Host Interface considers it to be up. If a station fails to respond to a message the number of times specified in the MAX TIMEOUTS field on HCF screen 1, the ANSI Host Interface considers the station to be down.

Echo Test Messages

Echo test messages are network management messages used to establish whether a DPC station is available for message processing. They are sent as 0800 messages and require 0800 messages in response.

Echo test messages are distinguished by a value of 301 in the NM-INFO-CDE field of the network management message. Echo test messages can be sent by either BASE24 or a host.

Echo Test Messages from BASE24

The ANSI Host Interface initiates echo tests under the following circumstances:

1. At initialization. The ANSI Host Interface sends an echo test to each send/receive station at initialization.
2. Upon receipt of a Logon command from the Network Control Supervisor (NCS) screen. The ANSI Host Interface sends an echo test to each station associated with the DPC included in the command.
3. Upon expiration of an extended network management message timer. This processing is described in section 2.
4. Upon expiration of a wait-for-traffic timer if there has been no traffic on the line. This processing is described in section 2.
5. When a DPC initiates a logon after a period of being logged off. The ANSI Host Interface sends an echo test to each send/receive station once the DPC has been marked as logged on.

Echo Test Messages from a DPC

DPCs can send echo test messages when they wish to start a logon sequence after a period of being logged off. Or, they may send echo test messages in response to BASE24-initiated echo test messages.

Upon receipt of an echo test message from a DPC, the ANSI Host Interface always returns a logon message.

Logon Messages

Logon messages are network management messages used to manage communications access between BASE24 and a DPC. They are sent as 0800 messages and require 0800 messages in response.

Logon messages are distinguished by a value of 001 in the NM-INFO-CDE field of the network management message.

Logon Messages from BASE24

The ANSI Host Interface sends logon messages to DPCs in return for echo test messages. This is done regardless of whether the DPC initiated the echo test message or was simply responding to a prior BASE24-initiated echo test message with an echo test message.

Logon Messages from a DPC

DPCs send logon messages to start a logon sequence after a period of being logged off. They can also send logon messages in response to BASE24-initiated echo test or logon messages.

When the ANSI Host Interface receives a logon message after the DPC has been logged off for a period, the ANSI Host Interface marks the DPC as logged on and begins sending echo test messages to that DPC's stations.

Note: If the ANSI Host Interface receives a logon message when the DPC is not logged off, it simply drops the message.

When the ANSI Host Interface receives a logon message in response to a BASE24-initiated message, the ANSI Host Interface marks the station up and begins processing.

Logoff Messages

Logoff messages are network management messages used to stop communications between BASE24 and a DPC. They are sent as 0800 messages and do not require a response.

Logoff messages are distinguished by a value of 002 in the NM-INFO-CDE field of the network management message. They can be sent by either BASE24 or a host. In either situation, once a DPC is logged off, BASE24 does not send echo test messages or non-network management messages to any of the DPC's stations. Any non-network management messages that must be forwarded to the DPC are placed in the SAF instead.

BASE24 only attempts to reopen the link to a DPC that is logged off if a logon command is received from the NCS screen. The DPC, on the other hand, can reopen the link at any time by sending an echo test or a logon message.

Note: When a DPC is logged off, any transactions in progress are allowed to complete.

Logoff Messages from BASE24

BASE24 only sends logoff messages to a DPC as a result of a logoff command issued from the NCS screen.

To log a DPC off, BASE24 simply sends a logoff message to an available send/receive station and marks the DPC as down and logged off. A response may be returned by the host, but is not necessary.

Logoff Messages from a DPC

When a host must log off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations. Upon receipt of a logoff message, the ANSI Host Interface marks the DPC down and logged off. It does not return a response to the DPC.

Dynamic Key Management Messages

Dynamic key management messages make up a subset of network management (0800) messages. There are four key management messages:

- Change key
- New key
- Repeat key
- Verify key

The change key and new key messages can be generated automatically as the result of a defined threshold being surpassed. In addition, all of these message can be generated manually by an operator issuing a command from the Network Control Supervisor (NCS) screen.

This section provides an overview of how dynamic key management messages are used. For more information on dynamic key management, refer to the ***BASE24 Transaction Security Manual***. For more information on using the NCS to generate key management messages, and for complete syntax of the NCS commands involved, refer to the ***BASE24 Text Command Reference Manual***.

Change Key Message

A change key message requests that a key processor generate a new PIN encryption key. The receiving processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting process in a new key message.

The ANSI Host Interface sends this type of message to a host when the host is responsible for generating a particular key and the ANSI Host Interface process has determined that that key needs to be changed. The ANSI Host Interface could make that determination based on a DKM threshold being passed or in response to an NCS KEY CHANGE command (the NCS KEY CHANGE command has the same effect as passing a DKM threshold).

The ANSI Host Interface process can receive this type of message from a host when the ANSI Host Interface is responsible for generating a particular key and the host wants that key to be changed.

New Key Message

A new key message carries a new PIN encryption key between processors. The receiving processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

The ANSI Host Interface sends this type of message to a host when the ANSI Host Interface is responsible for generating a particular key and has either determined that that key needs to be changed or been requested by the host to change or repeat the key. This could occur as a result of a DKM threshold being passed, notification from the host (via a change key or repeat key message), or could occur in response to an NCS KEY CHANGE command (the NCS KEY CHANGE command has the same effect as passing a DKM threshold).

The ANSI Host Interface process can receive this type of message from a host when the host is responsible for generating a particular key and the host has either determined that that key needs to be changed or been requested by the ANSI Host Interface to change or repeat the key (via a change key or repeat key message).

Repeat Key Message

A repeat key message requests that a processor that generated a key resend that key. The processor receiving the repeat key message obtains the requested key from its database and sends it to the requesting process in a new key message. The key itself is not changed, nor are any threshold counters affected.

The ANSI Host Interface only generates a repeat key message as the result of receiving an NCS KEY REPEAT command. It will respond to a repeat key message from the host with a new key message carrying the requested key.

Verify Key Message

A verify key message requests that a processor that generated a key verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the processor that sent the verify key message should use a change key message to request a new key.)

The ANSI Host Interface only generates the verify key message as the result of receiving an NCS KEY VERIFY command. It will respond to a verify key message from a host by comparing check digits for the specified key and returning the results.

Interactive Network Management Message Flows

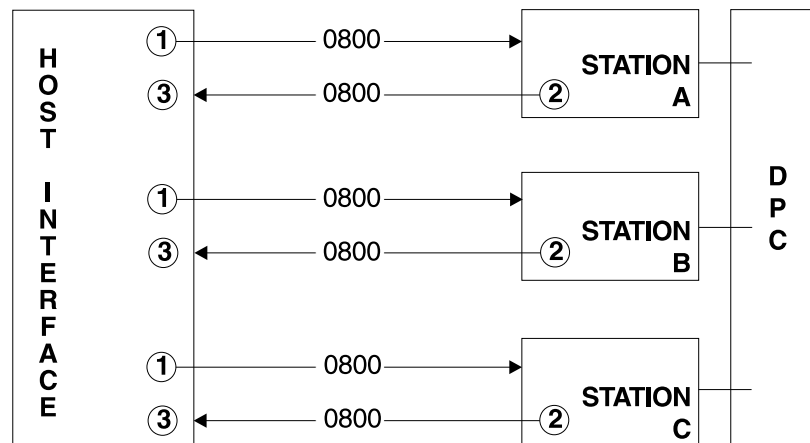
The remainder of this section contains examples of network management and DKM message flows, and describes the processing performed by BASE24 for each.

Echo Test/Logon Message Examples

The following are examples of the different echo test/logon sequences supported by BASE24.

Example #1

In this example, the ANSI Host Interface is echo testing each of the DPC's send/receive stations (as it does at initialization), and the DPC is set up to respond to echo test messages with logon messages.

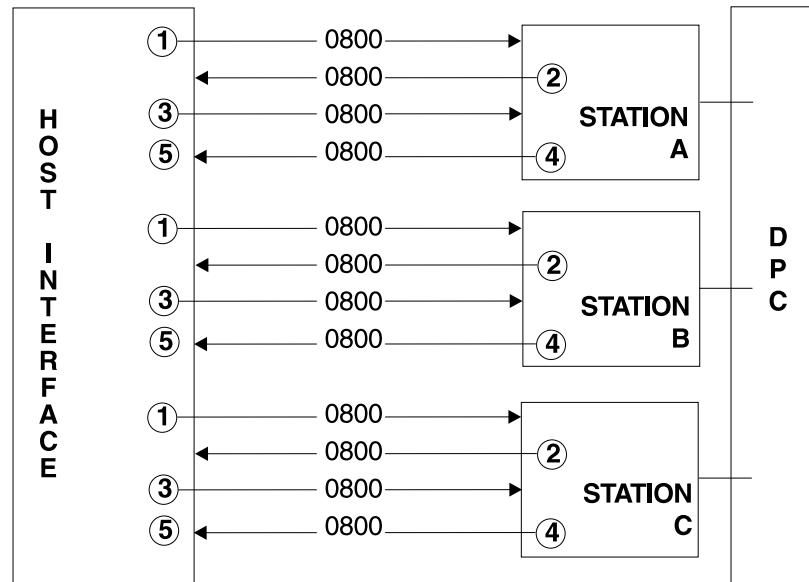


1. The ANSI Host Interface sends an echo test message to each of the DPC's send/receive stations and sets a network management message timer for each.
2. The host returns a logon message as a response from each station.
3. The ANSI Host Interface receives each logon message, marks the corresponding station up, and sets a wait-for-traffic timer for that station.

Upon receipt of the first logon message from the DPC's stations, the ANSI Host Interface also marks the DPC as up and logged on.

Example #2

In this example, the ANSI Host Interface is echo testing each of the DPC's send/receive stations (as it does at initialization), and the DPC is set up to respond to echo test messages with echo test messages.

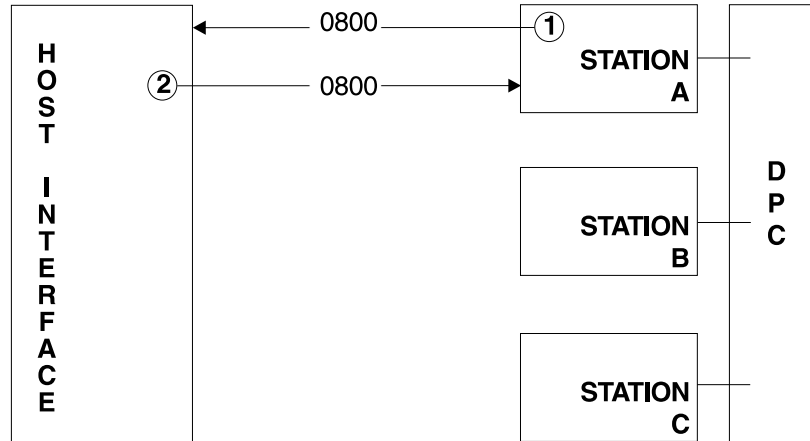


1. The ANSI Host Interface sends an echo test message to each of the DPC's send/receive stations and sets a network management message timer for each.
2. The host returns an echo test message as a response from each station.
3. The ANSI Host Interface receives the returning echo test messages; it then sends logon messages to the stations and sets a network management message timer for each.
4. The host returns a logon message as a response from each station.
5. The ANSI Host Interface receives each logon message, marks the corresponding station up, and sets a wait-for-traffic timer for that station.

Upon receipt of the first logon message from the DPC's stations, the ANSI Host Interface also marks the DPC as up and logged on.

Example #3

In this example, station A is the only station down, and the host sends an echo test.



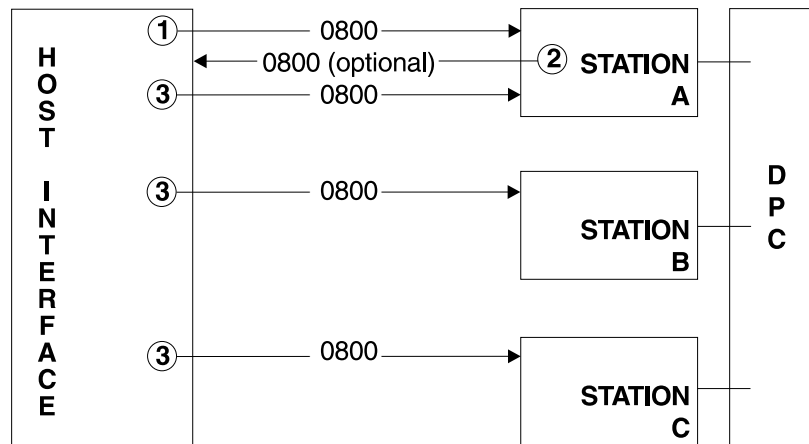
1. The host sends an echo test message from station A.
2. The ANSI Host Interface marks the station up, sets a wait-for-traffic timer for that station, and returns a logon message in response.

Logoff Message Examples

The following are two examples of logoff sequences supported by BASE24.

Example #1

In this example, the logoff and subsequent logon are both initiated by the ANSI Host Interface:

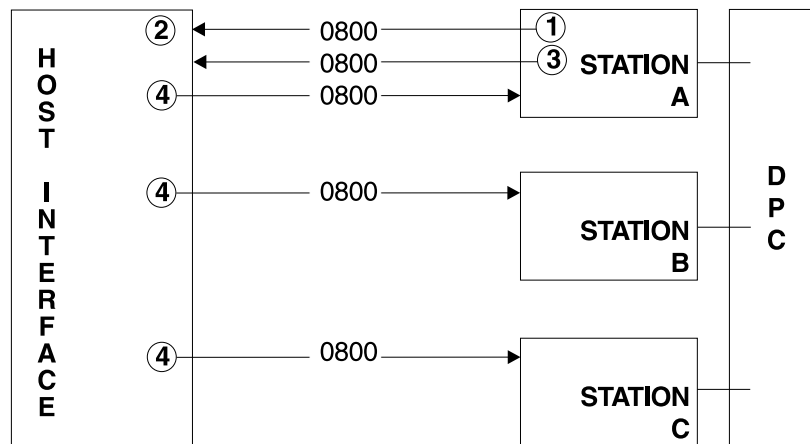


1. The ANSI Host Interface receives a logoff command from the NCS screen and:
 - a. Sends a logoff message to one of the DPC's send/receive stations.
 - b. Clears the following timers running for the DPC and its stations:
 - Network management message timers
 - Extended network management message timers
 - Wait-for-traffic timers
 - Performance timer
 - Key exchange timers
 - Clear old key timersAll other timers are left running to ensure that current transactions in progress complete.
 - c. Marks all of the DPC's stations down.
 - d. Marks the DPC as down and logged off.
2. The host may return a logoff message as a response, but it is not required.

3. The ANSI Host Interface receives a logon all command from the NCS screen and generates an echo test message to each of the DPC's send/receives stations.

Example #2

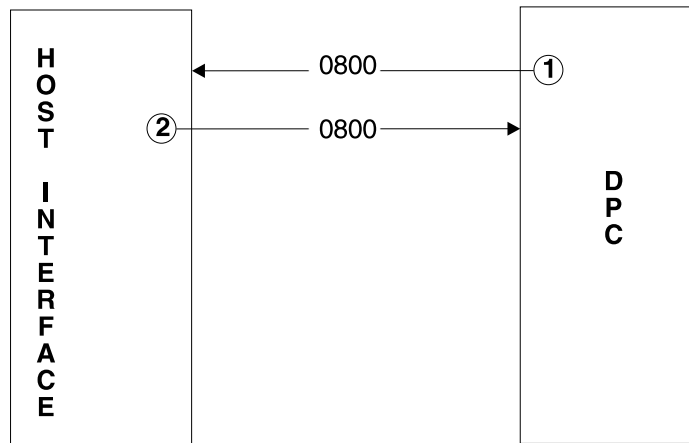
In this example, the logoff and subsequent logon are both initiated by the host:



1. The host sends a logoff message to BASE24.
2. The ANSI Host Interface receives the logoff message and:
 - a. Clears the following timers running for the DPC and its stations:
 - Network management message timers
 - Extended network management message timers
 - Wait-for-traffic timers
 - Performance timer
 - Key exchange timers
 - Clear old key timers
 All other timers are left running to ensure that current transactions in progress complete.
 - b. Marks all of the DPC's stations down.
 - c. Marks the DPC as down and logged off.
3. The host sends an echo test or logon message to reestablish communications.
4. The ANSI Host Interface receives the message and:
 - a. Marks the DPC as logged on.
 - b. Sends an echo test message to each of the DPC's send/receive stations.

Change Key Message Example

The diagram below illustrates the message flow between the host and ANSI Host Interface when the host initiates a change key request message. In this case, the ANSI Host Interface is responsible for generating the key being requested. The steps corresponding to the diagram are also described below.



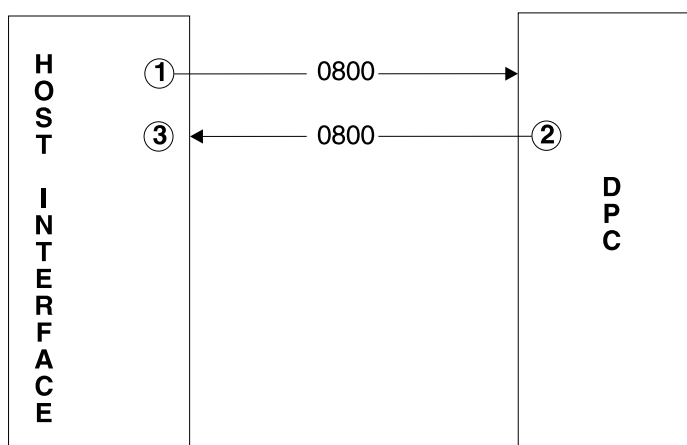
1. The host formats and sends an 0800 message with a network management information code of 161 to the ANSI Host Interface. This message includes the KEY-MGMT.X917-CSM-DATA field containing a Request Service Initiation message (RSI).
2. The ANSI Host Interface receives the 0800 message and validates it before responding. To validate the message, the ANSI Host Interface performs as follows:
 - a. Verifies that the RSI is in the correct format. If the RSI is not in the correct format, the ANSI Host Interface sets an error flag to F to indicate that a format error was found.
 - b. Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the ANSI Host Interface compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3, and the originator ID from the RSI to the value from the RECEIVING ID field on KEYF screen 3. If the receiver or originator does not match the corresponding KEYF values for this DPC, the ANSI Host Interface sets an error flag to H to indicate that the receiver of originator ID is incorrect.

- c. Verifies that there is not already a change key action in progress for the type of key identified in the RSI. The ANSI Host Interface checks an internal flag to verify that a change is not currently in progress. If a change key is in progress, the ANSI Host Interface sets an error flag to C to indicate that the current change key request cannot be processed.
- d. Formats an 0800 response message and returns it to the host. To format the response message, the ANSI Host Interface performs as follows:
 - 1) If one or more of the above checks failed with an error, the ANSI Host Interface formats an 0800 response message with a network management information code of 361. This message includes the KEY-MGMT.X917-CSM-DATA field containing an Error Service Message (ESM). The ERF field in the ESM includes any error flag values set during the checks. No further processing is performed.
 - 2) If no errors were encountered in the above checks, the ANSI Host Interface formats an 0800 response message with a network management information code of 261. This message includes the KEY-MGMT.X917-CSM-DATA field containing a copy of the original RSI.
- e. Creates a new key and sends it to the host in a new key message.

For a description of this processing, refer to the “New Key Message Example” immediately following.

New Key Message Example

The diagram below illustrates the message flow between the ANSI Host Interface and the host when the ANSI Host Interface sends a new key message. In this case, the ANSI Host Interface is responsible for generating the key being sent. The steps corresponding to the diagram are also described below.

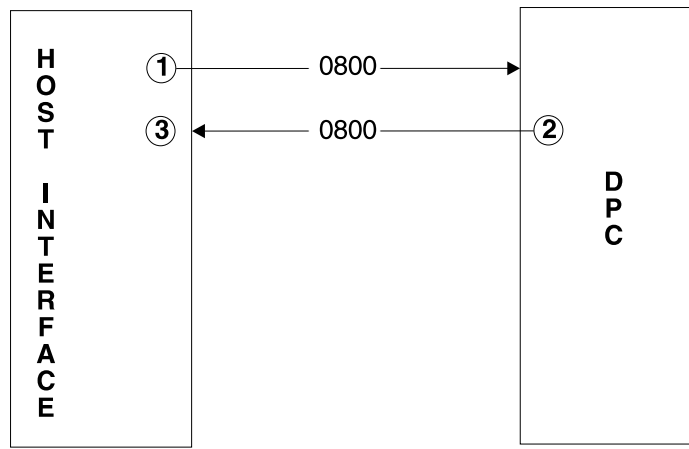


1. The ANSI Host Interface creates a new key message and sends it to the host. To do this, the ANSI Host Interface performs as follows:
 - a. Generates the new key and check digits.
 - b. Formats a Key Service Message (KSM) containing the new key and the check digits.
 - c. Sends an 0800 message to the host with the network management information code set to 162. This message includes the KEY-MGMT.X917-CSM-DATA field containing the KSM.
 - d. Sets an internal flag indicating that a key exchange is in progress.
 - e. Starts a network management message timer for the message. If this timer expires before an 0800 response is received from the host, the ANSI Host Interface starts an extended network management message timer for the message. When the extended network management message timer expires, the ANSI Host Interface resends the new key message.
2. The host receives the new key message, processes it, and returns an 0800 response message to the ANSI Host Interface. This message includes the KEY-MGMT.X917-CSM-DATA field containing either a Response Service Message (RSM) or an Error Service Message (ESM).

3. The ANSI Host Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out.
 - 1) If the 0800 message timed out before the 0800 response was received, the ANSI Host Interface has already set an extended network management message timer for the new key message (see step 1e). No further steps are performed.
 - 2) If the 0800 message has not timed out, processing continues with the next step.
 - b. Clears the internal flag indicating that a key exchange is in progress.
 - c. Continues processing depending on the network management information code sent in the 0800 response message.
 - 1) If the transaction was denied (the network management information code was 362), the ANSI Host Interface parses the KEY-MGMT. X917-CSM-DATA field to determine why the new key was denied. In this case, the Cryptographic Service Message data element contains an Error Service Message (ESM).
 - a) If the ESM indicates the message was denied because BASE24's key counter does not match the key counter maintained by the host, the ANSI Host Interface resets its key counter in the KEYF to the value sent by the host. The ANSI Host Interface then resends the new new key message to the host. Processing returns to step 2.
 - b) If the ESM indicates that any other error occurred, the ANSI Host Interface logs a message indicating the error and drops the 0800 response.
 - 2) If the transaction was approved (the network management information code was 262), the ANSI Host Interface updates the KEYF with the new key, check digit, and counter information. The ANSI Host Interface also reinitializes its internal key change threshold counters and timers.

Repeat Key Message Example

The diagram below illustrates the message flow between the ANSI Host Interface and the host when the ANSI Host Interface initiates a repeat key message. In this case, the host is responsible for generating the key being requested. The ANSI Host Interface process will only initiate a repeat key message if it receives a KEY REPEAT command from the Network Control Supervisor (NCS). The processing described below starts from the point where the ANSI Host Interface receives the NCS command. The steps corresponding to the diagram are also described below.

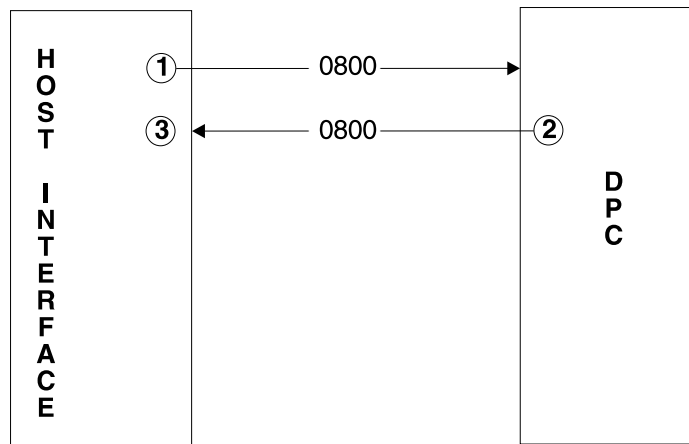


1. When the ANSI Host Interface receives the Repeat Key command from the NCS, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the ANSI Host Interface logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the ANSI Host Interface continues with the next step (step 1b).
 - b. Checks for an available station to which to send the message.
 - 1) If no stations are available, the ANSI Host Interface logs a message indicating that it is unable to send the repeat key message and drops the repeat key command. No further processing is performed.
 - 2) If a station is available, processing continues with the next step (step 1c).
 - c. Sends an 0800 message to the host with the network management information code set to 163. This message includes a blank KEY-MGMT.X917-CSM-DATA field.

- d. Starts a network management message timer for the message. If this timer expires before an 0800 response is received from the host, the ANSI Host Interface logs a message indicating the repeat key message timed out and no further processing is performed.
2. The host receives the repeat key message, processes the message, and returns an 0800 response with Key Service Message (KSM) data to the ANSI Host Interface.
3. The ANSI Host Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out. If the 0800 message timed out before the 0800 response was received, the ANSI Host Interface drops the message, and no further steps are performed. If the 0800 message has not timed out, processing continues with the next step.
 - b. Continues processing depending on the network management information code sent in the 0800 response message.
 - 1) If the transaction was denied (the network management information code was 363), the ANSI Host Interface logs a message and drops the 0800 message.
 - 2) If the transaction was approved (the network management information code was 263), the ANSI Host Interface performs as follows:
 - a) Verifies that the receiver and originator of the message are correct. To verify the originator and receiver of the message, the ANSI Host Interface compares the receiver ID from the RSI to the value from the ORIGINATING ID field on KEYF screen 3, and the originator ID from the KSM data to the value from the RECEIVING ID field on KEYF screen 3. If the receiver or originator does not match the corresponding KEYF values for this DPC, the ANSI Host Interface logs a message and drops the 0800 response.
 - b) Verifies the key. To do this, the ANSI Host Interface process translates the key sent by the host and verifies the check digits. The check digits generated as a result of translating the key must match the check digits sent by the host. If the ANSI Host Interface cannot translate the key or the check digits do not match, the ANSI Host Interface logs a message and drops the 0800 response.
 - c) Updates the KEYF with the new key, check digit, and counter information.

Verify Key Message Example

The diagram below illustrates the message flow between the ANSI Host Interface and the host when the ANSI Host Interface initiates a verify key message. In this case, the host is responsible for generating the key being verified. The ANSI Host Interface process will only initiate a verify key message if it receives a KEY VERIFY command from the Network Control Supervisor (NCS). The processing described below starts from the point where the ANSI Host Interface receives the NCS command. The steps corresponding to the diagram are also described below.



1. When the ANSI Host Interface receives the Verify Key command from the NCS, it performs as follows:
 - a. Verifies that the syntax is correct and that dynamic key management is supported. If the syntax is incorrect or dynamic key management is not supported, the ANSI Host Interface logs a message and drops the command. If the syntax is correct and dynamic key management is supported, the ANSI Host Interface continues with the next step (step 1b).
 - b. Checks for an available station to which to send the message.
 - 1) If no stations are available, the ANSI Host Interface logs a message indicating that it is unable to send the repeat key message and drops the repeat key command. No further processing is performed.
 - 2) If a station is available, processing continues with the next step (step 1c).
 - c. Sends an 0800 message to the host with the network management information code set to 164. This message includes a blank KEY-MGMT.X917-CSM-DATA field.

- d. Starts a network management message timer for the message. If this timer expires before an 0800 response is received from the host, the ANSI Host Interface logs a message indicating that the verify key message timed out. No further processing is performed.
2. The host receives the verify key message, processes the message, and returns an 0800 response to the ANSI Host Interface.
3. The ANSI Host Interface receives the 0800 response and processes it as follows:
 - a. Verifies that the original 0800 message did not time out. If the 0800 message timed out before the 0800 response was received, the ANSI Host Interface drops the message, and no further steps are performed. If the 0800 message has not timed out, processing continues with the next step.
 - b. Continues processing depending on the network management information code sent in the 0800 response message.
 - 1) If the transaction was denied (the network management information code was 364), the ANSI Host Interface logs a message indicating the host denied the request and drops the 0800 message.
 - 2) If the transaction was approved (the network management information code was 264), the ANSI Host Interface logs a message indicating that the command was successful and drops the 0800 message.

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Section 4

Store-and-Forward Processing

Store-and-forward processing allows for storing transactions that BASE24 has processed when an issuer host is down or unavailable.

The ANSI Host Interface process keeps track of each station and DPC. If a message bound for a DPC cannot be transmitted, the ANSI Host Interface places the message in the Store-and-Forward File (SAF) for later transmission. When communications with the host resume, the messages in the SAF are sent to the host.

This section describes store-and-forward processing. It includes information on what messages can be placed in the SAF, when messages are placed in the SAF, and when messages from the SAF are sent to the host. It also contains message flows that illustrate store-and-forward processing.

The Store-and-Forward File (SAF)

The Store-and-Forward File (SAF) temporarily stores messages bound for a host until those messages can be sent to the intended DPCs. The SAF is a key-sequenced file, and the messages are processed by DPC as first-in, first-out. In other words, the oldest SAF records are sent to the host first.

Messages in the SAF are stored in the format that they are to be sent to the host.

Issuer hosts can have store-and-forward messages sent to them via communications lines, or following long periods of down-time, the SAF can be extracted to tape in order to keep from clogging communications once communications have been restored.

Messages Placed in the SAF

The ANSI Host Interface places the following message types in the SAF:

- 0220—Financial Completion Advice
- 0221—Financial Completion Advice Repeat
- 0400—Reversal
- 0401—Reversal Repeat
- 5400—Adjustment
- 5401—Adjustment Repeat

0220, 0400, and 5400 messages are placed in the SAF if the intended DPC is marked down at the time the message is to be sent. If the ANSI Host Interface receives one of these messages to be sent to a DPC, but discovers the DPC is marked down, the ANSI Host Interface places the message in the SAF without attempting to send it to the DPC.

If the ANSI Host Interface does send a 0220, 0400, or 5400 message to a DPC, but the message times out, the ANSI Host Interface changes the message to a 0221, 0401, or 5401 message and places it in the SAF.

Messages Not Placed in the SAF

The ANSI Host Interface never places the following message types in the SAF:

0200—Financial Transaction Request
0210—Financial Transaction Request Response
0230—Financial Transaction Advice Response
0410—Reversal Response
5410—Interchange Adjustment Response
9220—Store-and-Forward Message
9230—Store-and-Forward Response

0200 and 0210 messages are interactive messages and their timing is crucial. If either is not processed within a given period of time, the transaction they represent is denied. Therefore, neither is placed in the SAF.

0230, 0410, and 5410 messages are only used as acknowledgments or responses to other messages. Here again, timing is the key. If a host does not receive a required acknowledgment or response within a given time, it will simply assume that BASE24 never received the message and may then send it again as a duplicate.

9220 and 9230 are the message types used to send store-and-forward messages, but messages are never actually stored in the SAF using these message types.

Deleting Messages from the SAF

Records are deleted from the SAF under the following circumstances:

- When the ANSI Host Interface determines that the SAF record was processed successfully by the host.

If a host must send text-level acknowledgments, the ANSI Host Interface uses the text-level acknowledgments to determine when the host has processed the SAF records. If the host is not required to send text-level acknowledgments, the BASE24 Nucleus notifies the ANSI Host Interface that the message was sent successfully. In both of these cases, the ANSI Host Interface assumes the record was processed successfully by the host and deletes the record.

- When the ANSI Host Interface determines that the SAF record cannot be processed by the host.

If the message is returned as failed from the host or the same store-and-forward message has timed out the number of times specified in the MAX SAF RETRY field on HCF screen 1, the ANSI Host Interface assumes the host cannot process the message. In this case, the ANSI Host Interface writes the message to its log and deletes it from the SAF.

- When the record is extracted from the SAF via the SAF extract.

Once a record is extracted to tape, the Super Extract process deletes it from the SAF.

Sending Store-and-Forward Messages to a DPC

The ANSI Host Interface checks for the presence of store-and-forward messages to be sent to a host at the following times:

- After a successful logon of the host.
- After receiving and processing a 9230 or 9231 message from the host. These messages are acknowledgments to previously-sent store-and-forward messages.
- After receiving and processing a 0210 message from the DPC.
- Upon completion of a SAF extract.
- When a previously-sent store-and-forward message times out.

If there are store-and-forward messages to send, the ANSI Host Interface determines whether a SAF extract is in progress for the DPC. If a SAF extract is not in progress, the ANSI Host Interface begins sending the store-and-forward messages.

DPCs can have their store-and-forward messages sent to them all at once or interspersed with live transaction messages. This option is controlled via the STORE AND FORWARD field in the Base segment of the HCF.

Noninterspersed Store-and-Forward Messages

If the HCF specifies that all store-and-forward transactions must be sent prior to sending any live transaction messages, the ANSI Host Interface stops sending any other types of messages while it sends store-and-forward messages.

Once store-and-forward processing has started, no other message types are sent to the host until the SAF is empty. 0200 messages from an Authorization process are returned to the Authorization process as failed messages; 0210 messages from the Authorization process are reversed; 0220, 0400, and 5400 messages are placed in the SAF.

The ANSI Host Interface does accept 0210 messages returning from a host. This allows transactions in progress to complete.

Interspersed Store-and-Forward Messages

If the HCF specifies that live transaction messages are to be interspersed with store-and-forward transactions, the ANSI Host Interface continues sending 0200, 0220, and 5400 messages along with its store-and-forward messages. For single-station hosts, the 0200, 0220, and 5400 messages always take priority.

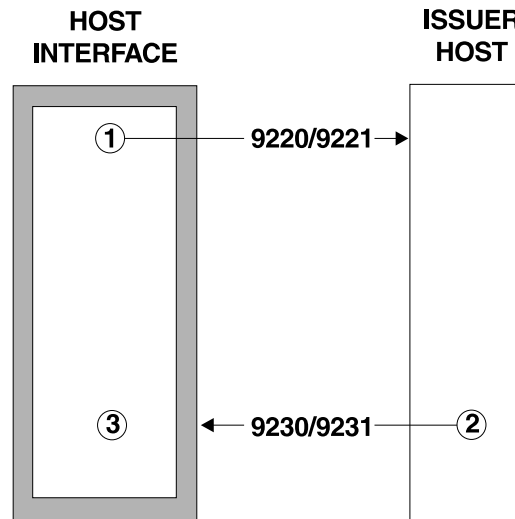
The only messages affected in this case are 0400 messages going to the host. Instead of being sent to the host, these are placed in the SAF to be sent at the end of the store-and-forward processing. This ensures that a reversal will not reach the host before a transaction.

Store-and-Forward Cycles

Store-and-forward cycles are single-threaded. The ANSI Host Interface never sends more than one store-and-forward message to a DPC at a time, regardless of the number of stations associated with the DPC and regardless of whether the DPC is single-threaded or multithreaded.

Store-and-Forward Message Processing

The following diagram illustrates the message exchange expected by BASE24 when store-and-forward messages are sent to an issuer host.



1. The ANSI Host Interface sends a store-and-forward message to the host by performing several steps:
 - a. Reads a record from the SAF.

The ANSI Host Interface selects the oldest record in the file for the DPC. The record is in the SEM format.
 - b. Sets a flag for the DPC to indicate that store-and-forward processing is in progress. The ANSI Host Interface performs this step only on the first store-and-forward message processed.
 - c. Sets the TYP-ORIG field in the SEM to 0220, 0400, or 5400 depending on the type of message.
 - d. Sets the TYP field in the SEM.
 - 1) If the TYP field in the SEM is 0220, 0400, or 5400, the ANSI Host Interface sets it to 9220.
 - 2) If the TYP field in the SEM is 0221, 0401, or 5401, the ANSI Host Interface sets it to 9221.
 - e. Locates an available send/receive station and sends the message.
 - f. Sets a store-and-forward timer for the message.

2. The host processes the 9220 or 9221 message and returns a 9230 or 9231 message if text-level acknowledgments are required.

If text-level acknowledgments are not required, the Nucleus notifies the ANSI Host Interface when it has successfully sent the message to the DPC.

Upon receipt of a 9230 or 9231 message or notification from the Nucleus that the 9220 message was sent, the ANSI Host Interface performs the following steps:

- a. Locates the corresponding record in the SAF and deletes it.
- b. Clears the SAF retries count (i.e., the number of times the SAF record had been previously sent to the host).
- c. Determines whether there are more SAF records to send.

If there are no more SAF records to send, the ANSI Host Interface terminates the store-and-forward processing and resets the flag for the DPC to indicate that store-and-forward processing is no longer in progress.

If there are more SAF records to send, the ANSI Host Interface begins again at step 1.

BASE24-atm Message Requirements

The SEM field requirements for each of the message types shown in the message flow are shown below.

Level and Field Name	9220/9221	9230/9231
02 ROUTE		
03 STAT	*	*
03 PATH	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 CARD-ACCT	*	*
02 TRAN-CDE	*	*
02 AMT-1	*	*

Level and Field Name	9220/9221	9230/9231
02 AMT-2	*	*
02 AMT-3	*	*
02 TRACE-NO	*	*
02 TRAN-DAT	*	*
02 POST-DAT	*	*
02 TRANSMISSION-DAT	*	*
02 TRAN-TIM	*	*
02 RTTN	*	*
02 ISSUER-ROUTING-NUMBER	*	*
02 CURRENCY-CDE	*	*
02 REVERSAL-CDE		
02 TYPE-ORIG	*	*
02 POINT-TRAN-SPEC	*	*
02 RESP-CDE	*	*
02 TERM-CITY-ST	*	*
02 ORIGL		
02 ISS-TRACE-NO		
02 PIN	*	*
02 FROM-ACCT	*	*
02 TO-ACCT	*	*
02 SHARING		
04 MBR-NUM	*	*
04 SHARING-GROUP	*	*

Level and Field Name	9220/9221	9230/9231
04 TERM-STATE	*	*
04 TERM-COUNTY	*	*
04 NOT-ON-US-MASK	*	*
04 FIID	*	*
02 TERM		
04 OWNER	*	*
04 TERM-NAME-LOC	*	*
04 TERM-OWNER-NAME	*	*
04 ATM-ID	*	*
04 BRANCH-ID	*	*
04 REGION-ID	*	*
04 REC-NUM	*	*
04 TYP	*	*
04 TIM-OFST	*	*
02 CARD		
04 FIID-LEN	*	*
04 CARD-FIID	*	*
04 PIN-OFST	*	*
02 MISC		
04 PAYEE-CDE	*	*
04 ICHG-SETT-DAT	*	*
02 DEPOSITORY		
04 DEP-BAL-CREDIT	*	*

Level and Field Name	9220/9221	9230/9231
04 AVAIL-DEP-TYPE	*	*
04 DEP-TYPE	*	*
02 HOST		
04 PROC-ACCT-IND	*	*
02 USER-FLD		X
02 TRACK2	*	*
02 TRACK3	*	*
02 FILLER	*	*

Key:

* = Initialized by BASE24-atm; must be copied as is.

X = Issuer host must provide.

b = (Blank) not applicable; zero- or blank-filled.

BASE24-pos PSEMA/PSEMAE Message Requirements

The fields required for BASE24-generated PSEMA or PSEMAE store-and-forward transaction messages and for issuer host DPC store-and-forward acknowledgment PSEMA or PSEMAE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMAE format. The PSEMA and PSEMAE are identical with the exception of the address verification fields.

Level and Field Name	9220/9221	9230/9231
02 ROUTE		
03 STAT	*	*
03 PATH	*	*
05 PRODUCT-IND	*	*

Level and Field Name	9220/9221	9230/9231
05 RELEASE-NUM	*	*
05 DPC-NUMBER	*	*
05 CARD-FIID	*	*
05 ORIGI-TYPE	*	*
05 CARD-LOG-NET	*	*
05 PSEM-TYP	*	*
05 CARD-TYP	*	*
05 AUTHORIZER	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 TRAN	*	*
04 CARD	*	*
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 TRAN-DAT	*	*
04 POST-DAT	*	*
04 TRAN-TIM	*	*
06 HH	*	*
06 MINUTE	*	*

Level and Field Name	9220/9221	9230/9231
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*
04 REVERSAL-CDE	*	*
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	*	*
04 RESP-CDE	*	*
04 RETRIEVAL-REF-NUM	*	*
04 PIN	*	*
04 APPROVAL-CODE	*	*
04 RETAILER	*	*
06 FIID	*	*
06 GRP	*	*
06 REGN	*	*
06 ID	*	*
06 DRAFT-CAPTURE	*	*
06 SETTLE-FLAG	*	*
04 MEMBER-NUMBER	*	*
04 REFERRAL-PHONE	*	*
04 PRE-AUTH-HOLD	*	*
04 ADDR-DATA	*	*
06 ADDR	*	*

Level and Field Name	9220/9221	9230/9231
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

Key:

* = Set by BASE24-pos; must be copied as is.

X = Issuer host must provide.

b = (Blank) not applicable; zero- or blank-filled.

BASE24-pos PSEMF/PSEMFE Message Requirements

The fields required for BASE24-generated PSEMF or PSEMFE store-and-forward transaction messages and for issuer host DPC store-and-forward acknowledgment PSEMF or PSEMFE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMFE format. The PSEMF and PSEMFE are identical with the exception of the address verification fields.

Level and Field Name	9220/9221	9230/9231
02 ROUTE		
03 STAT	*	*
03 PATH	*	*
05 PRODUCT-IND	*	*
05 RELEASE-NUM	*	*
05 DPC-NUMBER	*	*
05 CARD-FIID	*	*
05 ORIGI-TYPE	*	*
05 CARD-LOG-NET	*	*
05 PSEM-TYP	*	*

Level and Field Name	9220/9221	9230/9231
05 CARD-TYP	*	*
05 AUTHORIZER	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 TRAN	*	*
04 CARD	*	*
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 POST-DAT	*	*
04 TRAN-DAT	*	*
04 TRAN-TIM	*	*
06 HH	*	*
06 MINUTE	*	*
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*
04 REVERSAL-CDE	*	*
04 TYPE-ORIG	*	*

Level and Field Name	9220/9221	9230/9231
04 POINT-TRAN-SPEC	*	*
04 RESP-CDE	*	*
04 CARD-ACCEPTOR-NAME	*	*
04 CARD-ACCEPTOR-CITY-ST	*	*
04 RETRIEVAL-REF-NUM	*	*
04 PIN	*	*
04 FROM-ACCT-LEN	*	*
04 FROM-ACCT-NUM	*	*
04 APPROVAL-CODE	*	*
04 RETAILER	*	*
06 LOGICAL-NET	*	*
06 FIID	*	*
06 GRP	*	*
06 REGN	*	*
06 ID	*	*
06 TERM-ID	*	*
06 SHIFT	*	*
06 EMPLOYEE	*	*
08 ID	*	*
08 GRP	*	*
08 USER-ID	*	*
06 DRAFT-CAPTURE	*	*
06 SETTLE-FLAG	*	*

Level and Field Name	9220/9221	9230/9231
04 MEMBER-NUMBER	*	*
04 SHARING	*	*
04 MISC	*	*
06 ORIG	*	*
06 DEST	*	*
06 INVOICE-NUM	*	*
06 ORIG-INVOICE-NUM	*	*
06 CURRENCY-CDE	*	*
06 BATCH-NUM	*	*
06 REFERRAL-PHONE	*	*
06 PRE-AUTH-HOLD	*	*
06 PRODUCT	*	*
08 LEN	*	*
08 DAT	*	*
04 TRACK2-LEN	*	*
04 TRACK2-DATA	*	*
04 ADDR-DATA	*	*
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

Key:

- * = Set by BASE24-pos; must be copied as is.
- X = Issuer host must provide.
- Ø = (Blank) not applicable; zero- or blank-filled.

Rejected Store-and-Forward Messages

If a 9220 message cannot be processed by a host, the message must be rejected and returned with a value of 21 in the ROUTE.STAT field of the SEM.

When the ANSI Host Interface receives a rejected message back from the host, it:

1. Writes the message to its log
2. Deletes the record from the SAF.
3. Clears the SAF retries count.
4. Continues store-and-forward processing with the next record in the SAF.

Failed Store-and-Forward Messages

Messages failed by the Nucleus will contain zeros in the ROUTE.STAT field of the SEM. If a message is failed by the Nucleus, the ANSI Host Interface goes through basically the same processing as it does for a timeout, except that the timeout counter is not incremented.

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Section 5

BASE24-atm Financial Message Handling

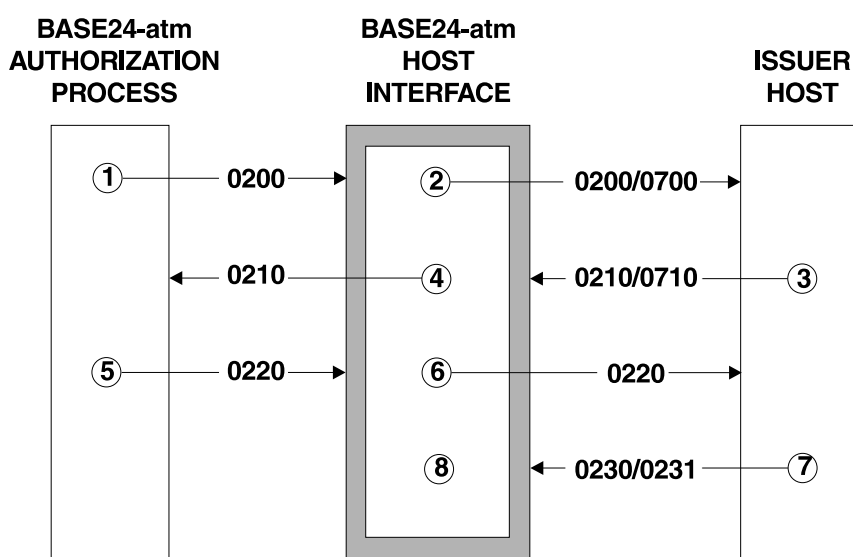
This section contains an overview of how BASE24-atm financial transaction messages are handled by the ANSI Host Interface. It includes message flows for both issuer and acquirer host processing.

Issuer Host Message Handling

This section describes the processing involved for messages to and from an issuer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected for financial transactions being sent to an issuer host by the BASE24-atm ANSI Host Interface process:



Processing Description

The following steps describe the processing for financial transactions being sent to an issuer host by the BASE24-atm ANSI Host Interface:

1. BASE24 receives a financial transaction request, which is sent to the BASE24-atm Authorization process.

The BASE24-atm Authorization process determines that the request must be sent to an issuer host for authorization and sends it to the ANSI Host Interface for forwarding to that host.

The BASE24-atm Authorization process gets the name of the ANSI Host Interface to which to send the message from the ATM ROUTING TABLE fields on IDF screen 9.

2. The ANSI Host Interface receives the 0200 message and:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the RQST.DPC-NUM field in the STM.
 - b. Creates an SEM from the STM.
 If the transaction is a balance inquiry, the ANSI Host Interface creates a 0700 message. Otherwise, the ANSI Host Interface creates a 0200 message.
 - c. Updates the EXIT-TIM field in the STM to the current time.
 - d. Determines its ability to send the message. The ANSI Host Interface cannot send the message if:
 - The DPC is marked down
 - There is a noninterspersed store-and-forward cycle in progress
 - No send/receive stations are available
 - The current number of outstanding messages to the DPC equals the MAXIMUM OUTSTANDING REQUESTS OUTBOUND value on HCF screen 1.

If any of the above conditions exist, the ANSI Host Interface:

- 1) Sets the RTE.STAT field in the STM to 11 to indicate that the DPC was unavailable (12 if the DPC was marked down).
 - 2) Sets the RQST.COMPL-REQ field in the STM to 2 to indicate that an advice is required if the request is approved offline.
 - 3) Returns the 0200 message (STM) to the process indicated in the ORIG-PRO-NAME field of the STM.
- e. Saves copies of the STM and SEM in extended memory.
 - f. Sets an outbound request timer for the number of seconds specified by the OUTBOUND LIMIT field on HCF screen 5.

If this timer expires before a 0210 or 0710 message is received, the ANSI Host Interface performs as follows:

- 1) Sets the RTE.STAT field in the STM to 11 to indicate that the DPC was unavailable (12 if the DPC was marked down).
- 2) Sets the RQST.COMPL-REQ field in the STM to 2 to indicate that an advice is required if the request is approved offline.
- 3) Returns the 0200 message (STM) to the process indicated in the ORIG-PRO-NAME field of the STM.

- g. Updates its internal counters. The ANSI Host Interface:
 - 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ANSI Host Interface.
- h. Sends the 0200 or 0700 message to the DPC.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using two values from HCF screen 5: QUEUE SUBTRACT and OUTBOUND LIMIT. The QUEUE SUBTRACT value is subtracted from the OUTBOUND LIMIT value and the resulting value is used.
- 3. The host validates the request, processes the transaction, and returns a response either approving or denying the transaction.
 - a. If the host received a 0200 message, it must return a 0210 message.
 - b. If the host received a 0700 message, it must return a 0710 message.
- 4. The ANSI Host Interface receives the 0210 or 0710 message and:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Retrieves the stored SEM and STM from extended memory.
 - c. Deletes the outbound request timer.
 - d. Updates its counters. The ANSI Host Interface:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Adds one to the number of requests processed for the DPC.
 - 3) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
 - 4) Subtracts one from the total number of outstanding messages.
 - e. Sets the RE-ENTRY-TIM field in the STM to the current time.
 - f. Sets the TYP field in the STM to 0210.
 - g. Updates the STM from the SEM using only that information which may have been changed by the host.

- h. Sends the 0210 message (STM) to the Authorization process.
- 5. The BASE24-atm Authorization process finishes processing the transaction, logs it to the TLF, and returns it to its originator.

If the host requires advices, the BASE24-atm Authorization process generates the appropriate 0220 message and sends it to the ANSI Host Interface.

- 6. The ANSI Host Interface receives the 0220 message and:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the RQST.DPC-NUM field in the STM.
 - b. Creates an SEM from the STM.
 - c. Checks the RESP-CDE field in the STM. If the transaction was denied, the ANSI Host Interface drops the message.
 - d. Checks the RQST.COMPL-REQ field in the STM. If that field is equal to 2 (host is offline—completion required), the ANSI Host Interface writes the 0220 message (SEM) to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - e. Determines whether the DPC is marked down. If so, the ANSI Host Interface writes the SEM (0220) message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - f. Determines whether there is a noninterspersed store-and-forward cycle in progress. If so, the ANSI Host Interface writes the SEM (0220) message to the SAF, and the message is handled in sequential order.
 - g. Locates an available send/receive station. If no send/receive stations are available, the ANSI Host Interface writes the SEM (0220) message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
 - h. Sends the 0220 message to the DPC.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using two values from HCF screen 5: QUEUE SUBTRACT and COMPLETION ACK. The QUEUE SUBTRACT value is subtracted from the COMPLETION ACK value and the resulting value is used.

- i. Determines whether the DPC is required to return a text-level acknowledgment.

DPCs are required to return text-level acknowledgments if the DPC's stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface:

- 1) Stores copies of the SEM and STM.
 - 2) Sets a completion acknowledgment timer.
 - 3) Adds one to the number of messages currently outstanding to the DPC.
 - 4) Adds one to the number of messages currently outstanding from the ANSI Host Interface.
7. The host processes the 0220 message and returns a 0230 or 0231 message if text-level acknowledgments are required.
 8. Upon receipt of a 0230 or 0231 message, the ANSI Host Interface:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Deletes the stored SEM and STM from extended memory.
 - c. Deletes the completion acknowledgment timer.
 - d. Subtracts one from the number of messages currently outstanding to the DPC.
 - e. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
 - f. Drops the 0230 or 0231 message.

Message Requirements

The SEM field requirements for each of the message types described in the message flow are shown below. The first four message type columns are for non-balance inquiry transactions; the last four message type columns are for balance inquiry transactions.

Level and Field Name	Non-Balance Inquiry				Balance Inquiry			
	0200	0210	0220	0230/0231	0700	0710	0220	0230/0231
02 ROUTE								
03 STAT	*	*	*	*	*	*	*	*
03 PATH	*	*	*	*	*	*	*	*
02 TYP	*	X	*	X	*	X	*	X
02 BYTE-MAP	*	*	*	*	*	*	*	*
02 CARD-ACCT	*	*	*	*	*	*	*	*
02 TRAN-CDE	*	*	*	*	*	*	*	*
02 AMT-1	*	*	*	*		X	*	*
02 AMT-2		X	*	*		X	*	*
02 AMT-3		X	*	*		X	*	*
02 TRACE-NO	*	*	*	*	*	*	*	*
02 TRAN-DAT	*	*	*	*	*	*	*	*
02 POST-DAT	*	*	*	*	*	*	*	*
02 TRANSMISSION-DAT	*	*	*	*	*	*	*	*
02 TRAN-TIM	*	*	*	*	*	*	*	*
02 RTTN	*	*	*	*	*	*	*	*

Level and Field Name	Non-Balance Inquiry				Balance Inquiry			
	0200	0210	0220	0230/0231	0700	0710	0220	0230/0231
02 ISSUER-ROUTING-NUMBER	*	*/X	*	*	*	*	*	*
02 CURRENCY-CDE	*	*	*	*	*	*	*	*
02 REVERSAL-CDE								
02 TYPE-ORIG	*	*	*	*	*	*	*	*
02 POINT-TRAN-SPEC	*	*	*	*	*	*	*	*
02 RESP-CDE		X	*	*		X	*	*
02 TERM-CITY-ST	*	*	*	*	*	*	*	*
02 ORIGL								
02 ISS-TRACE-NO		X			*	*	*	*
02 PIN	*	*	*	*	*	*	*	*
02 FROM-ACCT	*	*/X	*	*	*	*	*	*
02 TO-ACCT	*	*/X	*	*				
02 SHARING								
04 MBR-NUM	*	*	*	*	*	*	*	*
04 SHARING-GROUP	*	*	*	*	*	*	*	*
04 TERM-STATE	*	*	*	*	*	*	*	*
04 TERM-COUNTY	*	*	*	*	*	*	*	*
04 NOT-ON-US-MASK	*	*	*	*	*	*	*	*
04 FIID	*	*	*	*	*	*	*	*
02 TERM								

Level and Field Name	Non-Balance Inquiry				Balance Inquiry			
	0200	0210	0220	0230/0231	0700	0710	0220	0230/0231
04 OWNER	*	*	*	*	*	*	*	*
04 TERM-NAME-LOC	*	*	*	*	*	*	*	*
04 TERM-OWNER-NAME	*	*	*	*	*	*	*	*
04 ATM-ID	*	*	*	*	*	*	*	*
04 BRANCH-ID	*	*	*	*	*	*	*	*
04 REGION-ID	*	*	*	*	*	*	*	*
04 REC-NUM	*	*	*	*	*	*	*	*
04 TYP	*	*	*	*	*	*	*	*
04 TIM-OFST	*	*	*	*	*	*	*	*
02 CARD								
04 FIID-LEN	*	*	*	*	*	*	*	*
04 CARD-FIID	*	*	*	*	*	*	*	*
04 PIN-OFST	*	*	*	*	*	*	*	*
02 MISC								
04 PAYEE-CDE	*	*	*	*				
04 ICHG-SETT-DAT	*	*	*	*	*	*	*	*
02 DEPOSITORY								
04 DEP-BAL-CREDIT		X	*	*				
04 AVAIL-DEP-TYPE	*	*	*	*				
04 DEP-TYP	*	*	*	*				

Level and Field Name	Non-Balance Inquiry				Balance Inquiry			
	0200	0210	0220	0230/0231	0700	0710	0220	0230/0231
02 HOST								
04 PROC-ACCT-IND	*	*	*	*	*	*	*	*
02 USER-FLD		X	*	X		X	*	X
02 TRACK2	*	*	*	*	*	*	*	*
02 TRACK3	*	*	*	*				
02 FILLER	*	*	*	*				

Key:

* = Initialized by BASE24-atm; must be copied as is.

X = Issuer host must provide.

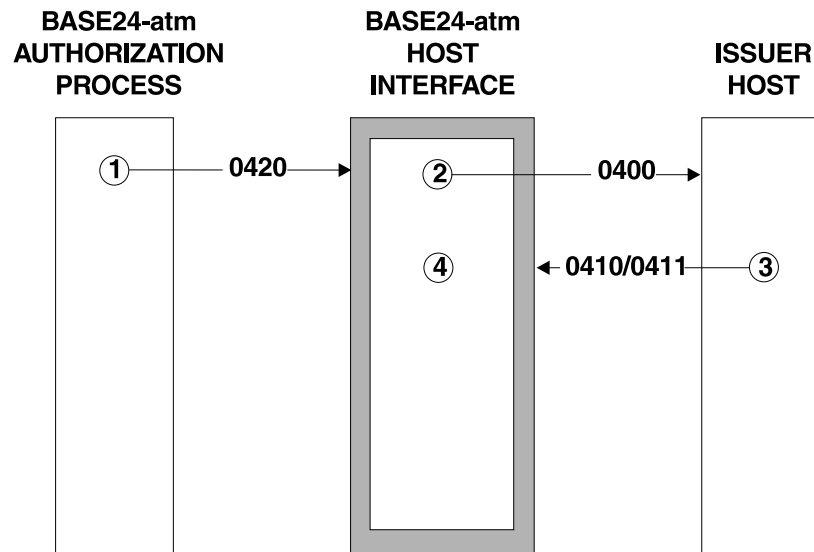
Ø = (Blank) not applicable; zero- or blank-filled.

/ = Optional.

Reversal Message Flow

The following diagram illustrates the message exchange expected for reversal transactions being sent to an issuer host by the BASE24-atm ANSI Host Interface.

A reversal indicates that a previously-approved transaction did not complete as was intended.



Processing Description

The following steps describe the processing for reversal transactions being sent to an issuer host by the BASE24-atm ANSI Host Interface:

1. BASE24 is advised that a transaction did not complete as previously authorized. (The notice may be from an ATM controlled by BASE24, an acquirer host, or an interchange.)

The BASE24-atm Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the TLF, and sends a 0420 message STM to the ANSI Host Interface.

2. The ANSI Host Interface receives the 0420 message and processes it as follows:
 - a. Locates the DPC record in its internal table using the RQST.DPC-NUM field in the STM.

- b. Creates an SEM from the STM.

During the creation of the SEM, the ANSI Host Interface changes the TYP field in the SEM to 0400.

- c. Determines whether the DPC is marked down. If so, the ANSI Host Interface writes the 0400 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
- d. Determines whether there is a store-and-forward cycle (interspersed or noninterspersed) in progress for the DPC. If so, the ANSI Host Interface writes the 0400 message to the SAF, and the message is handled in sequential order.
- e. Checks the RQST.COMPL-REQ field in the STM. If it equals 2 (host offline—completion required), the ANSI Host Interface writes the 0400 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
- f. Locates an available send/receive station. If no send/receive stations are available, the ANSI Host Interface writes the 0400 message to the SAF and takes no further action with the message until a store-and-forward cycle is started.
- g. Sends the 0400 message to the DPC.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using two values from HCF screen 5: QUEUE SUBTRACT and COMPLETION ACK. The QUEUE SUBTRACT value is subtracted from the COMPLETION ACK value and the resulting value is used.

- h. Determines whether the DPC is required to return a text-level acknowledgment.

DPCs are required to return text-level acknowledgments if the DPC's stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface:

- 1) Stores copies of the SEM and STM.
- 2) Sets a completion acknowledgment timer.
- 3) Adds one to the number of messages currently outstanding to the DPC.

- 4) Adds one to the number of messages currently outstanding from the ANSI Host Interface.
3. The host processes the 0400 message and returns a 0410 or 0411 message if text-level acknowledgments are required.
4. Upon receipt of a 0410 or 0411 message, the ANSI Host Interface:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Deletes the stored SEM and STM from extended memory.
 - c. Deletes the completion acknowledgment timer.
 - d. Subtracts one from the number of messages currently outstanding to the DPC.
 - e. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
 - f. Drops the 0410 or 0411 message.

Message Requirements

The SEM field requirements for each of the message types described in the message flow are shown below.

Level and Field Name	0400	0410/0411
02 ROUTE		
03 STAT	*	*
03 PATH	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 CARD-ACCT	*	*
02 TRAN-CDE	*	*
02 AMT-1	*	*
02 AMT-2	*	*

Level and Field Name	0400	0410/0411
02 AMT-3	*	*
02 TRACE-NO	*	*
02 TRAN-DAT	*	*
02 POST-DAT	*	*
02 TRANSMISSION-DAT	*	*
02 TRAN-TIM	*	*
02 RTTN	*	*
02 ISSUER-ROUTING-NUMBER	*	*
02 CURRENCY-CDE	*	*
02 REVERSAL-CDE	*	*
02 TYPE-ORIG	*	*
02 POINT-TRAN-SPEC	*	*
02 RESP-CDE	*	*
02 TERM-CITY-ST	*	*
02 ORIGL		
02 ISS-TRACE-NO	*	*
02 PIN	*	*
02 FROM-ACCT	*	*
02 TO-ACCT	*	*
02 SHARING		
04 MBR-NUM	*	*
04 SHARING-GROUP	*	*
04 TERM-STATE	*	*

Level and Field Name	0400	0410/0411
04 TERM-COUNTY	*	*
04 NOT-ON-US-MASK	*	*
04 FIID	*	*
02 TERM		
04 OWNER	*	*
04 TERM-NAME-LOC	*	*
04 TERM-OWNER-NAME	*	*
04 ATM-ID	*	*
04 BRANCH-ID	*	*
04 REGION-ID	*	*
04 REC-NUM	*	*
04 TYP	*	*
04 TIM-OFST	*	*
02 CARD		
04 FIID-LEN	*	*
04 CARD-FIID	*	*
04 PIN-OFST	*	*
02 MISC		
04 PAYEE-CDE	*	*
04 ICHG-SETT-DAT	*	*
02 DEPOSITORY		
04 DEP-BAL-CREDIT	*	*
04 AVAIL-DEP-TYPE	*	*

Level and Field Name	0400	0410/0411
04 DEP-TYPE	*	*
02 HOST		
04 PROC-ACCT-IND	*	*
02 USER-FLD		X
02 TRACK2	*	*
02 TRACK3	*	*
02 FILLER	*	*

Key:

* = Initialized by BASE24-atm; must be copied as is.

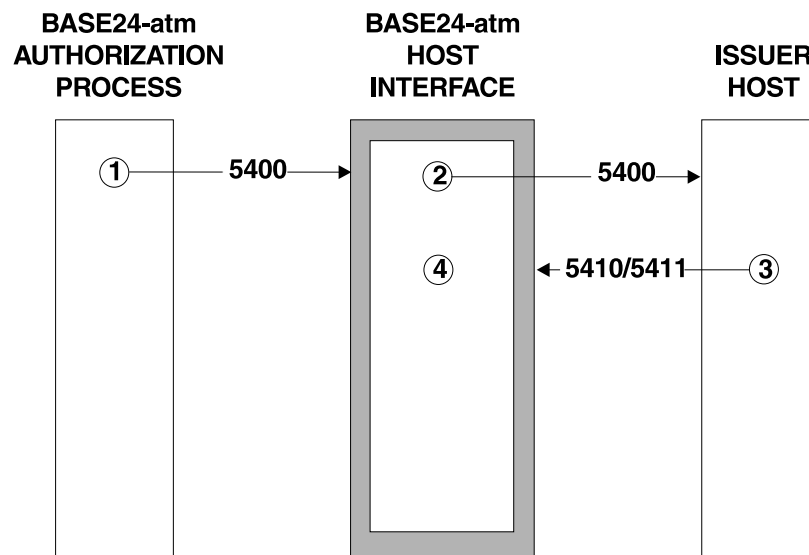
X = Issuer host must provide.

Ø = (Blank) not applicable; zero- or blank-filled.

Interchange Adjustment Transaction Message Flow

The following diagram illustrates the message exchange expected when BASE24-atm receives an interchange adjustment from a regional or national interchange that must be transmitted to an issuer host.

Note: Adjustments are used by some interchanges to advise BASE24 of previously-authorized financial transactions that do not complete as authorized. They are only allowed from interchanges.



Processing Description

BASE24-atm adjustment transactions are sent to the BASE24-atm Authorization process as 5400 messages by the Interchange Interface.

The Authorization process checks the HOST ADJ. PROCESSING flag on IDF screen 3 to determine whether the 5400 message must be sent to an issuer host. If the 5400 message must be sent to a host, the Authorization process forwards the message to the ANSI Host Interface.

The ANSI Host Interface handles the 5400 messages the same way it handles 0420 messages, with the exception of the message types.

Message Requirements

The SEM field requirements for each of the message types described in the message flow are shown below.

Level and Field Name	5400	5410/5411
02 ROUTE		
03 STAT	*	*
03 PATH	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 CARD-ACCT	*	*
02 TRAN-CDE	*	*
02 AMT-1	*	*
02 AMT-2	*	*
02 AMT-3	*	*
02 TRACE-NO	*	*
02 TRAN-DAT	*	*
02 POST-DAT	*	*
02 TRANSMISSION-DAT	*	*
02 TRAN-TIM	*	*
02 RTTN	*	*
02 ISSUER-ROUTING-NUMBER	*	*
02 CURRENCY-CDE	*	*
02 REVERSAL-CDE	*	*
02 TYPE-ORIG	*	*
02 POINT-TRAN-SPEC	*	*

Level and Field Name	5400	5410/5411
02 RESP-CDE	*	*
02 TERM-CITY-ST	*	*
02 ORIGL	*	*
02 ISS-TRACE-NO	*	*
02 PIN	*	*
02 FROM-ACCT	*	*
02 TO-ACCT	*	*
02 SHARING		
04 MBR-NUM	*	*
04 SHARING-GROUP	*	*
04 TERM-STATE	*	*
04 TERM-COUNTY	*	*
04 NOT-ON-US-MASK	*	*
04 FIID	*	*
02 TERM		
04 OWNER	*	*
04 TERM-NAME-LOC	*	*
04 TERM-OWNER-NAME	*	*
04 ATM-ID	*	*
04 BRANCH-ID	*	*
04 REGION-ID	*	*
04 REC-NUM	*	*
04 TYP	*	*

Level and Field Name	5400	5410/5411
04 TIM-OFST	*	*
02 CARD		
04 FIID-LEN	*	*
04 CARD-FIID	*	*
04 PIN-OFST	*	*
02 MISC		
04 PAYEE-CDE	*	*
04 ICHG-SETT-DAT	*	*
02 DEPOSITORY		
04 DEP-BAL-CREDIT	*	*
04 AVAIL-DEP-TYPE	*	*
04 DEP-TYPE	*	*
02 HOST		
04 PROC-ACCT-IND	*	*
02 USER-FLD		X
02 TRACK2	*	*
02 TRACK3	*	*
02 FILLER	*	*

Key:

* = Initialized by BASE24-atm; must be copied as is.

X = Issuer host must provide.

␣ = (Blank) not applicable; zero- or blank-filled.

Late Response from a DPC

If a DPC responds late to a request, the ANSI Host Interface performs as follows for the 0210 message:

1. Checks the RESP-CDE field in the SEM.
If the transaction was denied, the ANSI Host Interface drops the message.
2. Checks the TYP field in the SEM.
If the transaction was a balance inquiry, the ANSI Host Interface drops the message.
3. Creates a reversal from the 0210 message as follows:
 - a. Sets the TYP field to 0400.
 - b. Sets the REVERSAL-CDE field to 01 to indicate that the message timed out.
 - c. Sets the AMT-2 and AMT-3 fields to zeros.
 - d. Sets the PIN field to zeros.

4. Selects the station to which to send the message.

To do this, the ANSI Host Interface first checks the station from which the 0210 was received. If it is a send/receive station and is available, the ANSI Host Interface uses that one. Otherwise, it checks for any available send/receive stations.

If a station is available, the ANSI Host Interface uses it. If there are no available stations, the ANSI Host Interface writes the 0400 message to the SAF.

5. Sends the 0400 message to the selected station.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus may allow the message to queue to the host. This number of seconds is derived using two values from HCF screen 5: QUEUE SUBTRACT and COMPLETION ACK. The QUEUE SUBTRACT value is subtracted from the COMPLETION ACK value and the resulting value is used.

6. Determines whether the DPC is required to return a text-level acknowledgment.

DPCs are required to return text-level acknowledgments if the DPC's stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface:

- a. Stores a copy of the SEM.
- b. Sets a completion acknowledgment timer.
- c. Adds one to the number of messages currently outstanding to the DPC.
- d. Adds one to the number of messages currently outstanding from the ANSI Host Interface.

Failed Messages to Stations

If a message is forwarded to the Nucleus, but cannot be sent to a station, the Nucleus returns the message to the ANSI Host Interface as failed.

Upon receipt of a message failure, the ANSI Host Interface marks the station down and sets an extended network management message timer. It then performs based on message type.

Failed 0200/0700 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0200 or 0700 message from the Nucleus. This condition could be due to a station being unavailable or the transaction becoming stale while queued by the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored SEM and STM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters. The ANSI Host Interface:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. (For performance statistics, this value is divided by the number of requests processed for the DPC.)
 - d. Subtracts one from the total number of outstanding messages.
5. Sets the RTE.STAT field in the STM to 12 to indicate the line was down.
6. Sets the RQST.COMPL-REQ field in the STM to 2 to indicate that a 0220 message is required if the transaction is subsequently approved.
7. Returns the STM (as a failed 0200 message) to the BASE24-atm Authorization process.

Failed 0220/0400/5400 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0220, 0400, or 5400 message back from the Nucleus. This condition could be due to a station being unavailable or the transaction becoming stale while queued by the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored SEM and STM from extended memory.
3. Checks for an outbound request timer for the message and deletes it if found.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
6. Updates the TYP field in the SEM to 0221 or 0401.
7. Adds the message to the SAF.

Unintelligible Messages from a DPC

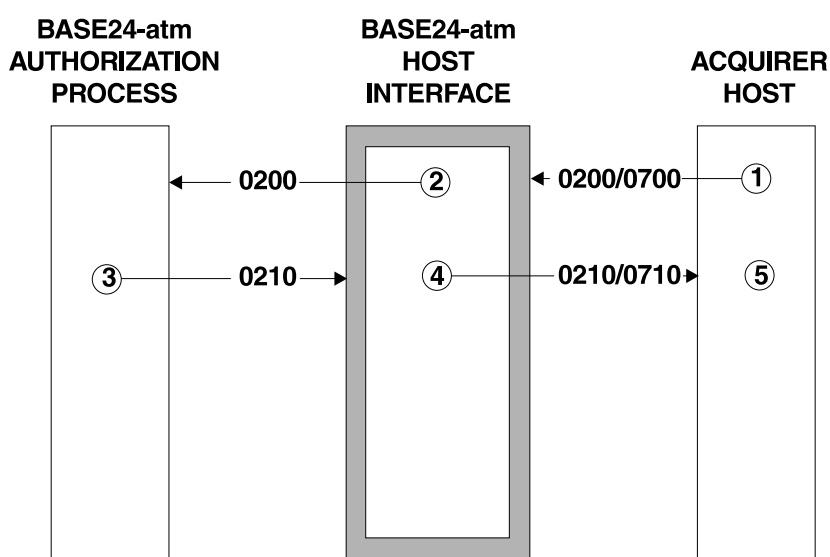
If the ANSI Host Interface receives a message with an invalid message type, or carrying information in an incorrect format, it changes the ROUTE.STAT field in the SEM to 21 to indicate an unknown message type and returns the message to the sending DPC.

Acquirer Host Message Handling

This section describes the processing involved for messages to and from an acquirer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-atm for financial transactions being sent to BASE24-atm by an acquirer host.



Processing Description

The following steps describe the processing for financial transactions being sent to the BASE24-atm ANSI Host Interface by an acquirer host:

1. A cardholder initiates a financial transaction at a terminal connected to and controlled by an acquirer host. The acquirer host then formats a 0200 or 0700 message and sends it to BASE24 (0700 messages may be used for balance inquiries).
2. The ANSI Host Interface receives the 0200 or 0700 message and performs as follows:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.

- b. Creates an STM from the SEM. (The TYP field in the STM is set to 0200.)
- c. Updates the following fields:
 - 1) RTE.ORIGINATOR—Set to 5 to indicate the transaction was initiated through an acquirer host.
 - 2) DPC-NUM—Set to the number of the DPC originating the transaction.
 - 3) ENTRY-TIM—Set to the current time.
- d. Checks the current number of outstanding 0200 messages to the Authorization process.

If it equals the MAXIMUM OUTSTANDING REQUESTS INBOUND value on HCF screen 1, the ANSI Host Interface sets the ROUTE.STAT field in the SEM to 11 to indicate that the destination was not available and returns the 0200 or 0700 message (SEM) to its originating DPC.

- e. Saves copies of the STM and SEM in extended memory.
- f. Sets an inbound request timer for the number of seconds specified by the INBOUND LIMIT field on HCF screen 5.

If this timer expires before a 0210 message is received, the ANSI Host Interface sets the ROUTE.STAT field in the SEM to 11 to indicate that the destination was not available and returns the 0200 or 0700 message (SEM) to its originating DPC.

- g. Updates its internal counters. The ANSI Host Interface:
 - 1) Adds one to the number of requests currently outstanding from the DPC.
 - 2) Adds one to the total number of requests currently outstanding from the ANSI Host Interface.
- h. Sends the 0200 message (STM) to the Authorization process indicated in the AUTH PROCESS field on HCF screen 5.

- 3. The BASE24-atm Authorization process processes the transaction, logs it to the TLF, and returns a 0210 message to the ANSI Host Interface.

Note: The transaction could also have been authorized by an interchange or issuer host, depending on the cardholder and the type of authorization processing performed.

- 4. The ANSI Host Interface receives the 0210 message and performs as follows:
 - a. Locates the stored SEM and STM.

- b. Deletes the inbound request timer.
 - c. Updates its internal counters. The ANSI Host Interface:
 - 1) Subtracts one from the number of requests currently outstanding from the DPC.
 - 2) Subtracts one from the total number of requests currently outstanding from the ANSI Host Interface.
 - d. Updates the SEM with only those fields from the 0210 message that may have been updated by the Authorization process.
 - e. If the transaction is a balance inquiry, sets the TYP field in the SEM to 0710. Otherwise, the TYP field in the SEM is set to 0210.
 - f. Locates an available send/receive station to which to send the 0210 message.

If there is not a send/receive station available, the ANSI Host Interface creates a reversal (0400) message using the STM and returns it to the Authorization process. (A reversal is only created if the transaction is approved and not a balance inquiry.)

Note: If the transaction is a balance inquiry and the REVERSE-BAL-INQ param in the LCONF is set to a “true” value, the ANSI Host Interface will not drop the transaction, but rather sends a reversal to the Authorization process.
 - g. Sends the 0210 or 0710 message (SEM) to the station.
5. The acquirer host then completes the transaction with the cardholder at the ATM.

Note: BASE24-atm does not accept 0220 advices from an acquirer host.

Message Requirements

The SEM field requirements for each of the message types described in the message flow are shown below. The first two message type columns are for non-balance inquiry transactions; the last two message type columns are for balance inquiry transactions.

Level and Field Name	Non-Balance Inquiry		Balance Inquiry	
	0200	0210	0700	0710
02 ROUTE				

Level and Field Name	Non-Balance Inquiry		Balance Inquiry	
	0200	0210	0700	0710
03 STAT	0	B	0	B
03 PATH	0	B	0	B
02 TYP	A	B	A	B
02 BYTE-MAP	A	*	A	*
02 CARD-ACCT	A	*	A	*
02 TRAN-CDE	A	*	A	*
02 AMT-1	A	*	A	*
02 AMT-2		B		B
02 AMT-3		B		B
02 TRACE-NO	0	B	0	B
02 TRAN-DAT	A	*	A	*
02 POST-DAT	A	*/B	A	*
02 TRANSMISSION-DAT	A	*	A	*
02 TRAN-TIM	A	*	A	*
02 RTTN	A	*	A	*
02 ISSUER-ROUTING- NUMBER	0	B	0	B
02 CURRENCY-CDE	A	*	A	*
02 REVERSAL-CDE				
02 TYPE-ORIG	A	*	A	*
02 POINT-TRAN-SPEC	A	*	A	*
02 RESP-CDE		B		B
02 TERM-CITY-ST	A	*	A	*

Level and Field Name	Non-Balance Inquiry		Balance Inquiry	
	0200	0210	0700	0710
02 ORIGL				
02 ISS-TRACE-NO		B		
02 PIN	A	*	A	*
02 FROM-ACCT	A	*/B	A	*
02 TO-ACCT	A	*/B	A	*
02 SHARING				
04 MBR-NUM	A	*	A	*
04 SHARING-GROUP	A	*	A	*
04 TERM-STATE	A	*	A	*
04 TERM-COUNTY	A	*	A	*
04 NOT-ON-US-MASK	A	*	A	*
04 FIID	A	*	A	*
02 TERM				
04 OWNER	A	*	A	*
04 TERM-NAME-LOC	A	*	A	*
04 TERM-OWNER-NAME	A	*	A	*
04 ATM-ID	A	*	A	*
04 BRANCH-ID	A	*	A	*
04 REGION-ID	A	*	A	*
04 REC-NUM	A	*	A	*
04 TYP	A	*	A	*
04 TIM-OFST	A	*	A	*

Level and Field Name	Non-Balance Inquiry		Balance Inquiry	
	0200	0210	0700	0710
02 CARD				
04 FIID-LEN	0	B	0	B
04 CARD-FIID	0	B	0	B
04 PIN-OFST	0	B	0	B
02 MISC				
04 PAYEE-CDE	A	*	A	*
04 ICHG-SETT-DAT	0	B	0	B
02 DEPOSITORY				
04 DEP-BAL-CREDIT	0	B	0	*
04 AVAIL-DEP-TYPE	A	*	A	*
04 DEP-TYPE	0	B	0	B
02 HOST				
04 PROC-ACCT-IND	0	B	0	B
02 USER-FLD	A	*	A	*
02 TRACK2	A	*	A	*
02 TRACK3	A	*	A	*
02 FILLER	A	*	A	*

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

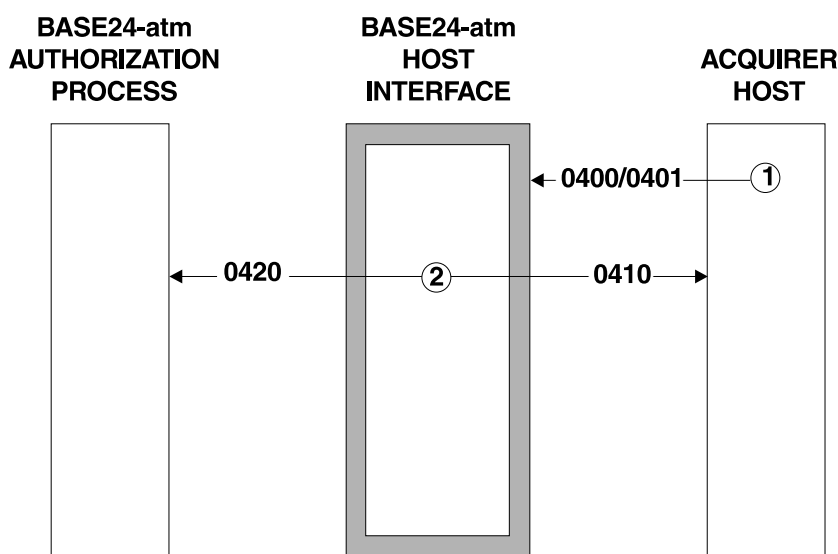
* = Copied as is.

b = (Blank) not applicable; zero- or blank-filled.

/ = Optional.

Reversal Transaction Message Flow

The following diagram illustrates the message exchanges expected by BASE24-atm for reversals being sent to BASE24-atm by an acquirer host. A reversal occurs when a BASE24-authorized transaction does not complete as authorized.



Processing Description

The following steps describe the processing for reversal transactions being sent to the BASE24-atm ANSI Host Interface by an acquirer host:

1. The transaction-originating terminal notifies the acquirer host that the transaction did not complete as required. The acquirer host then formats a 0400 message and sends it to BASE24. (0401 messages may be used to indicate a possible duplicate.)
2. The ANSI Host Interface receives the 0400 or 0401 message and performs as follows:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Creates an STM from the SEM.
 - c. Sets the TYP field in the STM to 0420.
 - d. Sends the 0420 message (STM) to the Authorization process indicated in the AUTH PROCESS field on HCF screen 5.

- e. Generates a 0410 message if necessary.

The ANSI Host Interface generates a 0410 message if the DPC's stations are single-threaded or the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If a 0410 message is required, the ANSI Host Interface changes the TYP field in the SEM to 0410 and sends the 0410 message to the DPC.

Message Requirements

The SEM field requirements for each of the message types described in the message flow are shown below.

Level and Field Name	0400/0401	0410
02 ROUTE		
03 STAT	*	B
03 PATH	*	B
02 TYP	A	B
02 BYTE-MAP	*	*
02 CARD-ACCT	*	*
02 TRAN-CDE	*	*
02 AMT-1	*	*
02 AMT-2	A	*
02 AMT-3	A	*
02 TRACE-NO	A	*
02 TRAN-DAT	*	*
02 POST-DAT	*	*
02 TRANSMISSION-DAT	*	*
02 TRAN-TIM	*	*
02 RTTN	*	*

Level and Field Name	0400/0401	0410
02 ISSUER-ROUTING-NUMBER	*	*
02 CURRENCY-CDE	*	*
02 REVERSAL-CDE	A	*
02 TYPE-ORIG	*	*
02 POINT-TRAN-SPEC	*	*
02 RESP-CDE	*	*
02 TERM-CITY-ST	*	*
02 ORIGL	A	*
02 ISS-TRACE-NO	*	*
02 PIN	*	*
02 FROM-ACCT	*	*
02 TO-ACCT	*	*
02 SHARING		
04 MBR-NUM	*	*
04 SHARING-GROUP	*	*
04 TERM-STATE	*	*
04 TERM-COUNTY	*	*
04 NOT-ON-US-MASK	*	*
04 FIID	*	*
02 TERM		
04 OWNER	*	*
04 TERM-NAME-LOC	*	*
04 TERM-OWNER-NAME	*	*

Level and Field Name	0400/0401	0410
04 ATM-ID	*	*
04 BRANCH-ID	*	*
04 REGION-ID	*	*
04 REC-NUM	*	*
04 TYP	*	*
04 TIM-OFST	*	*
02 CARD		
04 FIID-LEN	*	*
04 CARD-FIID	*	*
04 PIN-OFST	*	*
02 MISC		
04 PAYEE-CDE	*	*
04 ICHG-SETT-DAT	*	*
02 DEPOSITORY		
04 DEP-BAL-CREDIT	*	*
04 AVAIL-DEP-TYPE	*	*
04 DEP-TYPE	*	B
02 HOST		
04 PROC-ACCT-IND	*	B
02 USER-FLD	A	*
02 TRACK2	*	*
02 TRACK3	*	*
02 FILLER	*	*

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

* = Copied as is.

b = (Blank) not applicable; zero- or blank-filled.

Late Response from the Authorization Process

If the ANSI Host Interface receives a late 0210 message from the Authorization process, it:

1. Checks the RQST.RESP field in the STM to determine whether the transaction was denied. If it was, the ANSI Host Interface drops the message.
2. Checks the TRAN-CDE field in the STM to determine whether the transaction was a balance inquiry. If it is, the ANSI Host Interface drops the message.

Note: If the transaction is a balance inquiry and the REVERSE-BAL-INQ param in the LCONF is set to a “true” value, the ANSI Host Interface will not drop the transaction, but rather continue with step 3.

3. Changes the TYP field in the STM to 0420.
4. Sets the RQST.RVSL-CDE field in the STM to 01 to indicate that the Authorization process timed out.
5. Sets the AMT-2 field in the STM to zeros.
6. Sends the 0420 message back to the Authorization process that sent it.

Failed Messages to Stations

If a message is forwarded to the Nucleus, but cannot be sent to a station, the Nucleus returns the message to the ANSI Host Interface as failed.

Upon receipt of a message failure, the ANSI Host Interface marks the station down and sets an extended network management message timer. It then performs based on message type.

Failed 0210/0710 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0210 or 0710 message back from the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Checks the RQST.RESP field in the STM to determine whether the transaction was denied. If it was, the ANSI Host Interface drops the message.
3. Creates an STM from the SEM.
4. Sets the TYP field in the STM to 0420.
5. Sets the RQST.RVSL-CDE field in the STM to 01 to indicate that the Authorization process timed out.
6. Sets the AMT-2 field in the STM to zeros.
7. Sends the 0420 message back to the Authorization process that sent it.

Failed 0230/0410/5410 Messages

If the ANSI Host Interface receives a failed 0230, 0410, or 5410 message back from the Nucleus, it drops the message.

Unintelligible Messages from a DPC

If the ANSI Host Interface receives a message with an invalid message type, or carrying information in an incorrect format, it changes the ROUTE.STAT field in the SEM to 21 to indicate an unknown message type and returns the message to the sending DPC.

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Section 6

BASE24-pos Financial Message Handling

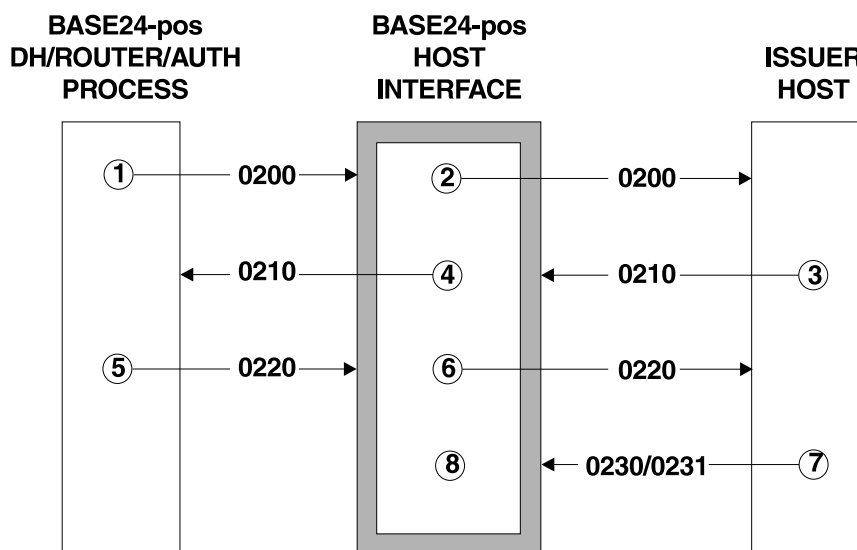
This section contains an overview of how BASE24-pos financial transaction messages are handled by the ANSI Host Interface. It includes message flows for both issuer and acquirer host processing.

Issuer Host Message Handling

This section describes the processing involved for messages to and from an issuer host. The SEM (Standard External Message) as discussed in this section is one of four formats, depending on how a particular BASE24-pos system is configured. The four formats include the PSEMA, PSEMF, PSEMAE, and PSEMFE.

Financial Transaction Message Flow

The following diagram illustrates the message exchange expected when the ANSI Host Interface sends financial transactions to an issuer host.



Processing Description

The following steps describe the processing for financial transactions being sent to an issuer host by the BASE24-pos ANSI Host Interface:

1. BASE24 receives a financial transaction request, which is sent to the BASE24-pos Device Handler/Router/Authorization process.

Authorization determines that the request must be sent to an issuer host for authorization and sends it to the ANSI Host Interface to be forwarded. The BASE24-pos Device Handler/Router/Authorization process retrieves the name of the ANSI Host Interface to which to send the message from the Institution Definition File (IDF).

2. The ANSI Host Interface receives the 0200 message and performs the following steps:
 - a. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the PSTM.
 - b. Verifies that the card type used in the transaction is supported based on the values in the ALLOWED SERVICES field on HCF screen 8.
 - c. Creates an SEM from the PSTM. The ANSI Host Interface determines the type of SEM to create based on the value in the PSEM TYPE field on HCF screen 7.
 - d. Updates the EXIT-TIM field in the PSTM to the current time.
 - e. Determines its ability to send the message. The ANSI Host Interface cannot send the message if any one of the following conditions exist:
 - The DPC is marked down.
 - A noninterspersed SAF is in progress.
 - The host is single-threaded and no send/receive stations are available.
 - The current number of outstanding messages to the DPC equals the MAXIMUM OUTSTANDING REQUESTS OUTBOUND value from HCF screen 1.

If any of the above conditions exist, the ANSI Host Interface takes the following actions:

- 1) Sets the RTE.STAT field in the PSTM to 11, indicating that the DPC was unavailable, or to 12 if the DPC was marked down.
 - 2) Sets the COMPL-REQ field in the PSTM to 2, indicating that an advice is required if the request is approved offline.
 - 3) Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.
- f. Saves a copy of the PSTM and SEM in extended memory.
 - g. Sets an outbound request timer for the number of seconds specified by the OUTBOUND LIMIT field on HCF screen 7.

If this timer expires before a 0210 message is received, the ANSI Host Interface takes the following actions:

- 1) Sets the RTE.STAT field in the PSTM to 11, indicating that the DPC was unavailable, or to 12 if the DPC was marked down.

- 2) Sets the COMPL-REQ field in the PSTM to 2, indicating that an advice is required if the request is approved offline.
 - 3) Returns the 0200 message (PSTM) to the process indicated in the ORIG-PRO-NAME field of the PSTM.
- h. Updates two of its internal counters as follows:
- 1) Adds one to the number of messages currently outstanding to the DPC.
 - 2) Adds one to the total number of messages currently outstanding from the ANSI Host Interface.
- i. Sends the 0200 message to the DPC.
- Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using the QUEUE SUBTRACT and OUTBOUND LIMIT fields on HCF screen 7. The QUEUE SUBTRACT value is subtracted from the OUTBOUND LIMIT value and the result is used.
3. The host validates the request, processes the transaction, and returns a 0210 message either approving or declining the transaction.
 4. The ANSI Host Interface receives the 0210 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Retrieves the stored PSTM and SEM from extended memory.
 - c. Deletes the outbound request timer.
 - d. Updates its counters as follows:
 - 1) Subtracts one from the number of messages currently outstanding to the DPC.
 - 2) Adds one to the number of requests processed for the DPC.
 - 3) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - 4) Subtracts one from the total number of messages outstanding from the ANSI Host Interface.
 - e. Sets the TYP field in the PSTM to 0210.

- f. Updates the PSTM from the SEM using only that information which may have been changed by the host.
 - g. Sets the RE-ENTRY-TIM field in the PSTM to the current time.
 - h. Sends the 0210 message (PSTM) to the Device Handler/Router/Authorization process.
 5. The BASE24-pos Device Handler/Router/Authorization process finishes processing the transaction, logs it to the PTLF, and returns it to its originator.

If the host requires advices, the BASE24-pos Device Handler/Router/Authorization process generates the appropriate 0220 message and sends it to the ANSI Host Interface.
 6. The ANSI Host Interface receives the 0220 message and performs the following steps:
 - a. Checks the RESP-CDE field in the PSTM. If the transaction was declined, the ANSI Host Interface drops the message.
 - b. Locates the appropriate DPC processing information in its internal table using the DPC number from the DPC-NUM field in the PSTM.
 - c. Creates an SEM from the PSTM. The ANSI Host Interface determines whether to create the SEM based on the value in the PSEM TYPE field on HCF screen 7.
 - d. Checks the COMPL-REQ field in the PSTM. If that field is set to 2 (host is offline—completion required), the ANSI Host Interface writes the external version of the 0220 (SEM) message to the SAF and takes no further action.
 - e. Determines whether the DPC is marked down. If so, the ANSI Host Interface writes the SEM (0220) message to the SAF and takes no further action.
 - f. Determines whether a noninterspersed SAF is in progress. If so, the ANSI Host Interface writes the external message to the SAF and takes no further action.
 - g. Locates an available send/receive station. If no send/receive stations are available, the ANSI Host Interface writes the external message to the SAF and takes no further action.
 - h. Sends the 0220 message to the DPC.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using the QUEUE

SUBTRACT and OUTBOUND LIMIT fields on HCF screen 7. The QUEUE SUBTRACT value is subtracted from the OUTBOUND LIMIT value and the result is used.

7. The host processes the 0220 message and returns a 0230 or 0231 message if text-level acknowledgments are required.

DPCs must return text-level acknowledgments if the DPC's stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface:

- a. Stores a copy of the PSTM and SEM.
 - b. Sets a outbound request timer.
 - c. Adds one to the number of messages currently outstanding to the DPC.
 - d. Adds one to the number of messages currently outstanding from the ANSI Host Interface.
8. Upon receipt of a 0230 or 0231 message, the ANSI Host Interface performs a number of steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Deletes the stored PSTM and SEM from extended memory.
 - c. Deletes the outbound request timer.
 - d. Subtracts one from the number of messages currently outstanding to the DPC.
 - e. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
 - f. Adds one to the number of requests processed for this DPC.
 - g. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - h. Drops the 0230 or 0231 message.

PSEMA/PSEMAE Message Requirements

The field requirements for BASE24 transaction request PSEMA or PSEMAE messages and for issuer host DPC-initiated PSEMA or PSEMAE transaction response messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMAE format. The PSEMA and PSEMAE are identical with the exception of the address verification fields.

Level and Field Name	0200	0210	0220/0221	0230/0231
02 ROUTE				
03 STAT	*	*	*	*
03 PATH				
05 PRODUCT-ID	*	*	*	*
05 RELEASE-NUM	*	*	*	*
05 DPC-NUMBER	*	*	*	*
05 CARD-FIID	*	*	*	*
05 ORIGI-TYPE	*	*	*	*
05 CARD-LOG-NET	*	*	*	*
05 PSEM-TYP	*	*	*	*
05 CARD-TYP	*	*	*	*
05 AUTHORIZER	*	*	*	*
02 TYP	*	X	*	X
02 BYTE-MAP	*	*	*	*
02 TRAN				
04 CARD				
06 LEN	*	*	*	*

Level and Field Name	0200	0210	0220/0221	0230/0231
06 NUM	*	*	*	*
04 TRAN-CDE	*	*	*	*
04 AMT-1	*	*/X	*	*
04 AMT-2	*	*	*	*
04 TRACE-NO	*	*	*	*
04 TRAN-DAT	*	*	*	*
04 POST-DAT	*	*	*	*
04 TRAN-TIM				
06 HH	*	*	*	*
06 MINUTE	*	*	*	*
06 SS	*	*	*	*
04 EXP-DAT	*	*	*	*
04 SIC	*	*	*	*
04 RTTN	*	*	*	*
04 REVERSAL-CDE			*	*
04 TYPE-ORIG	*	*	*	*
04 POINT-TRAN-SPEC	*	*	*	*
04 RESP-CDE		X	*	*
04 RETRIEVAL-REF-NUM	*	*	*	*
04 PIN	*	*	*	*
04 APPROVAL-CODE		X	*	*
04 RETAILER				
06 FIID	*	*	*	*

Level and Field Name	0200	0210	0220/0221	0230/0231
06 GRP	*	*	*	*
06 REGN	*	*	*	*
06 ID	*	*	*	*
06 DRAFT-CAPTURE	*	*	*	*
06 SETTLE-FLAG	*	*	*	*
04 MEMBER-NUMBER	*	*	*	*
04 REFERRAL-PHONE	*	X	*	*
04 PRE-AUTH-HOLD	*	*	*	*
04 ADDR-DATA				
06 ADDR	*	*	*	*
06 ZIP-CDE	*	*	*	*
06 ADDR-VRFY-STAT	*	X	*	*

Key:

- * = Set by BASE24-pos; must be copied as is.
- X = Issuer host must provide.
- ␣ = (Blank) not applicable; zero- or blank-filled.

PSEMF/PSEMFE Message Requirements

The field requirements for BASE24 transaction request PSEMF or PSEMFE messages and for issuer host DPC-initiated PSEMF or PSEMFE transaction response messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMFE format. The PSEMF and PSEMFE are identical with the exception of the address verification fields.

Level and Field Name	0200	0210	0220/0221	0230/0231
02 ROUTE				
03 STAT	*	*	*	*
03 PATH				
05 PRODUCT-IND	*	*	*	*
05 RELEASE-NUM	*	*	*	*
05 DPC-NUMBER	*	*	*	*
05 CARD-FIID	*	*	*	*
05 ORIGI-TYPE	*	*	*	*
05 CARD-LOG-NET	*	*	*	*
05 PSEM-TYP	*	*	*	*
05 CARD-TYP	*	*	*	*
05 AUTHORIZER	*	*	*	*
02 TYP	*	X	*	X
02 BYTE-MAP	*	*	*	*
02 TRAN				
04 CARD				
06 LEN	*	*	*	*

Level and Field Name	0200	0210	0220/0221	0230/0231
06 NUM	*	*	*	*
04 TRAN-CDE	*	*	*	*
04 AMT-1	*	*/X	*	*
04 AMT-2	*	*	*	*
04 TRACE-NO	*	*	*	*
04 POST-DAT	*	*	*	*
04 TRAN-DAT	*	*	*	*
04 TRAN-TIM				
06 HH	*	*	*	*
06 MINUTE	*	*	*	*
06 SS	*	*	*	*
04 EXP-DAT	*	*	*	*
04 SIC	*	*	*	*
04 RTTN	*	*	*	*
04 REVERSAL-CDE			*	*
04 TYPE-ORIG	*	*	*	*
04 POINT-TRAN-SPEC	*	*	*	*
04 RESP-CDE		X	*	*
04 CARD-ACCEPTOR-NAME	*	*	*	*
04 CARD-ACCEPTOR-CITY-ST	*	*	*	*
04 RETRIEVAL-REF-NUM	*	*	*	*
04 PIN	*	*	*	*

Level and Field Name	0200	0210	0220/0221	0230/0231
04 FROM-ACCT-LEN	*	*/X	*	*
04 FROM-ACCT-NUM	*	*/X	*	*
04 APPROVAL-CODE		X	*	*
04 RETAILER				
06 LOGICAL-NET	*	*	*	*
06 FIID	*	*	*	*
06 GRP	*	*	*	*
06 REGN	*	*	*	*
06 ID	*	*	*	*
06 TERM-ID	*	*	*	*
06 SHIFT	*	*	*	*
06 EMPLOYEE				
08 ID	*	*	*	*
08 GRP	*	*	*	*
08 USER-ID	*	*	*	*
06 DRAFT-CAPTURE	*	*	*	*
06 SETTLE-FLAG	*	*	*	*
04 MEMBER-NUMBER	*	*	*	*
04 SHARING				
04 MISC				
06 ORIG	*	*	*	*
06 DEST	*	*/X	*	*
06 INVOICE-NUM	*	*	*	*

Level and Field Name	0200	0210	0220/0221	0230/0231
06 ORIG-INVOICE- NUM	*	*	*	*
06 CURRENCY-CDE	*	*	*	*
06 BATCH-NUM	*	*	*	*
06 REFERRAL- PHONE	*	X	*	*
06 PRE-AUTH-HOLD	*	*	*	*
06 PRODUCT				
08 LEN	*	*	*	*
08 DAT	*	*	*	*
04 TRACK2-LEN	*	*	*	*
04 TRACK2-DATA	*	*	*	*
04 ADDR-DATA				
06 ADDR	*	*	*	*
06 ZIP-CDE	*	*	*	*
06 ADDR-VRFY- STAT	*	X	*	*

5

Key:

* = Set by BASE24-pos; must be copied as is.

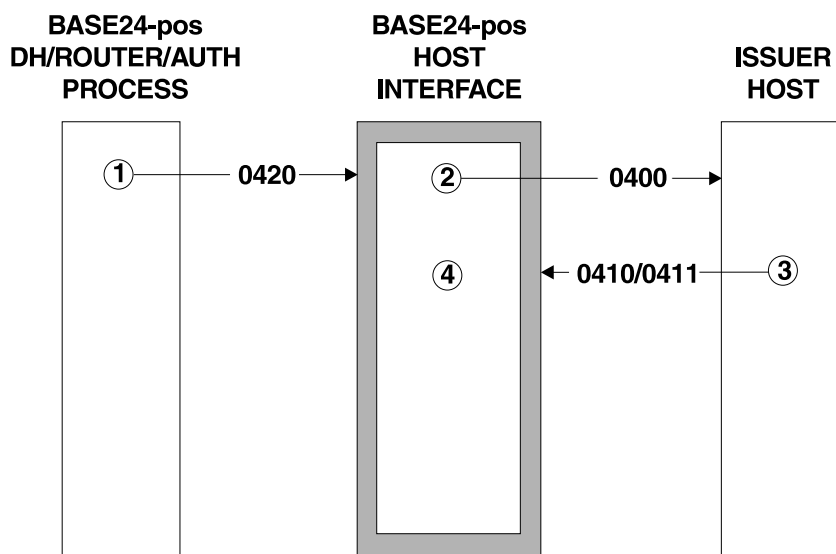
X = Issuer host must provide.

b = (Blank) not applicable; zero- or blank-filled.

Reversal Message Flow

The following diagram illustrates the message exchange expected when the ANSI Host Interface sends reversal transactions to an issuer host.

A reversal indicates that a previously-approved transaction did not complete as authorized.



Processing Description

The following steps describe the processing for reversal transactions being sent to an issuer host by the BASE24-pos ANSI Host Interface:

1. BASE24 is notified that a transaction did not complete as previously authorized. The notice can be from a POS device controlled by BASE24, an acquirer host, or an interchange.

The BASE24-pos Device Handler/Router/Authorization process backs out the original transaction, posts the revised transaction, logs the transaction to the PTLF, and sends a 0420 message PSTM to the ANSI Host Interface.

2. The ANSI Host Interface receives the 0420 message and processes it as follows:
 - a. Checks the TRAN.RESP-CDE field in the PSTM. If it is greater than 015, meaning the transaction was declined, the ANSI Host Interface drops the message.

- b. Locates the DPC record in its internal table using the DPC-NUM field in the PSTM.
- c. Creates an SEM from the PSTM. The ANSI Host Interface determines the type of SEM to create based on the value in the PSEM TYPE field on HCF screen 7. While creating the SEM, the ANSI Host Interface changes the message type to 0400.
- d. Determines whether a store-and-forward cycle (interspersed or noninterspersed) is in progress for the DPC. If so, the ANSI Host Interface writes the 0400 message to the SAF and takes no further action.
- e. Checks the COMPL-REQ field in the PSTM. If it is 2 (host offline—completion required), the ANSI Host Interface writes the 0400 message to the SAF and takes no further action.
- f. Determines whether the DPC is marked down. If so, the ANSI Host Interface writes the 0400 message to the SAF and takes no further action.
- g. Locates an available send/receive station. If no send/receive stations are available, the ANSI Host Interface writes the 0400 message to the SAF and takes no further action.
- h. Sends the 0400 message to the DPC.

Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using the QUEUE SUBTRACT and OUTBOUND LIMIT fields on HCF screen 7. The QUEUE SUBTRACT value is subtracted from the OUTBOUND LIMIT value and the resulting value is used.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface performs the following steps:

- 1) Stores copies of the PSTM and SEM.
- 2) Sets a outbound request timer.
- 3) Adds one to the number of messages currently outstanding to the DPC.
- 4) Adds one to the number of messages currently outstanding from the ANSI Host Interface.

3. The host processes the 0400 message and returns a 0410 or 0411 message if text-level acknowledgments are required.

DPCs must return text-level acknowledgments if the DPCs' stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.
4. Upon receipt of a 0410 or 0411 message, the ANSI Host Interface performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Deletes the PSTM and SEM from extended memory.
 - c. Deletes the outbound request timer.
 - d. Subtracts one from the number of messages currently outstanding to the DPC.
 - e. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
 - f. Adds one to the number of requests processed for this DPC.
 - g. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - h. Drops the 0410 or 0411 message.

PSEMA/PSEMAE Message Requirements

The fields required for BASE24-generated PSEMA or PSEMAE transaction reversal messages and for issuer host DPC PSEMA or PSEMAE reversal acknowledgment messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMAE format. The PSEMA and PSEMAE are identical with the exception of the address verification fields.

Level and Field Name	0400/0401	0410/0411
02 ROUTE		
03 STAT	*	*

Level and Field Name	0400/0401	0410/0411
03 PATH		
05 PRODUCT-IND	*	*
05 RELEASE-NUM	*	*
05 DPC-NUMBER	*	*
05 CARD-FIID	*	*
05 ORIGI-TYPE	*	*
05 CARD-LOG-NET	*	*
05 PSEM-TYP	*	*
05 CARD-TYP	*	*
05 AUTHORIZER	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 TRAN		
04 CARD		
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 TRAN-DAT	*	*
04 POST-DAT	*	*
04 TRAN-TIM		

Level and Field Name	0400/0401	0410/0411
06 HH	*	*
06 MINUTE	*	*
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*
04 REVERSAL-CDE	*	*
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	*	*
04 RESP-CDE	*	*
04 RETRIEVAL-REF-NUM	*	*
04 PIN	*	*
04 APPROVAL-CODE	*	*
04 RETAILER		
06 FIID	*	*
06 GRP	*	*
06 REGN	*	*
06 ID	*	*
06 DRAFT-CAPTURE	*	*
06 SETTLE-FLAG	*	*
04 MEMBER-NUMBER	*	*
04 REFERRAL-PHONE	*	*
04 PRE-AUTH-HOLD	*	*

Level and Field Name	0400/0401	0410/0411
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

Key:

* = Set by BASE24-pos; must be copied as is.

X = Issuer host must provide.

b = (Blank) not applicable; zero- or blank-filled.

PSEMF/PSEMFE Message Requirements

The fields required for BASE24-generated PSEMF or PSEMFE transaction reversal messages and for issuer host DPC PSEMF or PSEMFE reversal acknowledgment messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMFE format. The PSEMF and PSEMFE are identical with the exception of the address verification fields.

Level and Field Name	0400/0401	0410/0411
02 ROUTE		
03 STAT	*	*
03 PATH		
05 PRODUCT-IND	*	*
05 RELEASE-NUM	*	*
05 DPC-NUMBER	*	*
05 CARD-FIID	*	*
05 ORIGI-TYPE	*	*

Level and Field Name	0400/0401	0410/0411
05 CARD-LOG-NET	*	*
05 PSEM-TYP	*	*
05 CARD-TYP	*	*
05 AUTHORIZER	*	*
02 TYP	*	X
02 BYTE-MAP	*	*
02 TRAN		
04 CARD		
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 POST-DAT	*	*
04 TRAN-DAT	*	*
04 TRAN-TIM		
06 HH	*	*
06 MINUTE	*	*
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*

Level and Field Name		0400/0401	0410/0411
04	REVERSAL-CDE	*	*
04	TYPE-ORIG	*	*
04	POINT-TRAN-SPEC	*	*
04	RESP-CDE	*	*
04	CARD-ACCEPTOR-NAME	*	*
04	CARD-ACCEPTOR-CITY-ST	*	*
04	RETRIEVAL-REF-NUM	*	*
04	PIN	*	*
04	FROM-ACCT-LEN	*	*
04	FROM-ACCT-NUM	*	*
04	APPROVAL-CODE	*	*
04	RETAILER		
06	LOGICAL-NET	*	*
06	FIID	*	*
06	GRP	*	*
06	REGN	*	*
06	ID	*	*
06	TERM-ID	*	*
06	SHIFT	*	*
06	EMPLOYEE		
08	ID	*	*
08	GRP	*	*
08	USER-ID	*	*

Level and Field Name	0400/0401	0410/0411
06 DRAFT-CAPTURE	*	*
06 SETTLE-FLAG	*	*
04 MEMBER-NUMBER	*	*
04 SHARING	*	*
04 MISC		
06 ORIG	*	*
06 DEST	*	*
06 INVOICE-NUM	*	*
06 ORIG-INVOICE-NUM	*	*
06 CURRENCY-CDE	*	*
06 BATCH-NUM	*	*
06 REFERRAL-PHONE	*	*
06 PRE-AUTH-HOLD	*	*
06 PRODUCT		
08 LEN	*	*
08 DAT	*	*
04 TRACK2-LEN	*	*
04 TRACK2-DATA	*	*
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

s

Key:

- * = Set by BASE24-pos; must be copied as is.
- X = Issuer host must provide.
- Ø = (Blank) not applicable; zero- or blank-filled.

Late Response from a DPC

If a DPC responds late to a request, the ANSI Host Interface performs the following steps for the 0210 message:

1. Checks the RESP-CDE field in the SEM.
If the transaction was declined, the ANSI Host Interface drops the message.
2. Checks the TRAN-CDE field in the SEM.
If the transaction was a balance inquiry transaction, the ANSI Host Interface drops the message.
3. Creates a reversal from the 0210 message as follows:
 - a. Sets the TYP field to 0400.
 - b. Sets the REVERSAL-CDE field to 01, indicating that the message timed out.
 - c. Sets the AMT-2 field to zeros if the transaction is not an adjustment transaction.
4. Selects the station to which to send the message.
To do this, the ANSI Host Interface checks the station that sent the 0210 message. If it is a send/receive station and is available, the ANSI Host Interface sends the 0400 message to that station. Otherwise, it uses any available send/receive station.
If a station is not available, the ANSI Host Interface writes the 0400 message to the SAF and performs no further steps.
5. Sends the 0400 message to the selected station.
Along with the message, the ANSI Host Interface also sends the maximum number of seconds the Nucleus can allow the message to queue to the host. This number of seconds is derived using the QUEUE SUBTRACT and OUTBOUND LIMIT fields on HCF screen 7. The QUEUE SUBTRACT value is subtracted from the OUTBOUND LIMIT value and the resulting value is used.
6. Determines whether the DPC is required to return a text-level acknowledgment.
DPCs must return text-level acknowledgments if the DPCs' stations are single-threaded or if the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If text-level acknowledgments are not required, no more processing is performed.

If text-level acknowledgments are required, the ANSI Host Interface:

- a. Stores a copy of the SEM.
- b. Sets a completion acknowledgment timer.
- c. Adds one to the number of messages currently outstanding to the DPC.
- d. Adds one to the number of messages currently outstanding from the ANSI Host Interface.

Failed Messages to Stations

If the ANSI Host Interface forwards a message to the Nucleus, but the message cannot be sent to a station, the Nucleus returns the message to the ANSI Host Interface as failed.

Upon receiving a failed message, the ANSI Host Interface marks the station down and sets an extended network management message timer. It then performs based on message type.

Failed 0200 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0200 message back from the Nucleus. Failed messages occur when a station is unavailable or the transaction becomes stale while queued by the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored PSTM and SEM from extended memory.
3. Deletes the outbound request timer.
4. Updates its counters as follows:
 - a. Subtracts one from the number of messages currently outstanding to the DPC.
 - b. Adds one to the number of requests processed for the DPC.
 - c. Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - d. Subtracts one from the total number of outstanding messages from the ANSI Host Interface.
5. Sets the RTE.STAT field in the PSTM to 12, indicating the line was down.
6. Sets the COMPL-REQ field in the PSTM to 2, indicating that a 0220 message is required if the transaction is subsequently approved.
7. Returns the PSTM as a failed 0200 message to the BASE24-pos Device Handler/Router/Authorization process.

Failed 0220 or 0400 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0220 or 0400 message from the Nucleus. Failed messages occur when a station is unavailable or the transaction becomes stale while queued by the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Retrieves the stored PSTM and SEM from extended memory.
3. Checks for a outbound request timer for the message. If the ANSI Host Interface finds this timer, it is deleted.
4. Subtracts one from the number of messages currently outstanding to the DPC.
5. Adds one to the number of requests processed for the DPC.
6. Adds the amount of time taken to process this request to the total time taken for all requests for this DPC. For performance statistics, this sum is divided by the number of requests processed for the DPC.
7. Subtracts one from the number of messages currently outstanding from the ANSI Host Interface.
8. Adds the message to the Store-and-Forward File (SAF).

Unintelligible Messages from a DPC

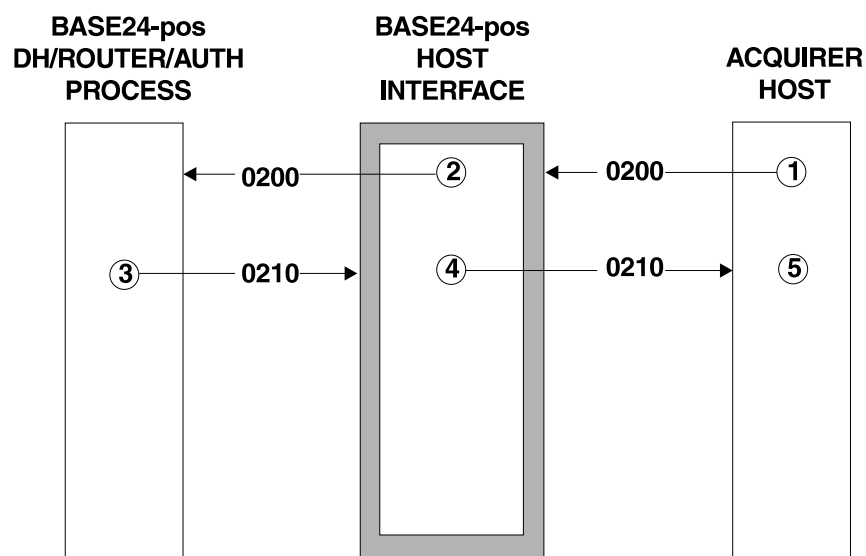
If the ANSI Host Interface receives a message with an invalid message type, or a message carrying information in an incorrect format, it changes the ROUTE.STAT field in the SEM to 21, indicating an unknown message type and returns the message to the sending DPC.

Acquirer Host Message Handling

This section describes the processing involved for messages sent to and from an acquirer host.

Financial Transaction Message Flow

The following diagram illustrates the message exchanges expected when an acquirer host sends financial transactions to BASE24-pos.



Processing Description

The following steps describe the processing for financial transactions being sent to an acquirer host by the BASE24-pos ANSI Host Interface:

1. A cardholder initiates a financial transaction at a terminal connected to and controlled by an acquirer host. The acquirer host then formats a 0200 message and sends it to BASE24.
2. The ANSI Host Interface receives the 0200 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.

- b. Creates a PSTM from the SEM and sets the TYP field in the PSTM to 0200.
- c. Updates the following fields in the PSTM:
 - 1) ORIGINATOR—Set to 5, indicating the transaction was initiated through an acquirer host.
 - 2) TRAN.ORIG—Set to the literal HOST.
 - 3) ORIG-PRO-NAME—Set to the symbolic name of the ANSI Host Interface process.
 - 4) HOST-TRACE-NUM—Set to a number generated by the ANSI Host Interface.
 - 5) DPC-NUM—Set to the number of the DPC originating the transaction.
 - 6) ENTRY-TIM—Set to the current time.
- d. Checks the current number of outstanding 0200 messages to the Device Handler/Router/Authorization process.

If the current number of outstanding 0200 messages equals the MAXIMUM OUTSTANDING REQUESTS INBOUND value on HCF screen 1, the ANSI Host Interface sets the ROUTE.STAT field in the SEM to 11, indicating that the destination was not available. The ANSI Host Interface then returns the 0200 message (SEM) to its originating DPC.
- e. Saves copies of the PSTM and SEM in extended memory.
- f. Sets an inbound request timer for the number of seconds specified by the INBOUND LIMIT field in the BASE24-pos segment of the HCF.

If this timer expires before a 0210 message is received, the ANSI Host Interface returns a 0210 message to the host with the RESP-CDE field set to 100.
- g. Updates its internal counters as follows:
 - 1) Adds one to the number of requests currently outstanding from the DPC.
 - 2) Adds one to the total number of requests currently outstanding from the ANSI Host Interface.
- h. Sends the 0200 message (PSTM) to the Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.

3. The BASE24-pos Device Handler/Router/Authorization process processes the transaction, logs it to the PTLF, and returns a 0210 message to the ANSI Host Interface.

Note: The transaction could also have been authorized by an interchange or issuer host, depending on the cardholder and the type of authorization processing performed.

4. The ANSI Host Interface receives the 0210 message and performs the following steps:
 - a. Locates the stored PSTM and SEM.
 - b. Deletes the inbound request timer.
 - c. Updates its internal counters as follows:
 - 1) Subtracts one from the number of requests currently outstanding from the DPC.
 - 2) Subtracts one from the total number of requests currently outstanding from the ANSI Host Interface.
 - 3) Adds one to the number of requests processed for this DPC.
 - 4) Adds the amount of time taken to process this request to the total time taken for all requests to the DPC. For performance statistics, this value is divided by the number of requests processed for the DPC.
 - d. Updates the SEM with only those fields from the 0210 message that have been updated by the Device Handler/Router/Authorization process.
 - e. Sends the 0210 message (SEM) to the station.

If a send/receive station is not available, the ANSI Host Interface creates a reversal (0420) message using the PSTM and returns the message to the Device Handler/Router/Authorization process. A reversal is created only if the transaction is approved and is not a balance inquiry.

5. The acquirer host then completes the transaction at the POS device.

PSEMA/PSEMAE Message Requirements

The fields required for acquirer host DPC-initiated PSEMA or PSEMAE transaction request messages and for BASE24 transaction response PSEMA or PSEMAE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMAE format. The PSEMA and PSEMAE are identical with the exception of the address verification fields.

Level and Field Name	0200	0210
02 ROUTE		
03 STAT	A = 00	B
03 PATH		
05 PRODUCT-IND		*
05 RELEASE-NUM	A = 30	*
05 DPC-NUMBER	0	B
05 CARD-FIID	A	B
05 ORIGI-TYPE	0	B
05 CARD-LOG-NET	A	B
05 PSEM-TYP	A = 00	*
05 CARD-TYP	A	B
05 AUTHORIZER	A	B
02 TYP	A	B
02 BYTE-MAP	A	*
02 TRAN		
04 CARD		
06 LEN	A	*

Level and Field Name	0200	0210
06 NUM	A	*
04 TRAN-CDE	A	B
04 AMT-1	A	*/B
04 AMT-2	A	*
04 TRACE-NO		*
04 TRAN-DAT	A	*
04 POST-DAT	A	*
04 TRAN-TIM		
06 HH	A	*
06 MINUTE	A	*
06 SS	A	*
04 EXP-DAT	A	B
04 SIC	A	*
04 RTTN	A	*
04 REVERSAL-CDE		
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	A	*
04 RESP-CDE		B
04 RETRIEVAL-REF-NUM	A	*
04 PIN	A	*
04 APPROVAL-CODE		B
04 RETAILER		
06 FIID	A	*

Level and Field Name	0200	0210
06 GRP	A	*
06 REGN	A	*
06 ID	A	*
06 DRAFT-CAPTURE	A	*
06 SETTLE-FLAG	A	*
04 MEMBER-NUMBER	A	B
04 REFERRAL-PHONE	A	B
04 PRE-AUTH-HOLD	A	*
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	A	B

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

* = Copied as is.

X = Issuer host must provide.

Ø = (Blank) not applicable; zero- or blank-filled.

/ = Optional.

PSEMF/PSEMFE Message Requirements

The fields required for acquirer host DPC-initiated PSEMF or PSEMFE transaction request messages and for BASE24 transaction response PSEMF or PSEMFE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMFE format. The PSEMF and PSEMFE are identical with the exception of the address verification fields.

Level and Field Name	0200	0210
02 ROUTE		
03 STAT	A = 00	B
03 PATH		
05 PRODUCT-IND		*
05 RELEASE-NUM	A = 30	*
05 DPC-NUMBER	0	B
05 CARD-FIID	A	B
05 ORIGI-TYPE	0	B
05 CARD-LOG-NET	A	B
05 PSEM-TYP	A = 01	*
05 CARD-TYP	A	B
05 AUTHORIZER	A	B
02 TYP	A	B
02 BYTE-MAP	A	*
02 TRAN		
04 CARD		
06 LEN	A	*

Level and Field Name	0200	0210
06 NUM	A	*
04 TRAN-CDE	A	*/B
04 AMT-1	A	*
04 AMT-2	A	*
04 TRACE-NO	A	*
04 POST-DAT	A	*
04 TRAN-DAT	A	*
04 TRAN-TIM		
06 HH	A	*
06 MINUTE	A	*
06 SS	A	*
04 EXP-DAT	A	B
04 SIC	A	B
04 RTTN	A	*
04 REVERSAL-CDE		
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	A	*
04 RESP-CDE		B
04 CARD-ACCEPTOR-NAME	A	*
04 CARD-ACCEPTOR-CITY-ST	A	*
04 RETRIEVAL-REF-NUM	A	*
04 PIN	A	*
04 FROM-ACCT-LEN	A	B

Level and Field Name		0200	0210
04	FROM-ACCT-NUM	A	B
04	APPROVAL-CODE		B
04	RETAILER		
06	LOGICAL-NET	A	*
06	FIID	A	*
06	GRP	A	*
06	REGN	A	*
06	ID	A	*
06	TERM-ID	A	*
06	SHIFT	A	*
06	EMPLOYEE		
08	ID	A	*
08	GRP	A	*
08	USER-ID	A	*
06	DRAFT-CAPTURE	A	*
06	SETTLE-FLAG	A	*
04	MEMBER-NUMBER	A	B
04	SHARING		
04	MISC		
06	ORIG	A	B
06	DEST	A	B
06	INVOICE-NUM	A	*
06	ORIG-INVOICE-NUM	A	*

Level and Field Name	0200	0210
06 CURRENCY-CDE	A	*
06 BATCH-NUM	A	*
06 REFERRAL-PHONE	A	B
06 PRE-AUTH-HOLD	A	*
06 PRODUCT		
08 LEN	A	*
08 DAT	A	*
04 TRACK2-LEN	A	*
04 TRACK2-DATA	A	*
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	A	B

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

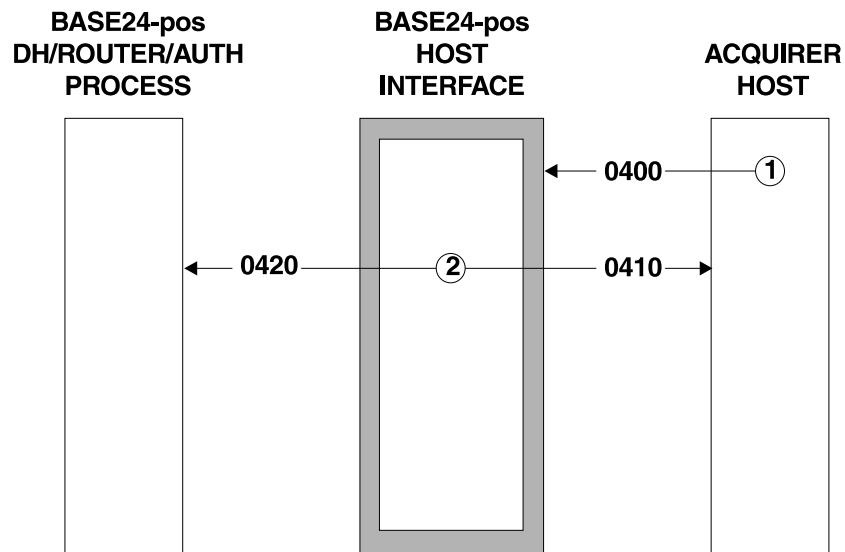
* = Copied as is.

Ø = (Blank) not applicable; zero- or blank-filled.

/ = Optional.

Reversal Transaction Message Flow

The following diagram illustrates the message exchange expected when an acquirer host sends reversal transactions to BASE24-pos. A reversal occurs when a BASE24-authorized transaction does not complete as authorized.



Processing Description

The following steps describe the processing for reversal transactions being sent to an acquirer host by the BASE24-pos ANSI Host Interface:

1. The terminal that originated the transaction notifies the acquirer host that the transaction did not complete as authorized. The acquirer host then formats a 0400 (reversal) message and sends it to BASE24.
2. The ANSI Host Interface receives the 0400 message and performs the following steps:
 - a. Locates the DPC in its internal table, identifying the necessary information for processing.
 - b. Creates a PSTM from the SEM.
 - c. Sets the following fields in the PSTM:
 - 1) ORIG-PRO-NAME—Set to the symbolic name of the ANSI Host Interface process.
 - 2) ORIGINATOR—Set to the value 5.

- 3) TRAN.ORIG—Set to the literal HOST.
- d. Changes the TYP field in the PSTM to 0420.
- e. Sends the 0420 message in PSTM format to the Device Handler/Router/Authorization process indicated in the AUTH PROCESS field on HCF screen 7.
- f. Generates a 0410 message if necessary.

The ANSI Host Interface generates a 0410 message if the DPC's stations are single-threaded or the ACK TO DPC field on HCF screen 1 indicates that text-level acknowledgments are required.

If a 0410 message is required, the ANSI Host Interface changes the TYP field in the SEM to 0410 and sends the 0410 message to the DPC.

PSEMA/PSEMAE Message Requirements

The fields required for acquirer host DPC-initiated PSEMA or PSEMAE transaction reversal messages and for BASE24 reversal acknowledgment PSEMA or PSEMAE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMAE format. The PSEMA and PSEMAE are identical with the exception of the address verification fields.

Level and Field Name	0400/0401	0410
02 ROUTE		
03 STAT	*	*
03 PATH		
05 PRODUCT-IND		
05 RELEASE-NUM	*	*
05 DPC-NUMBER	*	*
05 CARD-FIID	*	*
05 ORIGI-TYPE	*	*
05 CARD-LOG-NET	*	*

Level and Field Name	0400/0401	0410
05 PSEM-TYP	*	*
05 CARD-TYP	*	*
05 AUTHORIZER	*	*
02 TYP	A	B
02 BYTE-MAP	*	*
02 TRAN		
04 CARD		
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 TRAN-DAT	*	*
04 POST-DAT	*	*
04 TRAN-TIM		
06 HH	*	*
06 MINUTE	*	*
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*
04 REVERSAL-CDE	*	*

Level and Field Name	0400/0401	0410
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	*	*
04 RESP-CDE	*	*
04 RETRIEVAL-REF-NUM	*	*
04 PIN	*	*
04 APPROVAL-CODE	*	*
04 RETAILER		
06 FIID	*	*
06 GRP	*	*
06 REGN	*	*
06 ID	*	*
06 DRAFT-CAPTURE	*	*
06 SETTLE-FLAG	*	*
04 MEMBER-NUMBER	*	*
04 REFERRAL-PHONE	*	*
04 PRE-AUTH-HOLD	*	*
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

* = Copied as is from original request and response messages.

b = (Blank) not applicable; zero- or blank-filled.

PSEMF/PSEMFE Message Requirements

The fields required for acquirer host DPC-initiated PSEMF or PSEMFE transaction reversal messages and for BASE24 reversal acknowledgment PSEMF or PSEMFE messages are shown below.

Note: The ADDR-DATA, ADDR, ZIP-CDE, and ADDR-VRFY-STAT fields are used only for address verification and occur only in the PSEMFE format. The PSEMF and PSEMFE are identical with the exception of the address verification fields.

Level and Field Name		0400/0401	0410
02	ROUTE		
03	STAT	*	*
03	PATH		
05	PRODUCT-IND	*	*
05	RELEASE-NUM	*	*
05	DPC-NUMBER	*	*
05	CARD-FIID	*	*
05	ORIGI-TYPE	*	*
05	CARD-LOG-NET	*	*
05	PSEM-TYP	*	*
05	CARD-TYP	*	B
05	AUTHORIZER	*	B

Level and Field Name	0400/0401	0410
02 TYP	A	B
02 BYTE-MAP	*	*
02 TRAN		
04 CARD		
06 LEN	*	*
06 NUM	*	*
04 TRAN-CDE	*	*
04 AMT-1	*	*
04 AMT-2	*	*
04 TRACE-NO	*	*
04 POST-DAT	*	*
04 TRAN-DAT	*	*
04 TRAN-TIM		
06 HH	*	*
06 MINUTE	*	*
06 SS	*	*
04 EXP-DAT	*	*
04 SIC	*	*
04 RTTN	*	*
04 REVERSAL-CDE	*	*
04 TYPE-ORIG	*	*
04 POINT-TRAN-SPEC	*	*
04 RESP-CDE	*	*

Level and Field Name		0400/0401	0410
04	CARD-ACCEPTOR-NAME	*	*
04	CARD-ACCEPTOR-CITY-ST	*	*
04	RETRIEVAL-REF-NUM	*	*
04	PIN	*	*
04	FROM-ACCT-LEN	*	*
04	FROM-ACCT-NUM	*	*
04	APPROVAL-CODE	*	*
04	RETAILER		
06	LOGICAL-NET	*	*
06	FIID	*	*
06	GRP	*	*
06	REGN	*	*
06	ID	*	*
06	TERM-ID	*	*
06	SHIFT	*	*
06	EMPLOYEE		
08	ID	*	*
08	GRP	*	*
08	USER-ID	*	*
06	DRAFT-CAPTURE	*	*
06	SETTLE-FLAG	*	*
04	MEMBER-NUMBER	*	*
04	SHARING		

Level and Field Name	0400/0401	0410
04 MISC		
06 ORIG	*	*
06 DEST	*	*
06 INVOICE-NUM	*	*
06 ORIG-INVOICE-NUM	*	*
06 CURRENCY-CDE	*	*
06 BATCH-NUM	*	*
06 REFERRAL-PHONE	*	*
06 PRE-AUTH-HOLD	*	*
06 PRODUCT		
08 LEN	*	*
08 DAT	*	*
04 TRACK2-LEN	*	*
04 TRACK2-DATA	*	*
04 ADDR-DATA		
06 ADDR	*	*
06 ZIP-CDE	*	*
06 ADDR-VRFY-STAT	*	*

Key:

A = Supplied by acquirer host.

B = Supplied by BASE24.

0 = Should be zero-filled.

* = Copied as is from original request and response messages.

b = (Blank) not applicable; zero- or blank-filled.

Late Response from an Authorization Module

If the ANSI Host Interface receives a late 0210 message from the Device Handler/Router/Authorization process, it performs the following steps:

1. Checks the TRAN.RESP-CDE field in the PSTM to determine whether the transaction was declined. If it was declined, the ANSI Host Interface drops the message.
2. Checks the TRAN-CDE field in the PSTM to determine whether the transaction is a balance inquiry. If it is a balance inquiry, the ANSI Host Interface drops the message.
3. Changes the TYP field in the PSTM to 0420.
4. Changes the RTE.STAT field in the PSTM to 00.
5. Sets the RVSL-CDE field in the PSTM to 01, indicating that the Device Handler/Router/Authorization process did not complete processing before the message timed out.
6. Returns the 0420 message to the Device Handler/Router/Authorization process that sent the late response.

Failed Messages to Stations

If a message is forwarded to the Nucleus, but cannot be sent to a station, the Nucleus returns the message to the ANSI Host Interface as failed.

Upon receipt of a message failure, the ANSI Host Interface marks the station down and sets an extended network management message timer. It then performs based on message type.

Failed 0210 Messages

The ANSI Host Interface takes the following steps when it receives a failed 0210 message from the Nucleus.

1. Locates the DPC in its internal table, identifying the necessary information for processing.
2. Checks the RESP-CDE field in the SEM to determine whether the transaction was declined. If it was declined, the ANSI Host Interface drops the message.
3. Formats a PSTM from the SEM.
4. Sets the following fields in the PSTM:
 - a. ORIGINATOR—Set to the value 5.
 - b. TRAN.ORIG—Set to the literal HOST.
5. Changes the TYP field in the PSTM to 0420.
6. Sets the RVSL-CDE field in the PSTM to 03 to indicate that the destination is not available.
7. Returns the 0420 message to the Device Handler/Router/Authorization process that sent it.

Failed 0230/0410/5410 Messages

If the ANSI Host Interface receives a failed 0230, 0410, or 5410 message from the Nucleus, it drops the message.

Unintelligible Messages from a DPC

If the ANSI Host Interface receives a message with an invalid message type, or a message carrying information in an incorrect format, it changes the ROUTE.STAT field in the SEM to 21, indicating an unknown message type. The ANSI Host Interface then returns the message to the sending DPC.

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Appendix A

Processing Information

This section provides processing information on the following aspects of the ANSI Host Interface process:

- Initialization
- Warmboot
- Log messages
- Text Commands

Initialization

Initialization occurs when the ANSI Host Interface is started. During initialization, the ANSI Host Interface performs as follows.

Base ANSI Host Interface

During initialization, the ANSI Host Interface performs the steps listed below. If problems occur during initialization, the ANSI Host Interface logs a message and abends.

1. Reads the name of the BASE24 Logical Network Configuration File (LCONF) into memory.

The ANSI Host Interface obtains the name of the LCONF from the startup message sent by the Nucleus. This name allows the ANSI Host Interface to access the LCONF for the assigns and params it needs for processing.

2. Opens the LCONF, reads the following assigns and params, and closes the LCONF. If the ANSI Host Interface cannot locate the required assigns and params, the process abends.

assign HCF (required)
assign SAF (required)
assign KEYF (required)
assign LMCF
assign PLF

param LOGICAL-NET (required)

3. Opens and reads the Log Message Configuration File (LMCF) for its modified log messages. To do this, the ANSI Host Interface uses the file name specified in the LMCF assign from the LCONF.

The LMCF only contains records for modified log messages. If any records are found for the ANSI Host Interface, they are read into memory.

4. Closes the LMCF.
5. Opens the logging facility for writing log messages.
6. Initializes internal memory for the amount of space required to perform processing. The ANSI Host Interface allocates enough space for its internal tables, timers, and outstanding messages (each time the ANSI Host Interface sends a message, it stores a copy of the message in memory for processing if no response is returned).

7. Initializes its DPC table.

The DPC table is an internal table containing one record for each DPC the ANSI Host Interface supports. This table contains all of the information contained in the HCF for the DPC, along with other status flags and counters that the ANSI Host Interface uses for processing.

At this point, the ANSI Host Interface performs the following steps:

- a. Initializes all counters in the DPC table to zeros.
 - b. Opens the KEYF using the file name specified in the KEYF assign from the LCONF.
 - c. Opens and reads the HCF using the file name specified in the HCF assign from the LCONF.
 - d. Searches for its own records.
 - 1) If no records are found for the ANSI Host Interface, the process logs a message and abends.
 - 2) If more than eight records are found for the ANSI Host Interface, the process logs a message and abends.
 - e. Reads the HCF information for those DPCs it is to process into its DPC table.
 - f. Marks the DPC down.
 - g. Marks all of the DPC's stations down.
 - h. Reads the KEYF and adds the information from the KEYF to the DPC table.
 - i. Closes the HCF.
 - j. Initializes its timer table.
8. Opens and reads the Store-and-Forward File (SAF) using the file name specified in the SAF assign from the LCONF. If a SAF does not exist, the ANSI Host Interface creates one using the file name and template from the LCONF.
9. Counts the records in the SAF destined for each of its DPCs and places those counts in the DPC table.
10. Performs product-specific initialization steps for each product that is supported. Product-specific steps are performed in order by product ID. The table below identifies the product IDs for BASE24 products.
- 1 = BASE24-atm
 - 2 = BASE24-pos

11. Retrieves security module information from the LCONF and opens the security modules if necessary. To do this, the ANSI Host Interface process performs the following steps:
 - a. Opens the LCONF.
 - b. Retrieves the following params. These params are required to interface with the security module.

SECURE-DEV-TYP
SECURE-DEV-TIM-LMT
SECURE-DEV-FAIL-NOTIFY

If the SECURE-DEV-TYP param is not present, the ANSI Host Interface process closes the LCONF and skips to step 12, processing as if a security module is not being used.
 - c. Retrieves the assigns listed below that are required to interface with a security module.

SECURE-DEV1
SECURE-DEV2

If both of the above assigns are not present, the ANSI Host Interface process logs a message and sets an internal flag to prohibit the use of a security module.
 - d. Opens the security modules indicated in the SECURE-DEV1 and SECURE-DEV2 assigns.

Upon opening these devices, the operating system returns a file number for each device. The ANSI Host Interface process then tracks these file numbers.

If the ANSI Host Interface process does not, or cannot, open at least one device, it logs messages and abends.
 - e. Closes the LCONF.
12. Checks the NMM-ENABLE field taken from the HCF for each DPC.

If the NMM-ENABLE field indicates the DPC does not allow network management messages, the ANSI Host Interface marks the DPC and all of its stations up. Otherwise, the ANSI Host Interface checks the STA-TYP field taken from the HCF for each station and performs the following steps:
 - a. If the station is not a send/receive station, the ANSI Host Interface marks the station up and goes on to the next station.
 - b. If the station is a send/receive station, the ANSI Host Interface creates and sends an echo-test message and sets a Network Management Message timer for the station.

13. Sets a Performance timer for each DPC.
14. Generates the log message “ANSI Host Interface INITIALIZATION COMPLETE.”

BASE24-atm ANSI Host Interface

Because the ANSI Host Interface can handle multiple BASE24 products, those initialization actions specific to each product have been segregated. For BASE24-atm processing, the ANSI Host Interface:

1. Opens the LCONF, reads the following assigns and params, and closes the LCONF:

param ATM-LN-CUTOVER
param REVERSE-BAL-INQ
2. Derives the current BASE24-atm posting date.

To do this, the ANSI Host Interface checks the ATM-LN-CUTOVER param value it retrieved from the LCONF. If the current time is prior to logical network cutover, the ANSI Host Interface uses the current calendar date as the posting date. If the current time is after logical network cutover, the ANSI Host Interface adds one to the current calendar date to derive a posting date.
3. Updates the DPC table with BASE24-atm specific information from the HCF.

BASE24-pos ANSI Host Interface

Because the ANSI Host Interface can handle multiple BASE24 products, those initialization actions specific to each product have been segregated. For BASE24-pos processing, the ANSI Host Interface:

1. Opens the LCONF, reads the POS-LN-CUTOVER param, and closes the LCONF.
2. Derives the current BASE24-pos posting date.

To do this, the ANSI Host Interface checks the POS-LN-CUTOVER param value it retrieved from the LCONF. If the current time is prior to logical network cutover, the ANSI Host Interface uses the current calendar date as the posting date. If the current time is after logical network cutover, the ANSI Host Interface adds one to the current calendar date to derive a posting date.
3. Updates the DPC table with BASE24-pos specific information from the HCF.

Warmboot

Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the ANSI Host Interface in memory.

During warmboot, the ANSI Host Interface opens the following files:

- Logical Network Configuration File (LCONF)
- Log Message Configuration File (LMCF)
- Host Configuration File (HCF)
- Institution Definition File (IDF)
- Key File (KEYF)

The ANSI Host Interface also opens the SAF and recounts those messages pending for each of its DPCs. The ANSI Host Interface retains all of its current timers, processing flags, status indicators, and performance accumulators so that current processing is not affected. It does not, however, reallocate extended memory or initiate new echo test messages to any of its DPCs.

During a warmboot, the ANSI Host Interface retrieves the names of its security modules from the LCONF and attempts to open those modules. If the security modules named cannot be opened, the ANSI Host Interface continues using those that are already open.

Following a successful warmboot, the ANSI Host Interface sends a “WARMBOOT COMPLETE” message to its log.

Log Messages

The ANSI Host Interface generates log messages to assist network operators in performing their daily tasks. Log messages provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The ANSI Host Interface generates log messages using a standard routing code of zero.

Log Message Documentation

Log message documentation is contained in files on the BA50LOGM subvolume. Each file contains log messages for a specific BASE24 module. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 module. PMIDs identify a specific module in the BASE24 system (for example, BAHISF identifies the base portion of the ANSI Host Interface and ATHISF identifies the BASE24-atm portion of the ANSI Host Interface).

Documentation for the log messages generated by the ANSI Host Interface is located in the files BAHISF, ATHISF, and PSHISF. For a complete explanation of log message documentation, refer to the ***BASE24 Log Message Reference Manual***.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains log message information, such as message text, routing code, and severity, that has been modified by the user. At initialization, the ANSI Host Interface builds a table in memory from the LMCF records. When a log message is generated, the ANSI Host Interface accesses the table to verify whether that log message has been modified. If the log message has been modified, the ANSI Host Interface logs the modified information instead of the standard log message.

The LMCF is accessed via operator functions. Each log message's routing code, severity, and text can be modified via the LMCF. Because LMCF records are read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized. For more information on accessing the LMCF, refer to the ***BASE24 Network Control Operators Manual***.

Network Environment Source File Information

The ANSI Host Interface is generally set up in the Network Environment Source File (NEFS) to start automatically when the resource node is initiated. Because of the length of time required for initialization, the ANSI Host Interface should be assigned a startup logic of BOJ (beginning of job) in the Network Environment Source File (NEFS). BOJ means that the ANSI Host Interface is started automatically when the resource node is initiated. This ensures that the ANSI Host Interface is ready when it receives its first transaction.

Multiple copies of the ANSI Host Interface can be defined in the NEFS to control or increase transaction throughput. Each ANSI Host Interface can support up to eight DPCs. Note that one DPC can be serviced by more than one ANSI Host Interface. ANSI Host Interface processes are tied to their DPCs via the HCF.

In addition to one process declaration for each ANSI Host Interface, the NEFS must also include station and line declarations for the stations and communications lines used to communicate with the host. For more information on setting up the required process, station, and line declarations in the NEFS, refer to the ***BASE24 Network Environment Source File Manual***.

BASE24 Text Commands

The ANSI Host Interface process can receive a number of BASE24 text commands that cause it to perform specialized types of activities. These commands can be separated into three categories: process-generated commands, general use operator commands, and restricted use operator commands.

Process-generated commands are generated and sent by other running processes in the system. In general, these commands notify the ANSI Host Interface process about events occurring elsewhere in the system.

General use operator commands are entered by an operator in the daily use of the system. These commands enable the operator to control or change the way the ANSI Host Interface process is operating.

Restricted use operator commands should be entered by an operator only at the request of ACI personnel. These commands allow programmers to test functionality and determine how the ANSI Host Interface process is running when changes are made to the code.

The table below lists all of the commands that the ANSI Host Interface process can receive and identifies the category under which the command is described. For more information on issuing these commands to the ANSI Host Interface and on the processing that the ANSI Host Interface performs when it receives one of these commands, refer to the ***BASE24 Text Command Reference Manual***.

Process-Generated Commands	General Use Operator Commands	Restricted Use Operator Commands
Cutover Store-and-Forward Extract Complete Store-and-Forward Extract Request	Device State Change/New Key Logon Logoff Performance Repeat Key Status Start Store-and-Forward Stop Store-and-Forward Verify Key Warmboot	Debug Dump Trace On Trace Off

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Appendix B

Network Management Messages

This appendix contains field-by-field descriptions of the following ANSI-based network management messages used by BASE24.

- Network management message used for logons, logoffs, and echo tests (NMM format)
- Network management message used for dynamic key management (NMMKEY format).

NMM Format

The following pages describe the fields that make up the Network Management Message (NMM) format.

Message Length: 94 bytes

Position	Level	Field Name and Description	Data Type
01–24	02	ROUTE A set of subfields indicating the status of the message at the system level and also defining the message origin.	
01–02	03	STAT Indicates the status of the message at the system level. Valid codes are: 00 = Default, no error condition 11 = Destination not available 12 = Communications line down 21 = Invalid message type	PIC 9(2)
03–24	03	PATH (NOT USED)	PIC X(22)
25–28	02	TYP The message type associated with the message. A value of 0800 must be placed in this field.	PIC X(4)
29–44	02	BYTE-MAP A 16-byte representation of the X9.2 standard 64-bit map. The ANSI X9.2 message standard mandates a 64-bit map to indicate the presence or absence of particular data fields in the message format. Because such a bit map can lead to communications line transparency problems, BASE24 uses a 16-byte representation. Note: In the X9.2 standard, bits are numbered from 1 to 64 rather than from 0 to 63.	PIC X(16)

Position	Level	Field Name and Description	Data Type
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The hexadecimal value of each 4 bits of the bit map is represented by one 8-bit character. For example, the bit string 1011 (hexadecimal B), is represented by a the character B. Thus, 16 bytes are used to represent 64 bits.

BASE24 does not use this field for processing and will perform no edits on its values. BASE24 does, however, always place the following values in the field:

BYTE MAP = 04C0000101000000

The Bit Map extender bit (bit map position 64) is not checked by BASE24 and the extended bit map feature is not supported.

45–50	02	TRACE-NO	PIC 9(6)
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A trace number assigned by the Host Interface process to uniquely identify each transaction message. This field is initialized by the originating network and must be returned intact by the authorizing network (right-justified and zero-filled).

51–54	02	TRAN-DAT	PIC X(4)
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The month and day (MMDD) the message was forwarded to the host.

Position	Level	Field Name and Description	Data Type
55–60	02	TRAN-TIM The local time (HHMMSS) the message was forwarded to the host.	PIC X(6)
55–56	04	HH	PIC 99
57–58	04	MM	PIC 99
59–60	04	SS	PIC 99
61–63	02	NM-INFO-CDE Identifies the type of network management message being sent. Valid codes are: 001 = logon 002 = logoff 301 = echo-test	PIC 9(3)
64–79	02	PASSWORD (NOT USED)	PIC X(16)
80–94	02	FILLER	PIC X(15)

NMMKEY Format

The following pages describe the fields that make up the Network Management Message format used for dynamic key management (NMMKEY format).

Message Length: 271 bytes

Position	Level	Field Name and Description	Data Type
1–24	02	ROUTE A set of subfields indicating the status of the message at the system level and also defining the message origin.	
1–2	04	STAT Indicates the status of the message at the system level. Valid codes are: 00 = Default, no error condition 11 = Destination not available 12 = Communications line down 21 = Invalid message type	PIC 9(2)
3–24	04	PATH (NOT USED)	PIC X(22)
25–28	02	TYP The message type associated with the message. A value of 0800 must be placed in this field.	PIC 9(4)
29–44	02	BYTE-MAP A 16-byte representation of the X9.2 standard 64-bit map. The ANSI X9.2 message standard mandates a 64-bit map to indicate the presence or absence of particular data fields in the message format. Because such a bit map can lead to communications line transparency problems, BASE24 uses a 16-byte representation. Note: In the X9.2 standard, bits are numbered from 1 to 64 rather than from 0 to 63.	PIC X(16)

Position	Level	Field Name and Description	Data Type
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The hexadecimal value of each 4 bits of the bit map is represented by one 8-bit character. For example, the bit string 1011 (hexadecimal B), is represented by a the character B. Thus, 16 bytes are used to represent 64 bits.

BASE24 does not use this field for processing and will perform no edits on its values. BASE24 does, however, always place the following values in the field:

BYTE MAP = 04C0000100008000

The Bit Map extender bit (bit map position 64) is not checked by BASE24 and the extended bit map feature is not supported.

45–50	02	TRACE-NO	PIC 9(6)
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A trace number assigned by the Host Interface process to uniquely identify each transaction message. This field is initialized by the originating network and must be returned intact by the authorizing network (right-justified and zero-filled).

51–54	02	TRAN-DAT	PIC X(4)
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The month and day (MMDD) the message was forwarded to the host.

Position	Level	Field Name and Description	Data Type
55–60	02	TRAN-TIM The local time (HHMMSS) the message was forwarded to the host.	
55–56	04	HH	PIC 9(2)
57–58	04	MINUTE	PIC 9(2)
59–60	04	SS	PIC 9(2)
61–63	02	NM-INFO-CDE Identifies the type of network management message being sent. Valid codes are: 161 = Change PIN Key 162 = New PIN Key 163 = Repeat PIN Key 164 = Verify PIN Key 261 = Change PIN Key Positive Response 262 = New PIN Key Positive Response 263 = Repeat PIN Key Positive Response 264 = Verify PIN Key Positive Response 361 = Change PIN Key Negative Response 362 = New PIN Key Negative Response 363 = Repeat PIN Key Negative Response 364 = No check digits Negative Response 365 = Verify PIN Key Negative Response 366 = New Key Invalid Check Digits Response	PIC 9(3)
64–271	02	KEY-MGMT The following fields contain the X9.17-1985 Dynamic Key Management Data.	
64–65	04	KEY-DIR Identifies the direction of the key being exchanged. Valid values are: 01 = Outbound 02 = Inbound 03 = Both	PIC X(2)

Position	Level	Field Name and Description	Data Type
66–71	04	CHK-DIGITS The check digits for the key being exchanged. The check digits are generated at the same time as the key. They are used to verify that the key was not corrupted during transmission.	PIC X(6)
72–271	04	X917-CSM-DATA The Cryptographic Service Message (CSM). The contents of the CSM depends on the type of message being exchanged. For more information on the contents of this field, refer to the <i>BASE24 Transaction Security Manual</i> .	PIC X(200)

Appendix C

BASE24-atm Standard External Messages

This appendix contains a field-by-field description of the ANSI-based BASE24-atm Standard External Message (SEM).

Message Length: 600 bytes

Position	Level	Field Name and Description	Data Type
001–024	02	ROUTE A set of subfields indicating the status of the message at the system level and also defining the message origin.	
001–002	03	STAT Indicates the status of the message at the system level. This field is set by the ANSI Host Interface and used by the Authorization process. Valid codes are: 00 = Default, no error condition 11 = Destination not available 12 = Communications line down 20 = Originator not available 21 = Invalid message type 22 = Invalid card number If this field is set to 00, the response code field is used to determine the disposition of the message. This field corresponds to the RTE.STAT field in the STM.	PIC 9(2)
003–024	03	PATH A set of subfields indicating the BASE24 product and the message origin type.	

Position	Level	Field Name and Description	Data Type
003–004	05	PRODUCT-IND Determines the BASE24 product with which the message is associated. This field must be set to 0 to indicate BASE24-atm. This field corresponds to the PROD-ID field in the STM.	PIC X(2)
005–006	05	RELEASE-NUM The BASE24 software release number with which the message is associated. Valid values are: 40 = Previous release format 50 = Current release format This field is not carried in the STM.	PIC X(2)
007–010	05	DPC-NUMBER The DPC number of the DPC to which, or from which, the message is being sent. This field corresponds to the DPC-NUM field in the STM.	PIC 9(4)
011	05	AUTHORIZER	PIC X(1)
012–014	05	FILLER	PIC X(3)
015	05	ORIGI-TYPE Identifies the location from which the transaction originated. This field may be checked by issuer hosts to determine message origins. Valid values are: 0 = A device controlled by BASE24 1 = An acquirer host 2 = Not used 3 = Indicates that the transaction originated at, or was authorized by, an interchange.	PIC X

Position	Level	Field Name and Description	Data Type
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This field corresponds to the RTE.ORIGINATOR field in the STM. For incoming messages, the STM field is set as follows based on this flag:

- 0 = The RTE.ORIGINATOR field is set to 1
- 1 = The RTE.ORIGINATOR field is set to 5
- 3 = The RTE.ORIGINATOR field is set to 7

For outgoing messages, this field is set from the RTE.ORIGINATOR field as follows:

- 1, 2, or 3 = This field is set to 0
- 4 or 5 = This field is set to 1
- 6 or 7 = This field is set to 3

016–024	05	FILLER	PIC X(09)
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025–028	02	TYP	PIC 9(4)
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The message type associated with the message. Valid message types are:

- 0200 = Financial Transaction Request
- 0210 = Financial Transaction Request Response
- 0220 = Financial Transaction Advice
- 0221 = Financial Transaction Advice Repeat
- 0230 = Financial Transaction Advice Response
- 0400 = Transaction Reversal
- 0401 = Transaction Reversal Repeat
- 0410 = Transaction Reversal Response
- 0700 = Balance Inquiry
- 0710 = Balance Inquiry Response
- 5400 = Adjustment
- 5410 = Adjustment Response
- 9220 = Store-and-Forward Force Post
- 9221 = Store-and-Forward Force Post Repeat
- 9230 = Store-and-Forward Force Post Response

This field corresponds to the TYP field in the STM, although not all external message types are carried internally by BASE24.

Position	Level	Field Name and Description	Data Type
029–044	02	BYTE-MAP The ANSI X9.2 message standard mandates a 64-bit map to indicate the presence or absence of particular data fields in the message format. Because such a bit map can lead to communications line transparency problems, BASE24 uses a 16-byte representation of the X9.2 standard 64-bit map. Note: In the X9.2 standard, bits are numbered from 1 to 64 rather than from 0 to 63. The hexadecimal value of each 4 bits of the bit map is represented by one 8-bit character. For example, the bit string 1011 (hexadecimal B), is represented by the character B. Thus, 16 bytes are used to represent 64 bits. BASE24 does not use this field for processing and will perform no edits on its values. BASE24 does, however, always place the following values in the field: BYTE MAP = FFCA1802E2007FE6 The bit map extender bit (bit map position 64) is not checked by BASE24 and the extended bit map feature is not supported.	PIC X(16)
045–065	02	CARD-ACCT These subfields define the length of the card account number and the actual account number.	
045–046	03	LEN The actual number of characters of data (from 01 to 19) contained in the fixed-length CRD-ACCT.NUM field below.	PIC 9(2)
047–065	03	NUM The card number from Track 2 of the card used to initiate the transaction (left-justified and blank-filled).	PIC X(19)
066–071	02	TRAN-CDE The following fields contain a series of codes used to specify the type of transaction and cardholder account(s) to affect.	

Position	Level	Field Name and Description	Data Type
066–067	03	TYP Indicates the specific transaction. Valid values are: 03 = Check Guarantee 04 = Check Verification 10 = Withdrawal 20 = Deposit 30 = Balance Inquiry 40 = Transfer 50 = Payment 51 = Payment Enclosed 60 = Message Enclosed to Financial Institution This field corresponds to the TRAN-CDE field in the STM.	PIC X(2)
068–069	03	FROM-ACCT-TYP Indicates the primary account affected on a debit or inquiry transaction or the <i>from</i> account in the case of a transfer transaction. Valid values are: 00 = No account 01 = Checking 11 = Savings 31 = Credit This field corresponds to the FROM-ACCT-TYP field in the STM.	PIC X(2)
070–071	03	TO-ACCT-TYP Indicates the primary account affected on a credit transaction or the <i>to</i> account in the case of a transfer transaction. Valid values are: 00 = No account 01 = Checking 11 = Savings 31 = Credit This field corresponds to the TO-ACCT-TYP field in the STM.	PIC X(2)

Position	Level	Field Name and Description	Data Type
072–079	02	AMT-1 The requested total amount of the original transaction in U.S. dollars and cents for message types 0200, 0210, 0220, or 0400. For message type 5400, this field contains the amount originally completed (which might have been AMT-2 in a reversal). This field corresponds to the RQST.AMT-1 field in the STM.	PIC 9(6)V99
080–087	02	AMT-2 For message type 0400, this field contains the completion amount. For message type 0210, this field contains a balance amount depending on the transaction type. Field position 0 contains a minus sign if the amount is negative. Otherwise, it contains a zero. For message type 5400, this field contains the amount actually completed. This field corresponds to the RQST.AMT-2 field in the STM.	PIC 9(6)V99
088–095	02	AMT-3 For message type 0210, this field contains a balance amount depending on the transaction type. Field position 0 contains a minus sign if the amount is negative. Otherwise, it contains a zero. This field corresponds to the RQST.AMT-3 field in the STM.	PIC 9(6)V99
096–101	02	TRACE-NO An echo value trace number for matching a financial transaction request with an authorization response. This field is initialized by the originating network and must be returned intact by the authorizing network (right-justified and zero-filled). This field corresponds to the RQST.TRACE-NO field in the STM.	PIC 9(6)
102–105	02	TRAN-DAT The month and day that the transaction was initiated (in MMDD format). This field corresponds to the TRAN-DAT field in the STM.	PIC X(4)

Position	Level	Field Name and Description	Data Type
106–109	02	POST-DAT The BASE24 transaction settlement month and day (in MMDD format). This date is normally the same as the transaction date unless the originating terminal device has already been settled for the business day. In this case, this field will be set to the next settlement business date. This field corresponds to the POST-DAT field in the STM.	PIC X(4)
110–113	02	TRANSMISSION-DAT The month and day forwarded to the host (in MMDD format). During periods when the host is offline, BASE24 will set this field to the date that a store-and-forward message is sent to the host. This field corresponds to the EXIT-TIM field in the STM.	PIC X(4)
114–119	02	TRAN-TIME The local time the transaction began (hour, minute, and second in HHMMSS format). This field corresponds to the TRAN-TIM field in the STM.	
114–115	04	HH	PIC 99
116–117	04	MM	PIC 99
118–119	04	SS	PIC 99
120–129	02	RTTN An identifier for the terminal-owning institution. In the United States, this field is used for the route/transit number of the institution. This field corresponds to the ACQ-INST-ID-NUM field in the STM.	PIC 9(10)
130–139	02	ISSUER-ROUTING-NUMBER A 1-digit code followed by a 9-digit route and transit identifier for the card-issuing institution, as previously defined for the RTTN field (positions 120–129). This field corresponds to the RCV-INST-ID-NUM field in the STM.	PIC 9(10)

Position	Level	Field Name and Description	Data Type
140–142	02	CURRENCY-CDE A required value of 840 in Release 3, indicating General U.S. Funds. This field corresponds to the RQST.ORIG-CRNCY-CDE field in the STM.	PIC 9(3)
143–144	02	REVERSAL-CDE Indicates the reason for a reversal or adjustment; used for reversal (0400) and adjustment (5400) messages only. This field corresponds to the RQST.RVSL-CDE field in the STM. Valid reversal codes are: 00 = Not applicable 01 = Timeout 02 = Command reject 03 = Destination not available 08 = Customer cancellation 10 = Hardware error 20 = Suspect reversal 21 = Message authentication code (MAC) failure 22 = MAC key (KMAC) synchronization error 23 = Message replay error 24 = Invalid MAC Valid adjustment codes are: 12 = Original amount incorrect 13 = Terminal malfunction 14 = Suspicious reversal override 15 = Misdispense reversal override 16 = Duplicate transaction 17 = Reconciliation error 18 = PLUS add cash withdrawal or advance	PIC 99

Position	Level	Field Name and Description	Data Type
145–148	02	TYPE-ORIG The type of the original message, if this message is a store-and-forward message. For original transaction messages, this field is set to 0000. For store-and-forward transaction messages, this field contains 0400 for reversal messages and 0220 for completion messages.	PIC 9(4)
149–150	02	POINT-TRAN-SPEC A code indicating a special condition which existed at the time the transaction was initiated. This field is not carried in the STM. It is always set to 03 by the ANSI Host Interface to indicate the transaction originated at a terminal device.	PIC 99
151–158	02	RESP-CDE-R	PIC X(8)
151–158	02	RESP-CDE A code set by the Authorization process identifying the results of a Financial Transaction Request message.	REDEFINES RESP-CDE-R
151–155	04	FILLER	PIC X(5)
156–158	04	RESP This field is split into a one-position field and a two-position field. Position 1 contains the card return/retain action assigned to the message by the authorizer. Valid codes are: 0 = Return card (BASE24-authorized) 1 = Retain card (BASE24-authorized) 5 = Return card (host-authorized) 6 = Retain card (host-authorized) 7 = Return card (switch-authorized) 8 = Retain card (switch-authorized)	PIC X(3)

Position	Level	Field Name and Description	Data Type
		Positions 2 and 3 contain the transaction approval or denial code assigned to the message by the authorizer. Response codes are described in appendix A. This field corresponds to the RQST.RESP field in the STM.	
159–173	02	TERM-CITY-ST The terminal location city and state or country. For transactions originating at BASE24 terminals, this field is initialized from the Terminal Data File (TDF). Positions 1 through 13 of this field correspond to the RQST.TERM-CITY field in the STM; positions 14 and 15 of this field correspond to the RQST.TERM-CITY-ST field in the STM.	PIC X(15)
174–193	02	ORIGL These subfields contain original data elements applicable only for adjustment (5400) transactions.	
174–179	03	TRACE-NO The sequence number assigned to the original transaction. This field corresponds to the RQST.ORIG.SEQ-NUM field in the STM.	PIC X(6)
180–183	03	TRAN-DATE The date (MMDD) of the original transaction. This field corresponds to the RQST.ORIG.TRAN-DAT field in the STM.	PIC X(4)
184–189	03	TRAN-TIME The time (HHMMSS) of the original transaction. This field corresponds to the RQST.ORIG.TRAN-TIM field in the STM.	PIC X(6)

Position	Level	Field Name and Description	Data Type
190–193	03	B24-POST-DAT The BASE24 posting date (MMDD) for the original transaction. This field corresponds to the RQST.ORIG.B24-POST-DAT field in the STM.	PIC X(4)
194–199	02	ISS-TRACE-NO The interchange issuer-assigned sequence number for interchange transactions. This field corresponds to the ISS-TRACE-NO field in the STM.	PIC 9(6)
200–215	02	PIN The personal identification number (PIN) identifying an individual cardholder as a person authorized to access the card account. BASE24 allows from 1 to a maximum of 16 ASCII/EBCDIC alphanumeric characters, left-justified and padded on the right with F's (capital F's required). Because of potential communications line transparency problems, the 64-bit PIN is represented by 16 8-bit characters. The algorithm used to convert between 16-character format and 64-bit format is identical to that used for the bit/byte map (refer to description of the BYTE-MAP field, positions 29 through 44). This field can also be encrypted by a security device. This field corresponds to the RQST.PIN field in the STM.	PIC X(16)
216–236	02	FROM-ACCT These subfields contain the <i>from</i> account number and length.	
216–217	04	LTH The length of the actual account number contained in the fixed-length FROM-ACCT.NUM field that follows (right-justified and zero-filled).	PIC X(2)

Position	Level	Field Name and Description	Data Type
218–236	04	NUM The application account number to be affected by a debit or inquiry transaction, or the <i>from</i> application account number for electronic payment and transfer transactions. This field is initially set to zero by the Device Handler process and is then set by the Authorization process for primary account numbers (left-justified and blank-filled). This field corresponds to the RQST.FROM-ACCT field in the STM.	PIC X(19)
237–257	02	TO-ACCT These subfields contain the <i>to</i> account number and length.	
237–238	04	LTH The length of the actual account number contained in the fixed-length TO-ACCT.NUM field which follows (right-justified and zero-filled).	PIC X(2)
239–257	04	NUM The application account number to be affected by a credit transaction or the <i>to</i> application account number for electronic payments and transfer transactions. This field is set to zero by the Device Handler process and is set by the Authorization process for primary account numbers (left-justified and blank-filled). This field corresponds to the RQST.TO-ACCT field in the STM.	PIC X(19)
258–307	02	SHARING These subfields are filled in by an acquirer host when sending a request to BASE24. All sharing groups to which the transaction-originating terminal belongs to should be filled in.	

Position	Level	Field Name and Description	Data Type
258	04	MBR-NUM The member number (may be 0 through 9) identifying the specific cardholder initiating the transaction at the transaction-initiating terminal. This field corresponds to the RQST.MBR- NUM field in the STM.	PIC 9
259–282	04	SHARING-GROUP All sharing groups (up to the maximum of 24) to which the transaction-initiating terminal belongs. BASE24 sets this field for all transactions being requested at devices connected to and controlled by BASE24. This field corresponds to the SHRG-GRP field in the STM.	PIC X OCCURS 24 TIMES
283–284	04	TERM-STATE A 2-character state code indicating the state in which the transaction-initiating terminal is located. State codes are available in ANSI publication: <i>States and Outlying Areas of the United States (FIPS PUB 5-1)</i> , National Bureau of Standards, U.S. Department of Commerce, Washington, D.C., Government Printing Office, June 15, 1970. This field corresponds to the RQST.TERM-ST field in the STM.	PIC 9(2)
285–287	04	TERM-COUNTY A 3-character county code indicating the county in which the transaction-originating terminal is located. County codes are available in ANSI publication: <i>Counties and County Equivalents of the States of the United States (FIPS PUB 6-1)</i> , National Bureau of Standards, U.S. Department of Commerce, Washington, D.C., Government Printing Office, June 15, 1970.	PIC 9(3)

Position	Level	Field Name and Description	Data Type
		This field corresponds to the RQST.TERM-CNTY field in the STM.	
288–303	04	NOT-ON-US-MASK A 16-byte not-on-us-mask set by an acquirer host to define the classification of a transaction being initiated by a foreign cardholder (not-on-us) using a device connected to and controlled by the acquirer host. Bytes 1 through 9 identify, in sequence, the following classification of transactions. (Bytes 10 through 16 are filler bytes and must be blank-filled.) Byte 1 = Withdrawal Byte 2 = Withdrawal credit Byte 3 = Deposit Byte 4 = Inquiry/transaction screening Byte 5 = Transfer Byte 6 = Electronic Payment Byte 7 = Payment Enclosed Byte 8 = Check verification/guarantee Byte 9 = Message to institution The following codes are used in the first 9 bytes of the “not-on-us-mask” to indicate whether not-on-us transactions are allowed: 0 = Not allowed 1 = Allowed within the County 2 = Allowed within the State 3 = Allowed Nationally 4 = Allowed Internationally On outgoing messages, the ANSI Host Interface fills in the first nine positions of this field with the value taken from the RQST.TERM-TRAN-ALLOWED field in the STM. On incoming messages, the ANSI Host Interface sets the RQST.TERM-TRAN-ALLOWED field in the STM from this field, based on the type of transaction.	PIC X(16)
304–307	04	FIID Not used; zero-filled by the ANSI Host Interface.	PIC X(4)

Position	Level	Field Name and Description	Data Type
308–397	02	TERM These subfields contain terminal-related information.	
308–311	04	OWNER The FIID of the terminal-owning institution from the Terminal Data File (TDF). This field corresponds to the TERM-OWNER-FIID field in the STM.	PIC X(4)
312–336	04	TERM-NAME-LOC The transaction-originating terminal’s descriptive name and location. This field corresponds to the RQST.TERM-NAME-LOC field in the STM.	PIC X(25)
337–361	04	TERM-OWNER-NAME The terminal owner’s descriptive name. This field corresponds to the RQST.TERM-OWNER-NAME field in the STM.	PIC X(25)
362–377	04	ATM-ID The terminal’s symbolic name as specified in the Terminal Data File (TDF). This field corresponds to the TERM-ID field in the STM.	PIC X(16)
378–381	04	BRANCH-ID The terminal’s Branch ID as specified in the Terminal Data File (TDF). This field corresponds to the BRCH-ID field in the STM.	PIC X(4)
382–385	04	REGION-ID The terminal’s Region ID as specified in the Terminal Data File (TDF). This field corresponds to the REGN-ID field in the STM.	PIC X(4)

Position	Level	Field Name and Description	Data Type
386–391	04	REC-NUM The terminal receipt number as set by the terminal and used to identify the transaction and for printing on a customer-related statement receipt and any BASE24-generated reports. This field corresponds to the SEQ-NUM field in the STM.	PIC 9(6)
392–393	04	TYP The terminal type code as specified in the Terminal Data File (TDF). For values applicable to BASE24-atm, refer to the DEVICE TYPE field description in the BASE24-atm Files Maintenance Manual .	PIC 99
394–397	04	TIM-OFST The time difference (in minutes) between the terminal location and the BASE24 transaction processor. This field contains the signed number of minutes to be added to the Tandem time in order to obtain the terminal local time. This field corresponds to the TIM-OFST field in the STM.	PIC S999
398–419	02	CARD These subfields contain card issuer information and the PIN offset value.	
398–399	04	FIID-LEN The actual number of characters of data (from 1 to 4) in the fixed-length CARD.CARD-FIID field below (right-justified). BASE24 sets the correct length before routing the transaction message to the card-issuing host. Acquirer hosts are not required to enter this field.	PIC 99
400–403	04	CARD-FIID The card issuer's FIID. This field corresponds to the CRD-FIID field in the STM.	PIC X(4)

Position	Level	Field Name and Description	Data Type
404–419	04	PIN-OFST The PIN offset value. This field corresponds to the RQST. PIN-OFST field in the STM.	PIC X(16)
420–427	02	MISC These subfields contain miscellaneous data required in some cases for processing.	
420–423	04	PAYEE-CDE The 4-character identifier of the company to receive the payment amount in a payment transaction. This field is not carried in the STM.	PIC 9(4)
424–427	04	ICHG-SETT-DAT The interchange settlement month and day (in MMDD format) for a transaction that passed through an interchange. This field corresponds to the ACQ-ICHG-SETL-DAT and ISS-ICHG-SETL-DAT fields in the STM.	PIC X(4)
428–439	02	DEPOSITORY These subfields contain depository data related to deposit transaction.	
428–437	04	DEP-BAL-CREDIT The calculated amount of deposit credit used for deposit reversal processing. The amount is in U.S. dollars and cents. A decimal point is implied in all cases with the last two positions being the actual cents. This field corresponds to the RQST.DEP-BAL-CR field in the STM.	PIC 9(8)V99

Position	Level	Field Name and Description	Data Type
438	04	AVAIL-DEP-TYPE The type of depository that is attached to the terminal which initiated a depository transaction. Valid codes are: 0 = Normal envelope depository 1 = Securomatic depository 2 = Envelope and Securomatic depository This field corresponds to the RQST.AVAIL-DEP-TYP field in the STM.	PIC 9
439	04	DEP-TYPE The type of depository specified for the card used to initiate the transaction. Valid codes are: 0 = Normal envelope depository 1 = Securomatic depository This field corresponds to the RQST.DEP-TYP field in the STM.	PIC 9
440	02	HOST A subfield specifying the accounts that are processed by the host.	
440	04	PROC-ACCT-IND Indicates the account(s) to be processed by the host in the case of two-sided transactions (e.g., transfers). Valid codes are: 0 = Process both the <i>from</i> and <i>to</i> accounts. 1 = Process the <i>from</i> account only. 1 = Process the <i>to</i> account only. This field is set from the PROC-ACCT-IND field in the STM.	PIC X(1)
441–444	02	STMT The following fields contain information for statement print transactions.	

Position	Level	Field Name and Description	Data Type
441	04	FRMT A code indicating whether the Device Handler process supports the BASE24-atm Statement Print Data Compression format on statement print transactions. The Device Handler sets this value from the STMT-PRNTR-CNTL.FRMT field in the Terminal Data File (TDF). The default value is 0. Valid values are: 0 = Columnar format (data compression is not supported) 1 = BASE24-atm Statement Print Data Compression format (data compression is supported)	PIC X(1)
442–444	04	PRNT-SIZ The maximum number of characters per line supported by the Device Handler process when BASE24-atm Statement Print Data Compression is implemented for statement print transactions. The Device Handler sets this value by subtracting the STMT-PRNTR-CNTL.LEFT-MARGIN field from the STMT-PRNTR-CNTL.RIGHT-MARGIN field in the Terminal Data File (TDF). If the Device Handler process does not support data compression, this field is zero-filled.	PIC X(3)
445–451	02	USER-FLD (NOT USED)	PIC X(7)
452–490	02	TRACK2 These subfields contain Track 2 data from the card.	
452–453	03	LEN The actual number of characters of data (from 00 through 37) that are contained in the fixed-length TRACK2.T2DATA field below (right-justified and zero-filled).	PIC 99
454–490	03	T2DATA The information encoded on Track 2 of the card originating the transaction, including field separators, but excluding beginning and ending sentinels and longitudinal redundancy	PIC X(37)

Position	Level	Field Name and Description	Data Type
		check characters (left-justified). BASE24 supports Track 2 data and uses the PAN, member number, card expiration date, and PIN verification fields. This field corresponds to the RQST.TRACK2 field in the STM.	
491–597	02	TRACK3 These subfields contain Track 3 data from the cardholder card.	
491–493	03	LEN The actual number of characters of data (from 000 through 104) that are contained in the fixed-length TRACK3.T3DATA field below (right-justified and zero-filled).	PIC 999
494–597	03	T3DATA The information encoded on Track 3 of the card originating the transaction, including field separators, but excluding beginning and ending sentinels and longitudinal redundancy check characters (left-justified). BASE24 does not support Track 3 data, but if Track 3 data is present, it is forwarded to the host for processing. This field corresponds to the RQST.TRACK3 field in the STM.	PIC X(104)
598–600	02	FILLER Reserved for 3780 protocol use.	PIC X(3)

Appendix D

BASE24-pos Standard External Messages

This appendix contains field-by-field descriptions of the following ANSI-based external messages used by BASE24-pos:

- BASE24-pos Standard Short External Message (PSEMA)
- BASE24-pos Extended Short External Message (PSEMAE)
- BASE24-pos Standard Long External Message (PSEMF)
- BASE24-pos Extended Long External Message (PSEMF E)

PSEMA and PSEMAE Formats

The following pages describe the fields that make up the BASE24-pos Standard Short External Message (PSEMA) and BASE24-pos Extended Short External Message (PSEMAE) formats.

PSEMA Length: 217 bytes

PSEMAE Length: 247 bytes

Position	Level	Field Name and Description	Data Type
1–24	02	ROUTE The following fields contain routing data.	
1–2	03	STAT The status of a message at the system level. If this field is set to 00, the response code field determines the disposition of the message. The ANSI Host Interface sets the value in this field. It is used by Authorization. Valid values are: 00 = No error 11 = Destination not available 12 = Line down 20 = Originator not available 21 = Unknown message type 22 = Unknown card number	PIC 9(2)
3–24	03	PATH The following fields define the BASE24 product and the originator of this message.	
3–4	05	PRODUCT-IND The BASE24 product to which this message is to be sent. For BASE24-pos messages, the valid value is 04.	PIC X(2)

Position	Level	Field Name and Description	Data Type
5–6	05	RELEASE-NUM The assigned BASE24-pos software release number of the host destination DPC.	PIC X(2)
7–10	05	DPC-NUMBER The assigned DPC location number used for routing to this host.	PIC 9(4)
11–14	05	CARD-FIID The FIID of the card issuer.	PIC X(4)
15	05	ORIGI-TYPE A code indicating where this transaction originated. Valid values are: 0 = At a terminal connected to BASE24-pos—set by the Device Handler receiving the transaction. 1 = At an intercept processor—set by the ANSI Host Interface receiving the transaction. 3 = Originated on or was authorized by an interchange processor—set by the Interchange Interface process handling the transaction.	PIC X
16–19	05	CARD-LOG-NET The logical network to which the card issuer belongs.	PIC X(4)
20–21	05	PSEM-TYP The PSEM format used for this transaction. Valid values are: 00 = PSEMA (short form) 01 = PSEMF (long form) 02 = PSEMAE (extended short form, with address verification) 03 = PSEMFE (extended long form, with address verification)	PIC X(2)

Position	Level	Field Name and Description	Data Type
22–23	05	CARD-TYP The type of card used to initiate this transaction.	PIC X(2)
24	05	AUTHORIZER The authorizer that authorized this transaction. The ANSI Host Interface sets the value in this field. It is used by Router when reversals are generated. The value in this field must be returned with all reversals.	PIC X
25–28	02	TYP The specific message type. Valid values are: 0200 = Financial Transaction Request 0201 = Financial Transaction Request Repeat 0210 = Financial Transaction Request Response 0220 = Financial Transaction Advice 0221 = Financial Transaction Advice Repeat 0230 = Financial Transaction Advice Response 0400 = Reversal 0401 = Reversal Repeat 0410 = Reversal Response 0411 = Reversal Response Repeat 0500 = Acquirer Reconciliation Request 5400 = Online Adjustment 5401 = Online Adjustment Repeat 5410 = Online Adjustment Acknowledgment 5411 = Online Adjustment Acknowledgment Repeat 9220 = Force-post SAF (ANSI Host Interface to host) 9221 = Force-post SAF Repeat (ANSI Host Interface to host) 9230 = Response SAF (Host) 9231 = Response SAF (Host) 9500 = SAF Settlement 9501 = SAF Settlement Repeat 9510 = SAF Settlement Acknowledgment 9511 = SAF Settlement Repeat Acknowledgment	PIC 9(4)

Position	Level	Field Name and Description	Data Type
29–44	02	BYTE-MAP A 16-byte representation of the X9.2 standard 64-bit map. The ANSI X9.2 message standard mandates a 64-bit map to indicate the presence or absence of particular data fields in the message format. Because such a bit map can lead to communications line transparency problems, BASE24 uses a 16-byte representation. In the X9.2 standard, bits are numbered from 1 to 64 rather than from 0 to 63. The hexadecimal value of each four bits of the bit map is represented by one 8-bit character. For example, the bit string 1011 (hexadecimal B) is represented by ASCII/EBCDIC B. Thus, 16 bytes are used to represent 64 bits. BASE24 does not use this field for processing and will not perform edits on its values. However, BASE24 always places the following values in this field: BYTE MAP = FFCA1802E2007FE6 BASE24 does not check the Bit Map extender bit (bit map position 64), nor does it support the extended bit map feature.	PIC X(16)
45–217	02	TRAN The following fields contain information about the transaction. If the BASE24-pos Short External Message (PSEMA) format is being used, the position of these fields is from 45 to 217. If the BASE24-pos Extended Short External Message (PSEMAE) format is being used, the position of these fields is from 45 to 247.	
45–65	04	CARD The following fields contain information about the card used to initiate the transaction.	

Position	Level	Field Name and Description	Data Type
45–46	06	LEN The length of the account number used in the transaction. Currently, BASE24-pos supports account numbers up to 19 digits long.	PIC 9(2)
47–65	06	NUM The number of the account involved in this transaction.	PIC X(19)
66–70	04	TRAN-CDE The type of transaction. The field format is CCTAA, where CC is the transaction code, T is the card type, and AA is the account type. Valid transaction codes (CC) are: 10 = Normal purchase 11 = Preauthorization purchase 12 = Preauthorization completion 13 = Mail/phone order 14 = Merchandise return 15 = Cash advance 16 = Card verification 17 = Balance inquiry 18 = Purchase with cash back 19 = Check verification 20 = Check guarantee 21 = Purchase adjustment 22 = Merchandise return adjustment 23 = Cash advance adjustment 24 = Cash back adjustment Note: BASE24-pos ANSI Host Interface currently does not support values 19 and 20. Valid card types (T) are: 0 = None 1 = Credit 2 = Debit	PIC X(5)

Position	Level	Field Name and Description	Data Type
		Valid account types (AA) are: 00 = None 01–09 = DDA 11–19 = Savings 31–39 = Credit	
71–79	04	AMT-1 The transaction amount requested. For 0220 or 0400 messages, this field contains the amount originally requested. For 5400 messages, this field contains the original completed amount. For preauthorization transaction requests, this field contains the amount to be held. For preauthorization completions, this field contains the completed amount. For 0210 cash back messages, this field contains the total amount of the transaction (the actual cash back amount is contained in the AMT-2 field).	TYPE 9(7)V99
80–88	04	AMT-2 The completion amount or the cash back amount. For 5400 messages, this field contains the actual (new) completed amount. For 0210 cash back messages, this field contains the actual cash back amount (the AMT-1 field contains the total amount).	TYPE 9(7)V99
89–94	04	TRACE-NO A trace number used to match transactions to their correct response timers when responses are received. The ANSI Host Interface sets the value in this field.	PIC 9(6)

Position	Level	Field Name and Description	Data Type
95–98	04	TRAN-DAT The date (MMDD) this transaction was initiated at the POS terminal.	PIC X(4)
99–102	04	POST-DAT The date (MMDD) this transaction will be reported by BASE24-pos.	PIC X(4)
103–108	04	TRAN-TIM The time (HHMMSS) this transaction was initiated at the POS terminal.	
103–104	06	HH	PIC 99
105–106	06	MM	PIC 99
107–108	06	SS	PIC 99
109–112	04	EXP-DAT The expiration date of the card used to initiate this transaction.	PIC X(4)
113–116	04	SIC The Standard Industrial Code (SIC) number assigned to this retailer, as defined in the PRDF or PTDF.	PIC X(4)
117–126	04	RTTN The transit/routing number or identification number from this terminal's PTDF record.	PIC 9(10)
127–128	04	REVERSAL-CDE The reason for the reversal or adjustment. This reversal or adjustment code is used only for reversal (0400) and adjustment (5400) transactions, respectively.	PIC 9(2)

Position	Level	Field Name and Description	Data Type
		Valid reversal codes are:	
		01 = Time-out	
		02 = Command reject	
		03 = Destination not available	
		08 = Customer canceled	
		10 = Hardware error	
		19 = System error	
		20 = Suspect reversal	
		21 = Message authentication code (MAC) failure	
		22 = MAC key (KMAC) synchronization error	
		23 = Message replay error	
		24 = Invalid MAC	
		Valid adjustment codes are:	
		00 = Reason unknown	
		12 = Original amount incorrect	
		14 = Suspicious reversal override	
		15 = Misdispense reversal override	
		16 = Duplicate transaction	
		17 = Reconciliation error	
		18 = Reserved	
		19 = System error	
129–132	04	TYP-ORIG	PIC 9(4)
		The message type identifier as it appeared in the original transaction. This field is used to identify reversals in SAF transactions.	
133–134	04	POINT-TRAN-SPEC	PIC 9(2)
		A special condition which existed at the time this transaction was initiated. Valid values are:	
		00 = Authorization request	
		01 = Customer not present	
		02 = Card acceptor suspicious	
		04 = Electronic cash register interface	
		05 = Customer present but card not present	
		06 = Preauthorization request transaction	
		07 = Telephone device request	

Position	Level	Field Name and Description	Data Type
		08 = Mail order/telephone order 09 = Security alert 10 = Customer identity verified 11 = Suspected fraud 12 = Security reasons 13 = Data capture item 14 = Credit 15 = Manually entered card	
135–137	04	RESP-CDE The response code generated by Authorization for this transaction. For a list of valid response codes, refer to the <i>BASE24-pos Transaction Processing Manual</i> .	PIC 9(3)
138–145	04	RETRIEVAL-REF-NUM The sequence number associated with the transaction.	PIC X(8)
146–161	04	PIN The PIN number entered at the POS terminal for this transaction. If the PIN has already been verified, this field contains blanks.	PIC X(16)
162–169	04	APPROVAL-CODE The approval code generated by the authorizer for this transaction request.	PIC X(8)
170–202	04	RETAILER The following fields define the retailer involved in this transaction. They are taken from the POS Retailer Definition File (PRDF). For more information on the PRDF, refer to the <i>BASE24-pos Files Maintenance Manual</i> .	

Position	Level	Field Name and Description	Data Type
170–173	06	FIID The FIID of the financial institution with which this retailer is associated. The FIID must be unique within the logical network.	PIC X(4)
174–177	06	GRP The retailer group to which this retailer belongs.	PIC X(4)
178–181	06	REGN The region to which this retailer belongs.	PIC X(4)
182–200	06	ID The retailer ID used to represent this retailer throughout the BASE24 system.	PIC X(19)
201	06	DRAFT-CAPTURE The draft capture value associated with the transaction.	PIC 9
202	06	SETTLE-FLAG This field currently is not used by BASE24-pos.	PIC 9
203	04	MEMBER-NUMBER The number identifying the specific cardholder who initiated this transaction.	PIC X
204–213	04	REFERRAL-PHONE The referral phone number for this transaction.	PIC X(10)

Position	Level	Field Name and Description	Data Type
214–217	04	PRE-AUTH-HOLD The maximum amount of time preauthorized funds are to be held. This field is used only for preauthorization hold transactions.	PIC 9(4)
218–247	04	ADDR-DATA The following fields are used only for address verification transactions. The use of these fields indicates that the message is in the BASE24-pos Extended Short External Message (PSEMAE) format. If these fields are not used, the message is in the BASE24-pos Standard External Message (PSEMA) format.	
218–237	06	ADDR The cardholder's billing address.	PIC X(20)
238–246	06	ZIP-CDE The ZIP code for the cardholder's billing address. The ZIP code can be either five or nine digits long. If the ZIP code is five digits long, it is left justified and blank-filled.	PIC X(9)
247	06	ADDR-VRFY-STAT A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's database. Valid values are: A = ADDRESS—Addresses matched, but ZIP codes did not match. E * = ERROR—The transaction was not eligible for address verification or an editing error occurred while attempting to process the message. N = NO—Addresses did not match and ZIP codes did not match. R * = RETRY—Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on BASE24-pos.	PIC X

Position	Level	Field Name and Description	Data Type
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- | | | | |
|--|--|---|--|
| S * | | SERVICE NOT SUPPORTED—BASE24-pos authorized the transaction, but did not have the add-on Address Verification module. | |
| U * | | UNAVAILABLE—Address information was not available to the processor performing address verification. | |
| W | | WHOLE ZIP—Nine-digit ZIP codes matched, but addresses did not match. | |
| X | | EXACT—Addresses and nine-digit ZIP codes matched. | |
| Y | | YES—Addresses and five-digit ZIP codes matched. | |
| Z | | ZIP—Five-digit ZIP codes matched, but addresses did not match. | |
| ␣ * | | Address verification information was not included in this transaction (␣ denotes a blank character). | |
| 0 * | | Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on BASE24-pos, but declined before address verification could be performed also carries this code. | |
| <p>* Identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. Interchange Interfaces substitute the interchange-specific value before sending the message from BASE24-pos to the interchange. Refer to the <i>BASE24-pos Address Verification Manual</i> for a listing of interchange-specific values. Interchange Interfaces do not change a value received from the interchange.</p> | | | |

PSEMF and PSEMFE Formats

The following pages describe the fields that make up the BASE24-pos Standard Long External Message (PSEMF) and BASE24-pos Extended Long External Message (PSEMFE) formats.

PSEMF Length: 617 bytes

PSEMFE Length: 647 bytes

Position	Level	Field Name and Description	Data Type
1–24	02	ROUTE The following fields contain routing data.	
1–2	03	STAT The status of a message at the system level. If this field is set to 00, the response code field is used to determine the disposition of the message. The ANSI Host Interface sets the value in this field. It is used by Authorization. Valid values are: 00 = No error 11 = Destination not available 12 = Line down 20 = Originator not available 21 = Unknown message type 22 = Unknown card number	PIC 9(2)
3–24	03	PATH The following fields define the BASE24 product and the originator of this message.	
3–4	05	PRODUCT-IND The BASE24 product to which this message is to be sent. For BASE24-pos messages, the valid value is 04.	PIC X(2)

Position	Level	Field Name and Description	Data Type
5–6	05	RELEASE-NUM The BASE24-pos software release number of the host destination DPC.	PIC X(2)
7–10	05	DPC-NUMBER The DPC location number used for routing to this host.	PIC 9(4)
11–14	05	CARD-FIID The FIID of the card issuer.	PIC X(4)
15	05	ORIGI-TYPE A code indicating where this transaction originated. Valid values are: 0 = At a terminal connected to BASE24-pos—set by the Device Handler receiving the transaction. 1 = At an intercept processor—set by the ANSI Host Interface receiving the transaction. 3 = Originated on or was authorized by an interchange processor—set by the Interchange Interface handling the transaction.	PIC X
16–19	05	CARD-LOG-NET The logical network to which the card issuer belongs.	PIC X(4)
20–21	05	PSEM-TYP The PSEM format used for this transaction. Valid values are: 00 = PSEMA (short form) 01 = PSEMF (long form) 02 = PSEMAE (extended short form, with address verification) 03 = PSEMFE (extended long form, with address verification)	PIC X(2)

Position	Level	Field Name and Description	Data Type
22–23	05	CARD-TYP The type of card used to initiate this transaction.	PIC X(2)
24	05	AUTHORIZER The authorizer which authorized this transaction. When reversals are generated, the ANSI Host Interface sets the value in this field. It is used by Router. The value in this field must be returned with all reversals.	PIC X
25–28	02	TYP The specific message type. Valid values are: 0200 = Financial Transaction Request 0201 = Financial Transaction Request Repeat 0210 = Financial Transaction Request Response 0220 = Financial Transaction Advice 0221 = Financial Transaction Advice Repeat 0230 = Financial Transaction Advice Response 0400 = Reversal 0401 = Reversal Repeat 0410 = Reversal Response 0411 = Reversal Response Repeat 0500 = Acquirer Reconciliation Request 0700 = Balance Inquiry 0710 = Balance Inquiry Response 5400 = Online Adjustment 5401 = Online Adjustment Repeat 5410 = Online Adjustment Acknowledgment 5411 = Online Adjustment Acknowledgment Repeat 9220 = Force-post SAF (ANSI Host Interface to host) 9221 = Force-post SAF Repeat (ANSI Host Interface to host) 9230 = Response SAF (Host) 9231 = Response SAF (Host) 9500 = SAF Settlement 9501 = SAF Settlement Repeat 9510 = SAF Settlement Acknowledgment 9511 = SAF Settlement Repeat Acknowledgment	PIC 9(4)

Position	Level	Field Name and Description	Data Type
29–44	02	BYTE-MAP A 16-byte representation of the X9.2 standard 64-bit map. The ANSI X9.2 message standard mandates a 64-bit map to indicate the presence or absence of particular data fields in the message format. Because such a bit map can lead to communications line transparency problems, BASE24 uses a 16-byte representation. In the X9.2 standard, bits are numbered from 1 to 64 rather than from 0 to 63. The hexadecimal value of each four bits of the bit map is represented by one 8-bit character. For example, the bit string 1011 (hexadecimal B) is represented by ASCII/EBCDIC B. Thus, 16 bytes are used to represent 64 bits. BASE24 does not use this field for processing and will not perform edits on its values. However, BASE24 always places the following values in this field: BYTE MAP = FFCA1802E2007FE6 BASE24 does not check the Bit Map extender bit (bit map position 64), nor does it support the extended bit map feature.	PIC X(16)
29–44	02	RETAILERS	REDEFINES BYTE-MAP
29–32	04	R-FIID	PIC X(4)
33–36	04	R-GRP	PIC X(4)
37–40	04	R-REGN	PIC X(4)
41–44	04	FILLER	PIC X(4)
45–617	02	TRAN The following fields contain information about the transaction. If the BASE24-pos Long External Message (PSEMF) format is being used, the position of these fields is from 45 to 617. If the BASE24-pos Extended Long External Message (PSEMAE) format is being used, the position of these fields is from 45 to 647.	

Position	Level	Field Name and Description	Data Type
45–65	04	CARD	
45–46	06	LEN The length of the account number.	PIC 9(2)
47–65	06	NUM The account number.	PIC X(19)
66–70	04	TRAN-CDE The type of transaction. The format is CCTAA, where CC is the transaction code, T is the card type, and AA is the account type. Valid transaction codes (CC) are: 10 = Normal purchase 11 = Preauthorization purchase 12 = Preauthorization purchase completion 13 = Mail/phone order 14 = Merchandise return 15 = Cash advance 16 = Card verification 17 = Balance inquiry 18 = Purchase with cash back 19 = Check verification 20 = Check guarantee 21 = Purchase adjustment 22 = Merchandise return adjustment 23 = Cash advance adjustment 24 = Cash back adjustment Note: Transaction codes 19 and 20 are not supported by the BASE24-pos ANSI Host Interface. Valid card types (T) are: 0 = None 1 = Credit 2 = Debit	PIC X(5)

Position	Level	Field Name and Description	Data Type
		Valid account types (AA) are: 00 = None 01–09 = Checking account 11–19 = Savings account 31–39 = Credit account	
71–79	04	AMT-1 The transaction amount requested. In the case of type 0220 or 0400 messages, this field contains the amount originally requested. For 5400 messages, this field contains the original completed amount. For preauthorization transaction requests, this field contains the amount to be held. For preauthorization completions, this field contains the completed amount. For 0210 cash back messages, this field contains the total amount of the transaction (the actual cash back amount is contained in the AMT-2 field).	TYPE 9(7)V99
80–88	04	AMT-2 For 5400 messages, this field contains the actual (new) completed amount. For 0210 cash back messages, this field contains the actual cash back amount (the AMT-1 field contains the total amount).	TYPE 9(7)V99
89–94	04	TRACE-NO The trace number. The ANSI Host Interface sets and uses the value in this field to match transactions to their correct response timers when responses are received.	PIC 9(6)
95–98	04	POST-DAT The date (MMDD) this transaction will be reported by BASE24-pos.	PIC X(4)

Position	Level	Field Name and Description	Data Type
99–102	04	TRAN-DAT The date (MMDD) the transaction was initiated.	PIC X(4)
103–108	04	TRAN-TIM The time (HHMMSS) the transaction was initiated, described by the following fields.	
103–104	06	HH	PIC 99
105–106	06	MM	PIC 99
107–108	06	SS	PIC 99
109–112	04	EXP-DAT The expiration date (YYMM) of the card that initiated this transaction.	PIC X(4)
113–116	04	SIC The Standard Industrial Code (SIC) number assigned to this retailer, as defined in the PRDF or PTDF.	PIC X(4)
117–126	04	RTTN The transit/routing number or identification number from this terminal's PTDF record.	PIC 9(10)
127–128	04	REVERSAL-CDE The reversal code or adjustment code indicating the reason for the reversal or adjustment, respectively. This field is used only for reversal (0400) and adjustment (5400) transactions. Valid reversal codes are: 01 = Time-out 02 = Command reject 03 = Destination not available 08 = Customer canceled 10 = Hardware error	PIC 9(2)

Position	Level	Field Name and Description	Data Type
		19 = System error 20 = Suspect reversal 21 = Message authentication code (MAC) failure 22 = MAC key (KMAC) synchronization error 23 = Message replay error 24 = Invalid MAC Valid adjustment codes are: 00 = Reason unknown 12 = Original amount incorrect 14 = Suspicious reversal override 15 = Misdispense reversal override 16 = Duplicate transaction 17 = Reconciliation error 18 = Reserved 19 = System error	
129–132	04	TYP-ORIG The message type identifier as it appeared in the original transaction. This field value is used to identify reversals in SAF transactions.	PIC 9(4)
133–134	04	POINT-TRAN-SPEC A special condition which existed at the time this transaction was initiated. Valid values are: 00 = Authorization request 01 = Customer not present 02 = Card acceptor suspicious 04 = Electronic cash register interface 05 = Customer present but card not present 06 = Preauthorization transaction request 07 = Telephone device request 08 = Mail order/telephone order 09 = Security alert 10 = Customer identity verified 11 = Suspected fraud 12 = Security reasons	PIC 9(2)

Position	Level	Field Name and Description	Data Type
		13 = Data capture item 14 = Credit 15 = Manually entered card	
135–137	04	RESP-CDE The response code generated by Authorization for this transaction. For a list of valid response codes, refer to the <i>BASE24-pos Transaction Processing Manual</i> .	PIC X(3)
138–162	04	CARD-ACCEPTOR-NAME The name of the retailer who accepted the card to perform this transaction.	PIC X(25)
163–177	04	CARD-ACCEPTOR-CITY-ST The city and state in which the card-accepting retailer is located.	PIC X(15)
178–185	04	RETRIEVAL-REF-NUM The sequence number associated with the transaction.	PIC X(8)
186–201	04	PIN The PIN entered at the POS terminal for this transaction. If the PIN has already been verified, this field contains blanks.	PIC X(16)
202–203	04	FROM-ACCT-LEN The length of the <i>from</i> account number used in this transaction.	PIC 9(2)
204–222	04	FROM-ACCT-NUM The <i>from</i> account number used in this transaction.	PIC X(19)

Position	Level	Field Name and Description	Data Type
223–230	04	APPROVAL-CODE The approval code generated by the authorizer for this transaction request.	PIC X(8)
231–302	04	RETAILER The following fields define the retailer involved in this transaction.	
231–234	06	LOGICAL-NET The logical network of the institution owning the terminal. This field value is taken from the TERM-OWNER. LOGICAL-NET field in the PTDF.	PIC X(4)
235–238	06	FIID The FIID, as defined in the Institution Definition File (IDF), of the financial institution with which this retailer is associated. The FIID is an identifier that must be unique within the logical network. This field value is taken from the RKEY.FI-ID field in the PRDF.	PIC X(4)
239–242	06	GRP The retailer group to which this retailer belongs. This field value is taken from RKEY.GRP field in the PRDF.	PIC X(4)
243–246	06	REGN The region group to which this retailer belongs. This field value is taken from RKEY.REGN field in the PRDF.	PIC X(4)
247–265	06	ID A unique identifier, retailer ID, used to represent this retailer throughout the BASE24 system. This field value is taken from RKEY.RETAILER-ID field in the PRDF.	PIC X(19)

Position	Level	Field Name and Description	Data Type
266–281	06	TERM-ID A unique identifier, terminal ID, for the terminal. This field value is taken from TERM-ID field in the PTDF.	PIC X(16)
282–284	06	SHIFT Taken from NUM-SHIFTS field in the PTDF.	PIC X(3)
285–300	06	EMPLOYEE The following fields identify the employee initiating this transaction.	
285–288	08	ID The ID of the employee who initiated the last request at this terminal. This field value is taken from the CLERK-ID field in the PTDF.	PIC X(4)
289–292	08	GRP The group number, as defined in the CRT Authorization Security File (CSEC), of the CRT Authorization operator. This field is used with CRT Authorization.	PIC X(4)
293–300	08	USER-ID The user number, as defined in the CRT Authorization Security File (CSEC), of the CRT Authorization operator. This field is used with CRT Authorization.	PIC X(8)
301	06	DRAFT-CAPTURE A flag that defines how draft-captures are handled. This field value is taken from the SRV.TRAN-PROFILE field in the PTDF.	PIC 9

Position	Level	Field Name and Description	Data Type
302	06	SETTLE-FLAG (NOT USED)	PIC 9
303	04	MEMBER-NUMBER The specific cardholder who initiated this transaction.	PIC X
304–327	04	SHARING (NOT USED)	PIC X (OCCURS 24 TIMES)
328—575	04	MISC	
328–331	06	ORIG A code identifying the entity on which this transaction originated. If the transaction originated at a terminal directly connected to BASE24, the literal B24 is used. If the transaction originated at an acquirer terminal not controlled by BASE24, the literal HOST is used. If the transaction originated at an interchange, the literal indicates the originating interchange.	PIC X(4)
332–335	06	DEST A code identifying the authorizing entity for this transaction.	PIC X(4)
336–345	06	INVOICE-NUM The transaction invoice number that is generated by POS terminals. Use of invoice numbers is optional by retailer. When a retailer does not use invoice numbers, this field contains zeros.	PIC X(10)
346–355	06	ORIG-INVOICE-NUM For merchandise returns, the invoice number of the original transaction in which this merchandise was purchased.	PIC X(10)

Position	Level	Field Name and Description	Data Type
356–358	06	CURRENCY-CDE The currency code from the PTDF for this terminal.	PIC X(3)
359–361	06	BATCH-NUM The batch to which this transaction belongs.	PIC X(3)
362–371	06	REFERRAL-PHONE The referral phone number for this transaction.	PIC X(10)
372–375	06	PRE-AUTH-HOLD The maximum amount of time preauthorized funds are to be held. This field is used only for preauthorization hold transactions.	PIC 9(4)
376—575	06	PRODUCT The following fields are used when information is passed up from the terminal. The length (LEN) indicates how much data (DAT) to use.	
376–378	08	LEN The amount (length) of data.	PIC 9(3)
379–575	08	DAT The data passed from the terminal.	PIC X(197)
576–577	04	TRACK2-LEN The length of Track 2 data from this card.	PIC 9(2)

Position	Level	Field Name and Description	Data Type
578–617	04	TRACK2-DATA The Track 2 data, as sent to the Device Handler by the POS terminal or to the ANSI Host Interface by the acquirer. If the card number was entered manually, the first byte contains an M, followed by the card number, a field separator, the expiration date (YYMM), and the member number.	PIC X(40)
618–647	04	ADDR-DATA The following fields are used only for address verification transactions. The use of these fields indicates that the message is in extended long form (PSEMFE). If these fields are not used, the message is in long form (PSEMF).	
618–637	06	ADDR The cardholder's billing address.	PIC X(20)
638–646	06	ZIP-CDE The ZIP code for the cardholder's billing address. The ZIP code must be either five or nine digits long. If the ZIP code is five digits long, it is left-justified and blank-filled.	PIC X(9)
647	06	ADDR-VRFY-STAT A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the processor's database. Valid values are: A = ADDRESS—Addresses matched, but ZIP codes did not match. E * = ERROR—The transaction was not eligible for address verification or an editing error occurred while attempting to process the message. N = NO—Addresses did not match and ZIP codes did not match. R * = RETRY—Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on BASE24-pos.	PIC X

Position	Level	Field Name and Description	Data Type
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|-----|--|---|--|
| S * | | = SERVICE NOT SUPPORTED—BASE24-pos authorized the transaction, but did not have the add-on Address Verification module. | |
| U * | | = UNAVAILABLE—Address information was not available to the processor performing address verification. | |
| W | | = WHOLE ZIP—Nine-digit ZIP codes matched, but addresses did not match. | |
| X | | = EXACT—Addresses and nine-digit ZIP codes matched. | |
| Y | | = YES—Addresses and five-digit ZIP codes matched. | |
| Z | | = ZIP—Five-digit ZIP codes matched, but addresses did not match. | |
| ␣ * | | = Address verification information was not included in this transaction (␣ denotes a blank character). | |
| 0 * | | = Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on BASE24-pos, but declined before address verification could be performed also carries this code. | |

* Identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. Interchange Interfaces substitute the interchange-specific value before sending the message from BASE24-pos to the interchange. Refer to section 1 of the **BASE24-pos Address Verification Manual** for a listing of interchange-specific values. Interchange Interfaces do not change a value received from the interchange.

45–617	02	BAL	REDEFINES TRAN
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45–50	04	TRACE-NO	PIC 9(6)
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The transaction trace number supplied by the ANSI Host Interface.

Position	Level	Field Name and Description	Data Type
51–53	04	FILLER	PIC X(3)
54–207	04	SETREC The following fields are used for settlement (0500) transactions.	
54–72	06	RETAILER-ID The ID of the retailer performing this settlement transaction.	PIC X(19)
73–88	06	TERMINAL-ID The ID of the POS terminal from which this settlement transaction is being performed.	PIC X(16)
89–107	06	RETAILER-ACCT-NUM The account number of this retailer’s account at the institution.	PIC X(19)
108–147	06	RETAILER-NAME The name of this retailer.	PIC X(40)
148–157	06	RETL-RTTN This retailer’s route/transit number.	PIC X(10)
158	06	SETTLE-TYP The type of settlement transaction reflected in this record. Valid values are: 0 = Shift terminal totals 1 = Daily terminal totals 2 = Network totals 5 = Batch terminal totals 9 = Services	PIC 9

Position	Level	Field Name and Description	Data Type
159–161	06	NUM-BATCHES (NOT USED)	PIC 9(3)
162	06	BAL-FLAG The method invoked to perform the last balancing of this terminal. Valid values are: 5 = Balanced; all transactions occurring at this terminal were written to the PTLF. 6 = Balanced; due to problems, some transactions were not written to the PTLF and, therefore, require verification. 8 = Force-balanced (incomplete batch); refer to the following description. 9 = Force-balanced; Settlement Initiator cut the terminal over at the time specified in the RETAILER-BAL-AND-CUTVR-END field in the PRDF. The terminal was cut over to a new reporting day and its totals were written to the PTLF, but only current network totals were reset to 0. Force-balance occurs when the terminal has not been balanced by a balancing transaction prior to the time specified in the RETAILER-BAL-AND-CUTVR-END field in the PRDF or when the Device Handler cuts a terminal over because the TERMINAL CUTOVER field in the PTDF contains the value 4.	PIC X
163–165	06	SHIFT-NUM The shift number that this transaction reflects.	PIC 9(3)
166–168	06	BATCH-NUM The number of the current batch.	PIC 9(3)
169–171	06	SEQ-NUM (NOT USED)	PIC 9(3)
172–177	06	TRAN-DAT The date this transaction occurred.	PIC X(6)

Position	Level	Field Name and Description	Data Type
178–183	06	TRAN-TIM The time this transaction occurred.	PIC X(6)
184	06	OB-FLAG When the terminal is out of balance with BASE24, this field indicates whether this balancing record has an associated balancing record. This field applies to records flagged as type 5 or 6 in the BAL-FLAG field. Valid values are: 0 = No associated record 1 = Has associated record	PIC X
185–194	06	ACH-COMP-ID This field is reserved for any information that may be required for ACH processing of transactions from this retailer. Information in this field is only passed along by BASE24-pos; it in no way affects BASE24-pos processing.	PIC X(10)
195–204	06	BILLING-INFO A free-form informational field used at the discretion of the institution. Typically, this field contains ACH information which BASE24-pos passes on to the host for use by the ACH system. Information in this field has no impact on BASE24-pos processing.	PIC X(10)
205–207	06	FILLER	PIC X(3)
208—617	04	STL The following fields contain balancing transaction counts and amounts from the PTDF for this terminal.	

Position	Level	Field Name and Description	Data Type
208–246	06	B-DC The following fields indicate batch draft capture counts and amounts for debits, credits, and adjustments.	
208–211	08	DB-CNT	PIC 9(4)
212–220	08	DEBIT	PIC 9(7)V99
221–224	08	CR-CNT	PIC 9(4)
225–233	08	CREDIT	PIC 9(7)V99
234–237	08	ADJ-CNT	PIC 9(4)
238–246	08	ADJ	PIC 9(7)V99
247–285	06	B The following fields indicate batch total counts and amounts for debits, credits, and adjustments.	
247–250	08	DB-CNT	PIC 9(4)
251–259	08	DEBIT	PIC 9(7)V99
260–263	08	CR-CNT	PIC 9(4)
264–272	08	CREDIT	PIC 9(7)V99
273–276	08	ADJ-CNT	PIC 9(4)
277–285	08	ADJ	PIC 9(7)V99
286–324	06	S-DC The following fields indicate shift draft capture total counts and amounts for debits, credits, and adjustments.	
286–289	08	DB-CNT	PIC 9(4)
290–298	08	DEBIT	PIC 9(7)V99
299–302	08	CR-CNT	PIC 9(4)
303–311	08	CREDIT	PIC 9(7)V99
312–315	08	ADJ-CNT	PIC 9(4)
316–324	08	ADJ	PIC 9(7)V99

Position	Level	Field Name and Description	Data Type
325–363	06	S	
		The following fields indicate shift total counts and amounts for debits, credits, and adjustments.	
325–328	08	DB-CNT	PIC 9(4)
329–337	08	DEBIT	PIC 9(7)V99
338–341	08	CR-CNT	PIC 9(4)
342–350	08	CREDIT	PIC 9(7)V99
351–354	08	ADJ-CNT	PIC 9(4)
355–363	08	ADJ	PIC 9(7)V99
364–402	06	D-DC	
		The following fields indicate daily draft capture total counts and amounts for debits, credits, and adjustments.	
364–367	08	DB-CNT	PIC 9(4)
368–376	08	DEBIT	PIC 9(7)V99
377–380	08	CR-CNT	PIC 9(4)
381–389	08	CREDIT	PIC 9(7)V99
390–393	08	ADJ-CNT	PIC 9(4)
394–402	08	ADJ	PIC 9(7)V99
403–441	06	D	
		The following fields indicate daily total counts and amounts for debits, credits, and adjustments.	
403–406	08	DB-CNT	PIC 9(4)
407–415	08	DEBIT	PIC 9(7)V99
416–419	08	CR-CNT	PIC 9(4)
420–428	08	CREDIT	PIC 9(7)V99
429–432	08	ADJ-CNT	PIC 9(4)
433–441	08	ADJ	PIC 9(7)V99

Position	Level	Field Name and Description	Data Type
442–480	06	C-N-DC The following fields indicate the current network draft capture total counts and amounts for debits, credits, and adjustments.	
442–445	08	DB-CNT	PIC 9(4)
446–454	08	DEBIT	PIC 9(7)V99
455–458	08	CR-CNT	PIC 9(4)
459–467	08	CREDIT	PIC 9(7)V99
468–471	08	ADJ-CNT	PIC 9(4)
472–480	08	ADJ	PIC 9(7)V99
481–519	06	C-N The following fields indicate the current network total counts and amounts for debits, credits, and adjustments.	
481–484	08	DB-CNT	PIC 9(4)
485–493	08	DEBIT	PIC 9(7)V99
494–497	08	CR-CNT	PIC 9(4)
498–506	08	CREDIT	PIC 9(7)V99
507–510	08	ADJ-CNT	PIC 9(4)
511–519	08	ADJ	PIC 9(7)V99
520–558	06	P-N The following fields indicate the previous network total counts and amounts for debits, credits, and adjustments.	
520–523	08	DB-CNT	PIC 9(4)
524–532	08	DEBIT	PIC 9(7)V99
533–536	08	CR-CNT	PIC 9(4)
537–545	08	CREDIT	PIC 9(7)V99
546–549	08	ADJ-CNT	PIC 9(4)
550–558	08	ADJ	PIC 9(7)V99

Position	Level	Field Name and Description	Data Type
559–617	06	FILLER	PIC X(59)
208–617	04	SRVC	OCCURS 10 TIMES REDEFINES STL
		The following fields identify the service type and break down transaction counts and amounts according to that service type. Each occurrence of SRVC is 41 positions in length.	
	06	TYP	PIC X(2)
	06	DB-CNT	PIC 9(4)
	06	DEBIT	PIC 9(7)V99
	06	CR-CNT	PIC 9(4)
	06	CREDIT	PIC 9(7)V99
	06	ADJ-CNT	PIC 9(4)
	06	ADJ	PIC 9(7)V99
	04	STL-TBL	REDEFINES STL
208–558	06	TOT	OCCURS 9 TIMES
		The following fields indicate the total counts and amounts for this terminal in terms of debits, credits, and adjustments. Each occurrence of TOT is 39 positions in length.	
	08	XDB-CNT	PIC 9(4)
	08	XDEBIT	PIC 9(7)V99
	08	XCR-CNT	PIC 9(4)
	08	XCREDIT	PIC 9(7)V99
	08	XADJ-CNT	PIC 9(4)
	08	XADJ	PIC 9(7)V99
559–617	06	FILLER	PIC X(59)

Position	Level	Field Name and Description	Data Type
208–617	04	P-D	REDEFINES STL
		The following fields are a product data redefinition area.	
208–210	06	LEN	PIC 999
211–407	06	D	PIC X(197)
408–617	06	FILLER	PIC X(210)
618–647	04	FILLER	PIC X(30)

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